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TFOS DEWS III Digest Report

Fiona Stapleton , Pablo Argüeso , Penny Asbell , Dimitri Azar , Charles Bosworth , Wei Chen , Joseph Ciolino , Jennifer P. Craig , Juana Gallar , Anat Galor , José A.P. Gomes , Isabelle Jalbert , Ying Jie , Lyndon Jones , Kenji Konomi , Yang Liu , Jesus Merayo-Lloves , Fabiola R. Oliveira , Victor A. Perez Quinones , Eduardo M. Rocha , Benjamin D. Sullivan , David A. Sullivan , Jelle Vehof , Susan Vitale , Mark Willcox , James Wolffsohn , Murat Dogru

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TFOS DEWS III Digest Report

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1. Abstract

This digest summarises the interdisciplinary research in dry eye disease (DED) published since the 2017 TFOS DEWS II reports. It comprises seven topics including Sex, Gender, and Hormones, Epidemiology, Pathophysiology, Tear Film, Pain and Sensation, Iatrogenic and Clinical Trial Design and explores how each of these inform diagnostic methodology, disease subtype and management of DED.

Sex- and gender-related differences significantly influence the ocular surface due to hormones, sex chromosomes, sex-specific autosomal factors, epigenetics, care-seeking behaviors, and service utilization. Epidemiological data reveal that DED prevalence varies by age and sex, influenced by diagnostic criteria and the multifactorial nature of the disease. New risk factors for DED include environmental, iatrogenic, systemic diseases and lifestyle domains.

Pathophysiological distinctions between Aqueous Deficient Dry Eye (ADDE) and Evaporative Dry Eye (EDE) have been clarified. EDE is characterized by a muted inflammatory response at the ocular surface, meibomian gland dysfunction and conceivably phenotypic changes in corneal epithelial cells. There is an expanding role for metabolic, hormonal, physical, neural and cellular stresses, including hyperosmolarity, mitochondrial stress, and neurogenic inflammation.

Advancements in tear film research recommend new approaches to understanding DED pathogenesis and identifying biomarkers, such as microRNAs. Ocular pain perception is linked to structural integrity of corneal nerves, functional capacities of neurons, and activity of the central and peripheral nervous systems. Iatrogenic DED can result from medications, contact lenses, and surgical procedures. Clinical trials now emphasize aligning design and endpoints with DED subtypes and therapeutic mechanisms, with new therapeutics and trial designs under consideration.

2. Keywords

Epidemiology, prevalence, risk factors, hormones, sex, gender, tear film, pathophysiology, neurogenic, clinical trials, iatrogenic

3. Abbreviations

ADDE	Aqueous-deficient dry eye disease
BALB/c	BALB/C wildtype (mice)
CXL	Corneal cross-linking
DED	Dry eye disease
DEQ-5	5-Item Dry Eye Questionnaire

DEWS	Dry eye workshops
DNA	Deoxyribonucleic acid
EDE	Evaporative dry eye disease
FDA	Food and drug administration
(f)MRI	(Functional) magnetic resonance imaging
GVHD	Graft versus host disease
HLA	Human leukocyte antigen
HPMC	Hydroxypropyl methylcellulose
ICAM-1	Intercellular Adhesion Molecule 1
IL	Interleukin
IGF	Insulin-like growth factor
IVCM	*In vivo* confocal microscopy
LASIK	Laser-assisted *in situ* keratomileusis
LIPCOF	Lid-parallel conjunctival folds
LLT	Lipid layer thickness
MGD	Meibomian gland dysfunction
MGYLS	Meibomian glands yielding liquid secretions
MMP	Matrix metalloproteinase
MUC4	Mucin 4
NGF	Nerve growth factor
NIBUT	Non-invasive tear film breakup time
NF-kB	Nuclear factor kappa-light-chain-enhancer of activated B cells
NRS	Numerical rating scale
OCT	Optical coherence tomography
OR	Odds ratio
OSDI	Ocular Surface Disease Index
PRK	Photorefractive keratectomy
QoL	Quality of Life
SANDE	Symptom Assessment iN Dry Eye questionnaire
SPEED	Standard Patient Evaluation of Eye Dryness questionnaire
TBUT	Tear film breakup time
TED	Thyroid eye disease
TFOS	Tear Film & Ocular Surface Society
TRPM8	Transient Receptor Potential cation channel subfamily M member 8
VAS	Visual analogue scale

4. Introduction

This review updates the evidence for interdisciplinary aspects of dry eye disease (DED). It considered novel, human and animal evidence-based research published between 2017 and 2024. The goal of this report was to identify key

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research published since the 2017 TFOS DEWS II Workshop reports to underpin the evidence described in the TFOS DEWS III Diagnostic Methodology [1] and Management and Therapy [2] reports. The topics include sex, gender, and hormones, epidemiology including prevalence and risk factors, novel concepts and findings associated with pathophysiology, relevant changes to the tear film and interactions, the mechanisms related to ocular pain and sensation, iatrogenic causes of DED and their unique management and a synthesis of clinical trial designs to inform exploration of new treatment modalities.

5. Sex, gender, and hormones

5.1 Introduction

The TFOS DEWS II report on Sex, Gender, and Hormones report [3] addressed many sex- and gender-related differences that significantly influence the ocular surface in health and DED. Many of these differences appeared to be due to the effects of hormones, sex chromosomes, sex-specific autosomal factors, epigenetics, care-seeking behaviors and service utilization. [3] The purpose of this section is to highlight some of the relevant research since the publication of that report. The focus is primarily on studies published after July 1, 2017.

Additional studies related to sexual health and DED may be found in the recent detailed review, entitled "TFOS Lifestyle: Impact of lifestyle challenges on the ocular surface". [4]

5.2 Sex-related differences in the ocular surface and adnexa

Significant sex-related differences in the lacrimal gland, meibomian gland, cornea and eyelid, reported since publication of the TFOS DEWS II report on Sex, Gender, and Hormones, [3] are briefly highlighted below. These differences may contribute to the increased prevalence of DED in females.

5.2.1 Lacrimal gland

Significant, sex-related differences exist in the gene expression, morphology, and pathophysiology of the lacrimal gland. The sex-associated differences in lacrimal gland gene expression may be very important in promoting lymphocyte accumulation in this tissue and contributing to the onset, progression, and/or severity of the inflammatory disease process in Sjögren disease. [5] This condition is an multisystem autoimmune disease affecting the exocrine glands including the salivary and lacrimal glands that occurs primarily in women, and is associated with aqueous-deficient DED. [6] Murine models of autoimmune lacrimal gland disease, such as MRL/MpJ-Tnfrsf6^{lpr} and non-obese diabetic/LtJ mice,

have provided critical insights into sex-related immune differences. The extent of lacrimal gland inflammation in MRL/MpJ-Tnfrsf6^{lpr} mice is, as in humans, far greater in females as compared with males, whereas the magnitude of lacrimal gland inflammation is far worse in non-obese diabetic/LtJ males. [7] Results showed that sex significantly influences the expression of thousands of genes, and that the immune nature of the glandular response is very dependent on the Sjögren disease model. Lacrimal tissues of female, as compared with male, MRL/MpJ-Tnfrsf6^{lpr} mice featured a significant increase in the expression of genes related to inflammatory responses, antigen processing, and chemokine pathways. In contrast, it was the lacrimal glands of non-obese diabetic/LtJ males, and not females, that presented with a significantly greater expression of immune-related genes. These data suggest that factors in the lacrimal gland microenvironment may be critically important in mediating these sex-associated immune effects and in promoting lacrimal gland inflammation. [5]

Analyses of single-cell transcriptomes from lacrimal glands of MRL/MpJ-Tnfrsf6^{lpr}, non-obese diabetic/LtJ and wild-type (BALB/c) mice have also defined the location of multiple cell-type-specific mRNA markers and proteins, the latter of which may be secreted into the tear film in a sex-specific manner. [8]

As concerns morphology and pathophysiology, significant, sex-linked differences occur during aging in the magnitude of acinar atrophy, periacinar fibrosis, periductal fibrosis, ductal dilation, ductal proliferation, fatty infiltration, and lymphocyte infiltration in human lacrimal glands. [9] Female tissues had a higher frequency of all observed degenerative changes, except for ductal dilation, which was significantly more prevalent in male glands. [9] The authors concluded that female lacrimal glands are more susceptible to degeneration, conceivably due to hormonal influences, estrogen withdrawal or genetic susceptibility, and that this susceptibility may play a significant role in the higher prevalence of DED in older women. [9]

5.2.2 Meibomian gland

Significant sex-related differences exist in meibomian gland gene expression, but the nature of these differences may be primarily species-dependent. [10] Analysis of the 500 most highly expressed genes from human and BALB/c mouse meibomian glands demonstrated that only 24.4% were the same. Further, analysis of 100 genes with the greatest sex-associated differences in the human and mouse meibomian glands showed that none were the same, indicating that mice are not optimal models for understanding sex-associated differences in gene expression of the human meibomian glands. [10]

The prevalence of meibomian gland dysfunction (MGD) may vary by sex but results are inconsistent. Population- and hospital-based studies report that MGD may occur more frequently in males, or in females, or in neither sex. A recent prospective cross-sectional study indicated that older males had more severe lid margin abnormities and decreased gland number, height, and area compared with females. [11] Some studies suggest that the influence of sex on MGD may depend on the type (e.g. obstructive versus hypersecretory forms, [12] the patient's age, [13] and/or the individual's medical condition. [14] The inconsistency in the sex-related prevalence of MGD is notable, given that DED is more common in females [15-24] and that MGD is a major cause of DED. [6,15,25,26] The prevalence of any MGD and clinically significant MGD are separately reported in a meta-analysis by age and sex (see Section 5, Figure 3 F and G). In brief, any MGD including asymptomatic gland changes is more prevalent in men than women in older age groups but sex-differences in clinically significant MGD are equivocal. There remains a need to explore the impact of more detailed diagnostic criteria to understand sex-related effects on MGD.

5.2.3 Cornea

Significant, sex-related differences exist in corneal thickness, sensitivity, re-epithelization and DED-induced damage. As has been found previously, males have greater corneal epithelial thickness in all but the peripheral nasal zone [27] and females have higher corneal sensitivity, [28] and corneal nerve regeneration [29] and slower corneal epithelial wound healing. [30] In addition, aqueous-deficient DED elicits more ocular pain, anxiety and severe corneal damage in female mice. [31]

5.2.4 Eyelid blinking

A sexual dimorphism has also been identified in eyelid blinking. DED appears to increase sex-related differences in blinking, including heightened exaggeration of excitability in males and enhanced modifiability of the female trigeminal complex. This latter modifiability is proposed to explain the female predominance in the development of focal dystonia and benign essential blepharospasm. [32]

5.3 Sex-related differences and immunity

As stated in the TFOS Sex, Gender, and Hormones report, [3] sex-related differences are well known to occur in both innate and adaptive immunity and lead to differences in the severity and frequency of infections (male > female) and the risk of developing autoimmune diseases (female > male). These sex-based variations appear to be due several factors, including sex steroid hormones, genetics, the microbiome and non-biological factors.

Since that report was published, almost 6,000 articles have been cited in PubMed addressing the phrase "sex differences and immune." These have continued to show that sex as a biological factor significantly influences the distribution of lymphocyte subsets, quality of T cell responses, development of regulatory T cells, formation of the germinal centers, and the epigenetic accessibility of B cell loci. [33,34] Sex also impacts transcriptional differences that are often highly immune cell-specific. [35] For example, more than 50 monocyte transcripts linked to the interferon pathway, inflammatory cytokines and chemokines display sex-associated expression, that are prominent in some female subjects. [35] In addition, changes in sex steroid hormone concentrations over the course of one's life contribute to sex-related differences in immune profiles and disease susceptibility patterns. [36]

There are also significant sex-related differences in regulatory processes between transcription factors and their target genes in multiple tissues. [37] Different transcription factors may regulate genes in males and females, irrespective of whether those target genes are differentially expressed. This sex-associated pattern of gene regulation may help to explain why males and females do not often manifest disease in the same way, or respond in the same way to treatment. [37]

All of these sex-related differences in immunity should have relevance to the ocular surface and adnexa in health and disease.

5.4 Sex and gender differences in pain assessment

The intersection of sex and pain has been widely studied at clinical, psychological, and social levels. However, pain research often conflates sex and gender, and little is known about how gender identity diversity, such as gender-affirming medical procedures (hormonal or surgical therapies), environmental exposures, and minority status, impact pain. [38]

Despite chronic pain and DED being more common in women, most pain mechanism studies are based on male rodents. [39] With the inclusion of sex as a biological variable in preclinical research, studies on the influence of sex on pain and analgesia have increased. Female sex and older age are still the main factors associated with chronic pain. Experimental pain response differences suggest a biological mechanism rather than sociocultural gender-related issues. [40]

Studies have investigated whether pain sensitivity influences DED symptoms differently between sexes, accounting for ocular parameters. In a cross-sectional

study of a young and healthy cohort (194 women and 93 men), intersex differences in ocular surface and pain sensitivity were linked to higher DED symptoms in women. [41] This finding helps explain the disparity between DED symptom intensity and signs previously described in females.

5.5 Hormonal regulation of the ocular surface and adnexa

5.5.1 Androgens

As detailed in the TFOS DEWS II report, [3] androgens are extremely important in the regulation of the ocular surface and adnexa and appear to mediate many of the sex-related differences in these tissues. Androgen deficiency, in turn, is associated with, and a risk factor for, both aqueous-deficient and evaporative DED. Recent studies are discussed below.

A one-month administration of the anti-androgen, finasteride, led to the development of rat lacrimal gland inflammation and aqueous tear deficiency. [42] Conversely, testosterone treatment of female MRL/MpJ-Tnfrsf6^{lpr} mice led to a striking down-regulation of the lacrimal gland expression of over 60 immune-associated biological process ontologies (≥ 20 genes/ontology), including those related to immune system processes, lymphocyte activation, cytokine production, and inflammatory response. [43] The nature of this androgen effect was dependent upon murine strain, and the data indicate a major role for the lacrimal gland microenvironment in mediating androgen effects on immune gene expression. [43]

In human meibomian gland epithelial cells *in vitro*,dihydrotestosterone administration suppressed proinflammatory gene expression, [44] a hormone action that may contribute to the typical absence of inflammation within the human glands. [6,25] Dihydrotestosterone also inhibited the hyperosmolar-induced expression of TNF-α, IL-8 and IL-6 mRNAs in human corneal epithelial cells, [45] but had no influence on proinflammatory gene expression in unchallenged human corneal epithelial cells. [44] In contrast, dihydrotestosterone significantly increased 33 gene ontologies linked to the immune system in human conjunctival epithelial cells. [44]

Anti-androgen therapy for the treatment of prostate cancer and benign prostate hyperplasia led to a significant increase in the signs and symptoms of MGD and DED. [46,47] Transdermal androgen therapy applied to the lower abdomen, in turn, alleviated DED signs and symptoms in androgen-deficient individuals, [48] as well as DED signs in rabbits with combined androgen deficiency and MGD. [49]

Androgenetic alopecia was associated with decreased tear film breakup times, increased meiboscores, ocular surface symptoms and MGD. [50] However, this

form of hair loss is not necessarily linked to androgen excess, but is associated with an increased risk of polycystic ovary syndrome. [51]

5.5.2 Estrogens

It has recently been reported that estrogen receptor-1 deficiency in mice induces inflammation and lipid deposition in the meibomian gland and lacrimal gland. [52] Estrogen receptor-1 loss however, does not block estrogen receptor-activity, but results in an abnormal endocrine environment (e.g. increased luteinizing hormone, estradiol, testosterone, and progesterone levels) and may lead to insulin resistance. [53-55]

Ovariectomy of monkeys [56] and rats [57] engenders the signs of DED, and that these can be reversed with estrogen administration. [57] Other studies show that ovariectomy leads to rat anxiety and depressive-like behavior [58] conditions that promote DED, [4], and that estrogen treatment may amplify rat ocular hyperalgesia. [59] Some of these disparate findings regarding the role of estrogens might be explained by differences in experimental design, hormone dosage or animal model. [60] However, complete estrogen absence does not cause lacrimal gland inflammation, gross alterations in meibomian gland histology, or aqueous-deficient DED in mice, and does not play a major role in the sex-related differences of the mouse meibomian gland. [60,61]

However, is it possible that this animal research does not reflect the situation in humans? This question is prompted by the results from aromatase inhibitor studies, which report an increased prevalence of DED. [62-65] Aromatase inhibitors block the synthesis of estrogens, induce estrogen deficiency, and are used as therapy in women with hormone receptor-positive breast cancer post initial chemotherapy. [66] However, these studies had no control groups. Treatment with aromatase inhibitors is known to promote anxiety (> 112 PubMed articles), depression (> 243 PubMed articles), and sleep disturbance (> 25 PubMed articles) (e.g., [67,68], all of which are associated with DED. [4] It may be that these mental health factors and sleep disorders contribute to, and possibly account for, these DED effects seen in those treated with aromatase inhibitors.

Investigators have also suggested that the increased estrogen levels following *in vitro* fertilization are responsible for the *in vitro* fertilization-associated DED symptoms [69] and/or signs. [69,70] However, there were no controls with these studies, and the effects may also have been linked to the known *in vitro* fertilization-induced anxiety, depression and sleep disturbance. [71]

Clinicians continue to test whether topical estradiol might serve as a DED treatment. One of the most recent clinical trials showed no significant differences in effects between any of the estrogen dosages and the placebo. [72]

Lastly, estradiol may inhibit the conjunctival goblet cell response *in vitro* to an inflammatory stimulus. [73] This effect may be dose-dependent, given that the estrogen concentration used (0.1 μM) was considerably higher than the physiological range. Very high doses of estrogen often suppress, whereas physiological doses often enhance, immune responses. [74]

5.5.3 Progestins

Progesterone appears to suppress ocular pain and discomfort, which are common features of DED. Within 30 minutes after application to the rat forehead, 1% progesterone gel (i.e. 10 mg/mL) produced corneal antinociception. [75] Forehead application of the same dose, twice per day, for ten weeks led to a significant decrease in the frequency and severity of ocular symptoms in ocular graft-versus-host patients. [76] Researchers speculated that this hormone effect may be mediated by the V1 branch of the trigeminal nerve that innervates the forehead skin, and that progesterone modifies the signal relay in the rostral and caudal trigeminal nucleus to dampen nociception. [76] However, given that the applied progesterone concentration in these studies was so much higher than that typically found in blood (e.g. ng/ml levels), it might be possible that hormone action also involved other receptors. Progestins are known to bind glucocorticoid, androgen and mineralocorticoid receptors, as well membrane receptors, oxytocin receptors and γ-aminobutyric acid ($GABA_A$). [77,78]

5.5.4 Sex steroids in the tear film

Investigators continue to try to measure 17β-estradiol, progesterone and testosterone in the human tear film. [79-81] However, as explained previously [82], such measurements are of questionable, or no, relevance, and do not necessarily reflect the concentration of sex steroids or their metabolites in any ocular tissue. The processes by which sex steroids are synthesized and metabolized in humans are addressed in detail in the TFOS DEWS II Sex, Gender, and Hormones report. [3]

5.6 Insulin-like growth factor and insulin

Insulin-like growth factor (IGF)-1 levels in tears may be an indicator of ocular surface health. Higher IGF-1 levels in tears were found in young adults compared to older adults, correlating positively with tear film break-up time (TBUT) and Schirmer test results. [83] In experimental studies, IGF binding protein-3 plays a

role in delivering IGF to target cells and works independently of IGF. IGF binding protein-3 levels decrease in response to hyperosmolarity, a marker of DED, potentially causing epithelial damage in DED. [84]

Insulin insufficiency, as found in diabetes mellitus, has adverse effects on the ocular surface. A study in South Africa found significantly worse tear film parameters and a higher frequency of DED in children with diabetes mellitus compared to healthy controls. [85] An investigation in Turkey found that obese children had lower tear meniscus parameters and worse TBUT and Schirmer scores than healthy children, and these results correlated with insulin resistance. [86] A study in India showed worse ocular surface parameters in those with diabetes mellitus compared to healthy controls. [87]

Additional research has linked the DED in diabetes mellitus to insulin impairment. For example, elevated serum opioid growth factor levels in diabetes are associated with DED and corneal damage, which can be mitigated by controlling glucose levels with insulin or opioid receptor antagonists. [88] Topical insulin therapy, in turn, has shown promise for treating the signs and symptoms of DED. [2,89-91]

For insulin topical therapy to be widely adopted for DED, questions about the optimal concentration, vehicle, and suitable DED subgroups need to be addressed. [92] Novel formulations using nanotechnology to improve insulin permeability and exposure to the ocular surface have shown promise in experimental models. [93,94]

5.7 Thyroid hormone regulation of the ocular surface and adnexa

Thyroid eye disease (TED) or thyroid autoimmune orbitopathy, includes a spectrum of conditions like Hashimoto thyroiditis, Graves' disease, and Schmidt's syndrome. [95]

The association between TED and DED is debated. A study in the USA found no link between TED and DED. [96] However, studies in India, Saudi Arabia, Spain, Taiwan, and Russia found TED to be a risk factor for DED. [97-100] The variable clinical manifestations of TED (Figure 1) and different definitions of DED likely contribute to these discrepancies.

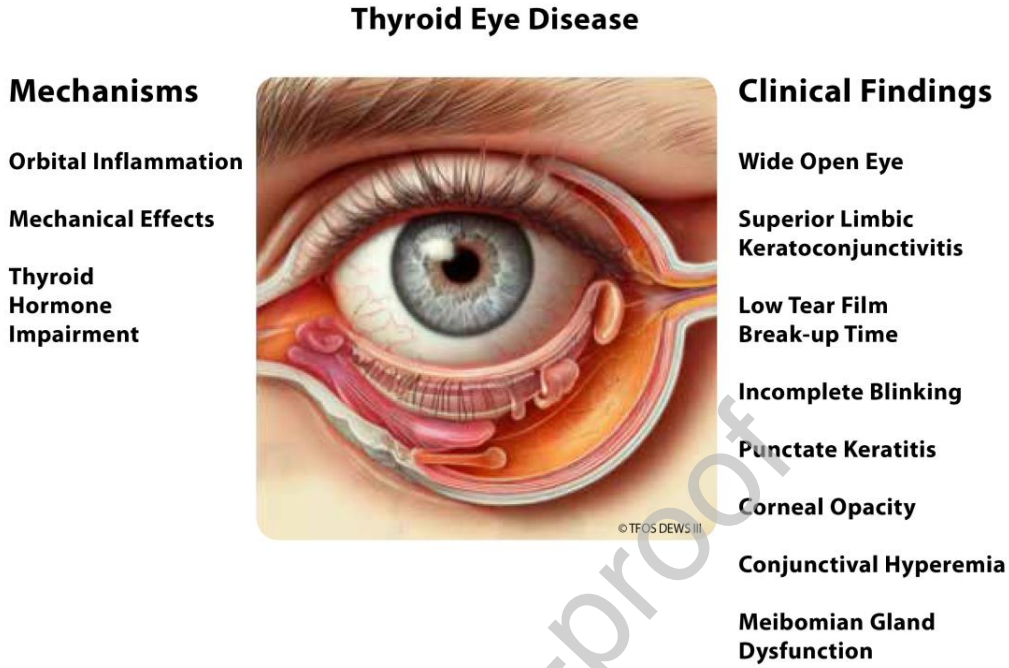


Figure 1. Thyroid eye disease

The mechanisms that cause DED in TED are broad and include anatomical changes in the orbit leading to proptosis and excessive corneal exposure, as well as inflammation of the ocular adnexa, including the main lacrimal gland and the eyelids. [101] Animal models deprived of thyroid hormone have shown clinical responses consistent with DED. As an example, rats made hypothyroid with methimazole developed higher tear film osmolarity and hypoesthesis. [102]

In TED, changes in the lacrimal gland have been observed via magnetic resonance imaging, with smaller lacrimal glands correlating with more severe clinical signs and higher inflammatory indices. [103] A study on the overlap of Hashimoto's thyroiditis and Sjögren disease found higher expression of four genes involved in both diseases, which may have diagnostic value. [104] These observations highlight the role of anatomical changes, genetic factors, and molecular inflammation mediators in linking TED, thyroid hormone dysfunction, and DED.

Recent studies have confirmed higher Ocular Surface Disease Index (OSDI) scores, lower tear TBUT, and MGD in TED patients, compared to healthy

controls. These symptoms and signs were correlated with worse proptosis, higher Clinical Activity Scores, incomplete blinking, as well as corneal damage. [100,105-108] Changes were also observed in euthyroid or inactive TED, indicating mixed inflammatory and anatomical factors. [109,110] Superior limbic keratoconjunctivitis was observed more frequently in TED patients (31%) and was associated with worse ocular surface conditions, younger age, and smoking. [111] Worse Schirmer test results, TBUT, and goblet cell density were found in children with Hashimoto's thyroiditis compared to healthy children, even without symptoms. [112] In a study of 38 individuals with moderate to severe TED in Iran, over 70% had DED. [113]

Non-invasive imaging techniques have recently improved the diagnosis and monitoring of TED. Magnetic resonance imaging and computed tomography, commonly used to assess orbital parameters in TED, are now used to study lacrimal gland volumetry and activity. [114,115] Evaluation of the lids and meibomian glands has shown an association between MGD scores and clinical activity scores in TED. [116]

5.8 Gender and DED

As reported in the Sex, Gender, and Hormones report of TFOS DEWS II, [3] "gender" refers to an individual's self-representation as a man or woman, and how social institutions respond to that person based on the person's gender presentation. [3] Gender is not the same as "sex." Sex distinguishes males and females based upon their biological characteristics, whereas gender reflects socially constructed characteristics such as behaviors related to being a woman, feminine, or being a man, masculine. [3] Both sex and gender affect health and disease, and gender also affects people's access to and interactions with the healthcare system. [3] Many health disparities are associated with gender, and both gender and biological sex influence DED risk and presentation, care-seeking behaviors, and service utilization.

Transgender individuals experience difficulties accessing appropriate health care.[117] Since that report, a case series reported ocular findings in a transgender and gender diverse population receiving gender-affirming hormone therapy. The treatments involved daily doses of estrogen and spironolactone or testosterone to 10 male-to-female and 7 female-to-male individuals, respectively. The major findings included intracranial hypertension in female-to-male and chorioretinal diseases in male-to-female. Regarding the ocular surface, one female-to-male patient developed dendritic keratitis, two male-to-female patients had prior DED,

and two others were diagnosed with *de novo* DED due to low tear film TBUT and punctate keratitis. [118]

Providing appropriate healthcare for transgender and gender diverse individuals is challenging due to a lack of specialized knowledge and social and cultural barriers. Clinical research and education for healthcare professionals are essential to address these challenges. [119-121] The interactions of sex hormones with ocular tissues and their documented effects on systemic health indicate that the impact of gender-affirming hormone therapy on vision and ocular health needs further investigation (Table 1). This includes exploring potential associations with DED and other ocular surface diseases and developing specific preventive and therapeutic strategies, such as perceived in geriatrics and gastroenterology conditions. [122,123]

Table 1. **Systemic effects of gender-affirming hormone therapy with potential impact on ocular surface health and DED.**

Reference	Gender-affirming hormone therapy type	Systemic manifestation	Relationship with DED and ocular surface disease
Betsi et al., 2024 [124]	Puberty suppression with GnRH	Bone mass retardation, mood fluctuation	Anxiety and Depression
Heng et al., 2024 [125]	Testosterone	Breast atrophy and reduction of breast epithelia	Potential risk of other exocrine glands atrophy
Hashemi et al., 2024 [126]	Testosterone	Higher risk for metabolic syndrome	Potential impact of Metabolic syndrome on DED
de Silva et al., 2024 [127]	Estrogen	Venous thromboembolism	Ischemic damage to ocular and adnexal tissues
Nieves-Rios et al.,2023 [118]	Estrogen and spironolactone for MTF or Testosterone for FTM		Corneal epitheliopathy attributed to HSK in FTM and DED and Punctate keratitis in MTF
Tienforti et al., 2024 [128]	Hormonal	Sex hormone imbalance, menopause, andropause	Sexual hormone imbalance and DED

DED: dry eye disease, FTM: female-to-male transgender; GnRH: gonadotrophin release hormone; HSK: herpes simplex keratitis; MTF: male-to-female transgender

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5.9 Future directions

There have been significant research advances linking sex, hormones and gender to DED. Aging, cancer and hormone therapy increasingly broaden the interdisciplinarity in this field over time. Despite the significant impact of gender-affirming hormone therapy on the entire endocrine system and its effects on physical and mental health, there is limited information on its impact on ocular health. Variations in age, health profile, gender-affirming hormone therapy compliance, and barriers to accessing regular healthcare limit the documentation of side effects. Clinicians and future research should consider these variations, as recommended in a recent systematic review on the medical aspects of the transgender and gender diverse population. [123]

6. Epidemiology

6.1 Scope of the update

This epidemiology update aimed to assess and summarize knowledge on the prevalence and incidence of DED from well-designed population studies and to perform a meta-analyses of existing study data to determine prevalence of DED using different diagnostic approaches, stratified by age and sex.

6.2 Operational definitions

As per the TFOS DEWS II report, [129] the subcommittee examined data from a large range of cohort studies and considered different methods of disease ascertainment and definition, including studies involving the type, frequency and severity of symptoms, patient self-report of a diagnosis of DED by an eyecare practitioner, and studies that involved a clinical examination.

An updated search of published peer-reviewed literature was conducted using PubMED (https://pubmed.ncbi.nlm.nih.gov/) which includes MEDLINE for articles that reported the prevalence or incidence of DED. The following terms (dry eye syndrome, OR dry eye disease, OR meibomian gland dysfunction, OR keratoconjunctivitis sicca, OR blepharitis) AND (prevalence, OR epidemiology, OR incidence) were used to identify potential additional articles. Human studies published since the date applied in the previous TFOS DEWS II update (18 September 2015 onwards) were considered for inclusion. For the meta-analyses, human studies captured in the TFOS DEWS II meta-analysis report and those published subsequently were included.

Eligible studies included those reporting prevalence of either or both dry eye symptoms and signs. Observational studies (cross-sectional or cohort) were included if they were population-based and presented the study outcome as DED versus non-DED. Studies were excluded if no variance in the measure of prevalence was available in the manuscript, if it was not possible to calculate it from the data presented, if the sampling criteria were not explicitly stated or if no denominator was reported. Detailed exclusion criteria are described in Figure 2. Article author and date, setting (region; population or hospital), numbers and characteristics of participants within each study group (age, sex, ethnicity), prevalence, and incidence data were extracted from each article. When required, data were extracted from manuscript figures using open-source software (available at https://plotdigitizer.com/app). [130] Where multiple studies were published from the same dataset, those with minimal overlap, distinctly different diagnostic criteria and age/sex disaggregation were preferentially chosen.

A meta-analysis was conducted to determine the prevalence of DED for different diagnostic criteria stratified by age and sex. Using the search strategy described above, population-based prevalence studies published since 1980 (per TFOS DEWS II 2015 search) were included. Prevalence data were extracted firstly by age group as follows; 6-9 years, 10-19 years, 20-29 years, 30-39 years, 40-49 years, 50-59 years, 60-69 years, 70-79 years and above 80 years and secondly by sex. For studies that had age categories that overlapped with the decade-based categorizations (e.g. 35-44, 45-54, 55-64 etc), the weighted averages of data from each contributing interval were computed for each decade. Confidence intervals for the measures of prevalence were computed by using standard methods for computing standard error of a proportion, or, if prevalence was zero, by computing the Poisson 95% confidence interval, dividing it by 2 to provide an estimate of the standard error as above. Studies were combined where the diagnostic criteria were broadly similar and in line with the 2017 approach and included a new category for studies consistent with the diagnostic methods described in the TFOS DEWS II Diagnostic Methodology Report [131] as follows:

1. Women's Health Study criteria
2. Symptomatic DED (OSDI above 13 or significant ocular or visual symptoms, where signs are not reported)
3. Insurance claims data supporting diagnosis or treatment of DED
4. Symptoms and signs (e.g., fluorescein staining, TBUT, and Schirmer score)
5. Diagnostic criteria broadly aligned with that described in the TFOS DEWS II Diagnostic Methodology Report. [131]

6. Clinical diagnosis or prior diagnosis of DED
7. Any MGD reported and separately, clinically significant (grade 2 and above, as reported) MGD

A random effects model was used to combine prevalence data. To compute the standard error, the formula SE = √p*q/n, where p was the proportion with DED was used. For studies where prevalence was 0, the exact Poisson confidence limits for the proportion computed and the width of that interval divided by 3.92 to approximate the standard error.

For descriptive analyses, box plots were produced using SAS version 9.5 (SAS Institute, Cary, North Carolina, USA). For most diagnostic categories, the number of studies available for each age category was small. When there was only a single study in the category, the prevalence is reported as originally reported by the study. If there were two or more studies in the category, a pooled rate was computed using a suite of SAS macros models developed by Weir and Senn. (Senn et al., 2011) These macros include summary statistics for the DerSimonian and Laird. (DerSimonian & Laird, 1986) The Weir and Senn macro forest was used to produce forest plots of the results.

The flowchart describes the studies identified since 2015 and reasons for exclusion (Figure 2). A total of 2687 articles were identified through PubMed search (updated 29 June 2024). The extracted data summary is available here (Supplementary Table 1. Studies included in the meta-analysis). For the overall prevalence estimation and meta-analysis, the final dataset combined both 2015 and 2024 results. Subsequently meta-analyses were conducted by age and sex by decade where data were available.

Estimates of the prevalence of DED from 76 large international cohort studies (24 and 52 from the 2015 and 2024 datasets, respectively) are summarized in Figure 2 and below. Supplementary Table 1 describes the characteristics of the additional included studies from the 2024 dataset.

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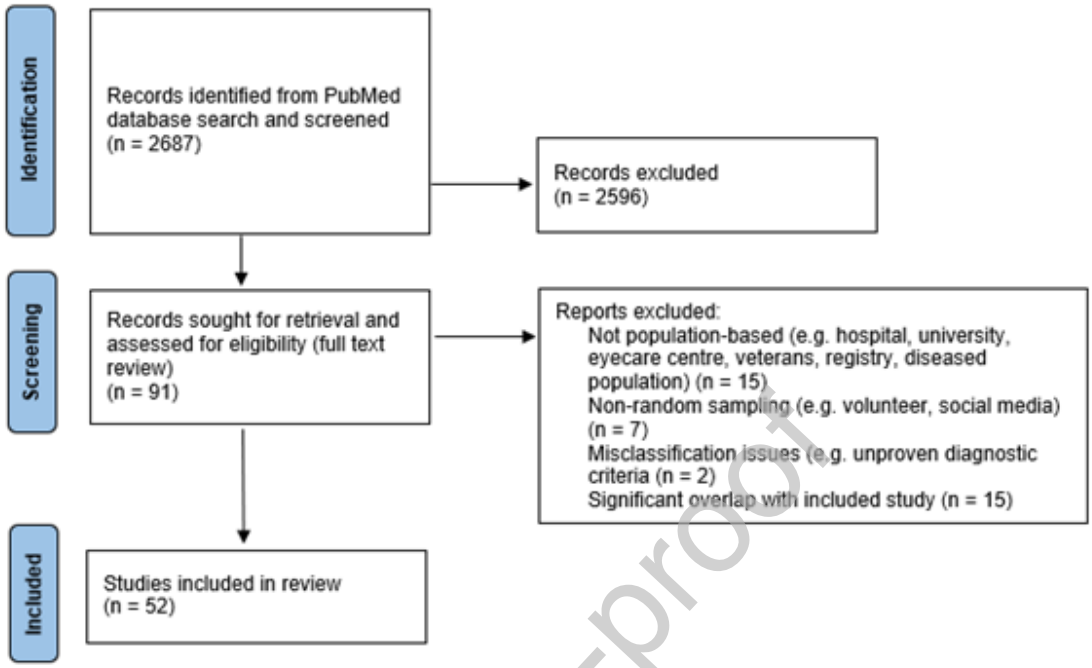


Figure 2. Prisma flowchart of the literature search outcome for prevalence studies published between 18 September 2015 and 29 June 2024.

Table 2 summarizes the overall prevalence range by diagnostic criteria and Figure 3 disaggregates the prevalence by age and sex where there were sufficient data and includes data from both the 2015 and 2024 datasets.

Table 2. Prevalence range for DED for each of the diagnostic criteria

Diagnostic criteria	Prevalence range (%)*
Women's Health Study criteria	2.7 - 30.1
Symptomatic DED	7.3 - 31.6
Claims data	2.8 - 8.5
Symptoms and signs of DED	4.7 - 62.9
TFOS DEWS II criteria	5.4 - 44.2
Clinical or prior diagnosis of DED	1.0 - 15.3
Any MGD	0.0 - 66.3
Clinically significant MGD	1.8 - 23.3

*Noting the lower end of the range relates to the rates in children. See Figure 3 below for prevalence disaggregated by age

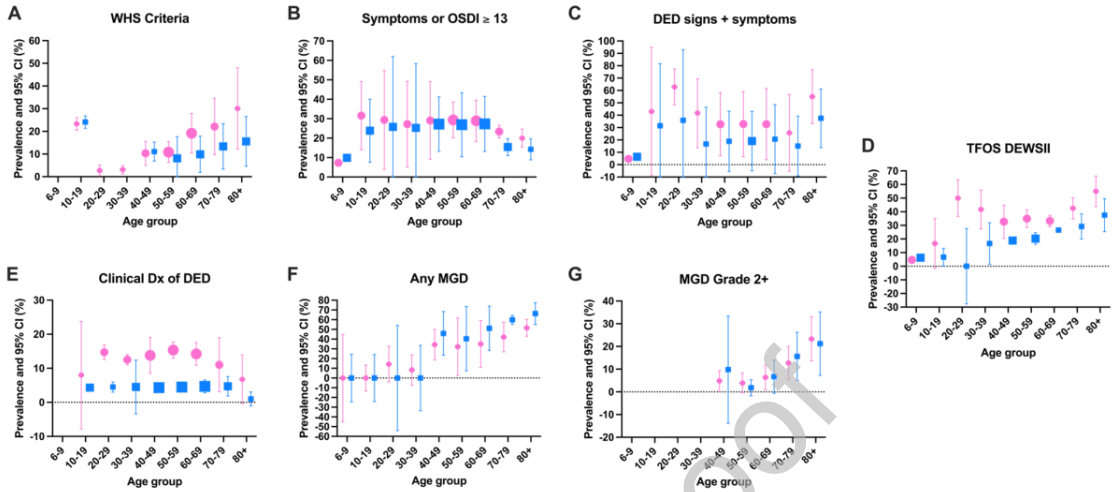


Figure 3. Prevalence of DED based on age and sex for different diagnostic criteria

6.3 Prevalence of DED

6.3.1 Prevalence of DED based on the Women's Health Study criteria

Figure 3, panel A represents prevalence by age and sex based on report of severe symptoms and/or a diagnosis of DED by a practitioner (n=12). [132-143] The prevalence ranges from 2.7% in those 20-29 to 30.1% in women over 80. The rate increases by age particularly after the age of 40 in both sexes and women have a higher rate of DED above the age of 50, with the sex difference in prevalence becoming more marked with age. One study has shown high prevalence rates in those 10-19 years old without sex related differences. (See supplementary Table 1 for individual study data).

6.3.2 Prevalence of symptomatic DED

Figure 3, panel B represents prevalence of symptomatic DED by age and sex. The prevalence ranges from 7.3% in those 6-9 years old to 31.6% in females aged 20-29 years, however confidence intervals are wide (n=33). [133-135,139-142,144-168] There were higher than expected rates of disease in the age groups under 30, except a low rate in children aged 6-9 years. Rates were consistent in the adult groups aged 40-80 years, with rates in men in the order of 20% and women 30%. Higher rates were observed in the over 80s age group and in all ages above the age of 10, women had a consistently higher prevalence.

6.3.3 Prevalence of DED based on signs and symptoms

Figure 3, panel C represents prevalence of DED based on the presence of signs and symptoms by age and sex (n=10). [97,132,133,142,146,152,158,162,163,165,169] The prevalence ranges from 4.7% in those 6-9 years old to 62.9% in females aged 20-29 years, however confidence intervals are wide. The rates of disease are reasonably consistent across the adult age groups with some reduction in symptomatic disease above the age of 70 in both sexes. Sex differences were not pronounced in most age groups except for in those aged over 70 where women have a higher rate than men.

6.3.4 Prevalence of DED based on TFOS DEWS II criteria

The overall prevalence is broadly similar to diagnoses based on signs and symptoms of DED (n=3) [132,133,170] with prevalence ranging from 5.4 (in 6-9 year olds) to 44.2%. Above the age of 30, the prevalence is higher amongst females than males, and males show a more obvious age-related change.

6.3.5 Prevalence based on claims data

Overall prevalence based on claims, either identifying DED based on the International Classification of Disease code or using insurance claims data based on diagnosis or treatment code was generally low (n=1) [171], ranging from 2.8-8.5%. There was insufficient data to disaggregate prevalence by age and/or sex.

6.3.6 Prevalence of DED based on clinical diagnosis

Figure 3, panel D represents prevalence of DED based clinical diagnosis by age and sex (n=8). [135,140,148,149,151,155,172,173] The prevalence ranges from 1.0% in men over 80, to 15.3% in women aged 50-59 years. Rates were reasonably consistent over age in adults with lower rates in those 10-15 and over 80 years of age. Women had a higher rate of clinically diagnosed DED at all ages.

6.3.7 Prevalence of any MGD

Figure 3, panel E represents prevalence of any MGD by age and sex (n=6). [97,132,174-177] The point estimates of prevalence ranges from 0% in those under 20 years old to 66.3% men over 80. Rates appear to markedly increase above the age of 40 and in 40 plus age groups, MGD is significantly more prevalent in older men aged 70 and above than in older women ($p<0.05$). Confidence intervals in most age groups are wide.

6.3.8 Prevalence of clinically significant MGD

Figure 3, panel F represents prevalence of clinically significant MGD (Grade 2 or above) by age and sex (n=2). [165,175] Rates appear to increase with age although sex differences are equivocal. There are no studies reporting rates of clinically significant MGD in younger individuals.

Thirty-three of 52 studies included in the systematic review from the 2024 dataset did not contribute to the meta-analysis due to non-availability of by age decade or by sex data, or due to duplicate data. [20,23,97,140,141,161,162,164,166,167,169,177-198] The overall prevalence rates reported include all international cohort studies (24 and 52 from the 2015 and 2024 datasets, respectively). Meta-analysis includes only those studies publishing disaggregated age and sex data. Supplementary Table 1 shows the complete list of studies and data extraction for those identified in the recent dataset. The 2015 dataset is included as supplementary data previously. [129]

6.4 Annual incidence of DED

A limited number of population studies have attempted to assess the incidence of DED since the last TFOS DEWS II update. In a retrospective analysis of approximately 6.7 million medical claims from the United States Department of Defense Military Health System, the annual DED incidence in those aged 2 years and above was low but to gradually increase over time from 0.55% in 2008 to 0.87% in 2012, respectively. [171] DED annual incidence was consistently higher in women than men and increased with age. [171]

A similar retrospective analysis of electronic medical records of 1,458,830 new patients presenting to Indian hospitals across four states was conducted and revealed an annualised incidence (average across 8 years) of DED signs and symptoms of 1.46%. [199] An analysis of the Taiwanese National Health Insurance Research Database between 2001 and 2015 (covering over 23 million inhabitants) yielded an age-adjusted annual incidence of DED ranging from 0.15% (in 2001) to 0.37% (in 2015). [200]

An ancillary cohort study of the Vitamin D and Omega-3 Trial (VITAL), which assessed the incidence of DED in 12,174 men aged 50 years and older and 11 349 women aged 55 years and older during a median (range) 5.3 (3.8-6.1) years of follow-up in the United States. All participants were initially disease free and 2% of participants experienced a clinically incident DED. [201] The annualised incidence (average across 5 years) of clinically diagnosed DED and of clinically diagnosed DED plus incident reports of severe DED were 3.8 and 16 per thousand, respectively. [201] The Salnés Eye Study 2 re-examined a cohort of 264 Spanish individuals from the Salnés Eye Study 1 now aged 51 years and older, 11 years later. [202] The annualised incidence of DED signs and symptoms in these individuals was 2.3% (95% confidence interval 1.8-2.8). [202]

Participants (n=1,682) from the Singapore Malay Eye Study (SiMES) were re-examined 6-years later, and symptoms of DED were evaluated using the Salisbury Eye Evaluation Study dry eye questionnaire. [203] The 6-year incidence of DED symptoms was 5.1% (95% confidence interval 4.1-6.4). [203]

The incidence of DED is typically difficult to estimate as some with the disease will report resolution or a reduction in symptoms over multiple sampling periods. This may particularly occur for milder cases of DED where symptom report may vary over time. Incidence from claims data may be influenced by changes in reimbursement or the introduction of new therapies during the study.

6.5 Natural history of DED

Longitudinal studies of DED are rare. A cohort of 784 patients from 1000 with DED from the Women's Health Study and Physicians' Health Study were surveyed about change in their disease one year after enrolment in the study and medical records were obtained for 261 of the participants. [204] The mean disease duration was 10.5 years, and most participants reported no change to their disease status over time. Ocular surface symptoms were unchanged in 32% of participants and improved in 44%. Visual symptoms were unchanged in 52% and improved in 19%. Social impact was unchanged in 71% and improved in 19%. Risk factors associated with progression of ocular surface symptoms included a higher spend on treatment (more than $20 per month), history of more severe DED symptoms and use of systemic beta-blockers. Worsening of visual symptoms was additionally associated with a history of ocular surgery, untreated depression and blepharitis or MGD. Disease presentation and treatments varied by sex where women were more likely to present with corneal staining but worsening of symptoms was not predicted by corneal staining, however treatments beyond level 1 [205], higher symptoms and having a tear breakup test performed were associated with progression of symptoms.

One retrospective study followed clinic-based participants with a diagnosis of either Sjögren (n=101) or non-Sjögren (n=101) DED over a 7-year period. [206] In this arguably more severe disease group, with escalation of treatment, predominantly with use of topical anti-inflammatory/immunomodulatory therapies or in office treatments, both groups experienced a significant reduction in ocular surface staining although no patient reported-outcomes were recorded.

In a small retrospective study of DED in the absence of treatment (n=73), [207] changes in signs were observed with an increased bulbar redness, reduced tear meniscus height and lipid layer thickness after 8 years. Change in symptoms over time was not reported.

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In a registry-based study, signs of MGD and evaporative DED were apparent at an earlier age than aqueous-deficient DED (Wang et al., 2020; Wang et al., 2019), although there may be some confounding due to ethnic background, where up to 2/3 of those with signs of MGD do not experience symptoms (non-obvious disease). [175] There is indirect evidence to suggest that signs of MGD may precede other disease markers by up to 10 years (Wang et al., 2020), which may have implications for the timing of treatment of non-obvious MGD. In a small treatment trial of adults with symptomatic MGD who were treated for 12 months, 1/3 of participants had improvement in MGD signs and improvement was greater in younger participants (those under 40) and with less ocular surface damage. [208]

In summary, there is reasonable evidence for development of signs before symptoms in untreated disease, which may impact on treatment considerations. There is evidence that treatment improves corneal staining but limited data on the sequence of improvement of signs and symptoms. Longitudinal natural history studies are much needed.

6.6 Risk factors for DED

Conclusive and probable risk factors for DED grouped into major categories described in Table 3 and have been divided into modifiable and non-modifiable factors.

In clinical practice, administering a questionnaire to the patient (such as via paper or a digital application) for completion before the consultation is recommended to save time and to ensure that all commonly associated factors have been addressed. The following section briefly highlights the most important risk / associated factors by category. Irrespective of a possible causal mechanism, all associated factors described may be useful in elucidating the etiology of DED and seeking to identify the possible driver(s) in an individual.

Table 3. Risk factors associated with DED.

Risk factor for DED	Evidence level* C= consistent, P= probable	Modifiable? M= possibly modifiable, N = non modifiable
1. Systemic disorders		
a. Autoimmune disorders		
Sjögren disease [209]	C	N

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Rheumatoid arthritis [209,210]	C	N
Systemic sclerosis [209]	C	N
Systemic lupus erythematosus [209,211]	C	N
Sarcoidosis [209]	C	N
Thyroid disease [210,212,213]	C	N
Inflammatory bowel disease (Crohn's disease and ulcerative colitis) [214,215]	C	N
Psoriasis [216]	C	N
b. Hormone disorders or status		
Diabetes [210,212,217]	C	N
Androgen deficiency [218-220]	C	M
Polycystic ovary syndrome [221]	P	M
c. Dermatological and atopic disorders		
Acne rosacea [212,222,223]	C	M
Acne vulgaris [214,224]	P	M
Eczema [212]	C	M
Asthma [225]	C	N
Allergy [226]	C	M
d. Pain disorders		
Irritable bowel syndrome [227]	C	N
Fibromyalgia [227]	C	N
Chronic pelvic pain [214,228,229]	C	N
Migraine [227,230]	C	N
Other headache disorders [230]	C	N
Osteoarthritis [212,214]	C	N
Back pain [227]	C	N
Temperomandibular joint disorder [228]	P	N
e. Psychiatric disorders		
Depression [210,212,231-233]	C	M

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Anxiety [231,232]	C	M
Stress (including post-traumatic stress disorder) [212]	C	M
Burnout [214]	P	M
Autism [214]	P	N
f. *Sleep disorders*		
Obstructive sleep apnea syndrome [234]	C	M
Insomnia and poor sleep quality [141,235]	C	M
g. Other disorders		
Osteoporosis [212]	C	M
Parkinson's disease [236]	C	M
Sinusitis [214]	P	M
2. Ophthalmic disorders		
Meibomian gland dysfunction [237,238]	C	M
Anterior blepharitis [239-241]	C	M
Allergic conjunctivitis [226]	C	M
Glaucoma [212]	C	M
Facial nerve paralysis (Bell's palsy) [214,242,243]	C	N
Ocular surface disorders interfering with the tear film or inducing inflammation (e.g. pterygium, pingueculum, conjunctivochalasis) [160,212,244-250]	C	M
Thyroid eye disease (including Graves' disease) [251-254]	C	M
Ocular rosacea [222,223]	C	M
3. *Surgery or procedures*		
Eye surgery (e.g. refractive surgery, cataract surgery, intravitreal injections, glaucoma surgery, retinal surgery) [212,214,255-261]	C	N
Eyelid and periorbital surgery [261,262]	C	N
Periocular botulinum toxin injections [261,262]	C	N
Cosmetic periocular and ocular procedures [261-263]	P	N
Hematopoietic stem cell transplantation (graft-versus-host disease) [264,265]	C	N

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4. Medications		
Anticholinergic medications (antihistamine, antiarrhythmic, bronchodilator, antidepressant, anti-Parkinson's, and antispasmodic medications) [261,262,266-268]	C	M
Eye drops (e.g. preservative-containing, anti-glaucoma, antihistamine, anaesthetics, some NSAID's) [262,266,267,269]	C	M
Anticancer drugs (e.g. chemotherapeutic agents and hormone therapy) [262,270-274]	C	M
Vitamin A derivatives (including isotretinoin) [275-277]	C	M
Dupilumab [278-280]	C	M
Hormone replacement therapy [213,218,262]	P	M
Anti-androgen therapy [218,281,282]	C	M
Proton pump inhibitors [214]	P	M
Psychostimulant agents used for ADHD [214]	P	M
5. Environment		
Air pollution from NO_2 and CO [283]	P	M
Low humidity [283]	C	M
High or low temperatures [283]	P	M
Air conditioning and wind [283]	P	M
Contact lens wear [212,262,284]	C	M
Screen use [212,285-287]	C	M
Cosmetics [263]	C	M
6. Demographic factors		
Increasing age [212,288-290]	C	N
Female sex and gender [212,288-290]	C	N
Asian race / non-white race [212,291]	C	N
7. Nutrient / gene related		
Vitamin A deficiency [292]	C	M
Vitamin B12 deficiency [292]	P	M
Vitamin C deficiency [292]	C	M

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Vitamin D deficiency [292-294]	P	M
Omega-3 fatty acids deficiency [292]	C	M
Altered gut microbiome [292,295-297]	P	M
Genetic predisposition [136,298,299]	P	N

ADHD = Attention deficit hyperactivity disorder.

*Consistent evidence implies the existence of multiple adequately powered and otherwise well-conducted studies published in a peer-reviewed journal, along with the existence of a plausible biological rationale and corroborating basic research or clinical data . Probable evidence implies the existence of at least one adequately-powered and otherwise well-conducted study published in a peer-reviewed journal. Risk factors with mixed or inconclusive results from multiple studies or from a systematic review or meta-analysis are described in the Diagnostic Methodology Report. Non-modifiable = a lack of evidence that change will impact DED.

6.6.1 Systemic disorders

Numerous systemic disorders (Table 3) have been associated with an increased risk of DED, emphasising the multifactorial origin of the disease.

Autoimmune disorders are well-established risk factors of DED. Up to 90% of patients with Sjögren disease develop DED [209,300]. Rheumatoid arthritis, systemic lupus erythematosus, and systemic sclerosis are also associated with a vastly increased risk of DED [209,211,301]. Other autoimmune diseases that have been consistently linked with an increased risk of DED are sarcoidosis, thyroid disease, inflammatory bowel disease (Crohn's disease and ulcerative colitis), and psoriasis [209,210,212,214,215,302,303]. Less commonly investigated autoimmune disorders, also associated with DED in some studies, include lichen planus, lichen sclerosus, granulomatosis with polyangiitis, polyarteritis nodosa, primary biliary cirrhosis, mixed connective tissue disease, antiphospholipid syndrome, and dermatomyositis [209,214,240,301,303-305]. Infiltration of the lacrimal gland and accessory lacrimal glands by lymphocytes and other immune cells that leads to impaired tear secretion and aqueous-deficient dry eye has been suggested as a core mechanism. However, there is no peer-reviewed evidence for inflammation in the meibomian gland in obstructive MGD [240,306].

The endocrine system plays an important role in maintaining ocular surface homeostasis. Hormone disorders such as androgen deficiency, polycystic ovarian syndrome, and thyroid disease have been linked to an increased risk of DED [212,213,218,221]. Sex hormones exert effects on all major components of the tear film [218]. In thyroid disease, immune-mediated lacrimal gland dysfunction, and exposure keratopathy due to orbitopathy (such as in TED), are important mechanisms. [213,218,307] Diabetes has also been consistently, although mildly, associated with DED in systematic reviews and meta-analyses (odds ratio (OR) of around 1.2) [210,212,217].

Decreased corneal sensitivity and impaired reflex tear secretion have been suggested to be linked [308] alongside factors such as hyperglycemia, advanced glycated end product accumulation, oxidative stress, metabolic disease and vascular disease [309]. Menopause has also been proposed as a risk factor for DED, but definitive evidence is lacking [129,218,310].

In addition to psoriasis, other dermatological disorders may also be associated with an increased risk of DED. Rosacea without systemic involvement may present with ocular symptoms alone and can lead to chronic blepharoconjunctivitis and MGD (see Section 3.5.3.3.3)[212,222,223]. Acne vulgaris has been linked to DED, possibly by an associated risk of MGD or as a result of side-effects of isotretinoin therapy [214,224]. Atopic disorders such as eczema, asthma, and virtually all types of allergy have been consistently linked with an increased risk of DED [212,214,225,226,283]. Allergic eye disease may aggravate DED, and suggested mechanisms include increased inflammation at the ocular surface, altered epithelial barrier and corneal innervation, and tear film instability. Antihistamine use, which may result in anticholinergic side-effects, may also partly explain the association. Periocular eczema has also been linked to blepharitis. Alternatively, DED may aggravate allergic eye disease, since allergens may be less efficiently removed from the ocular surface in the presence of reduced tear turnover and DED is associated with an upregulation of inflammatory mediators, including some of those common to the allergic pathway, such as complement. Complement also plays a role in DED in the absence of allergy [311]. Complicating the association between DED and allergy is the overlap in symptoms such as burning, itching, and tearing. [225,226,312] It is therefore important to consider allergic conjunctivitis in patients with DED, and particularly in patients with a history of atopic disease. Specific questioning about eyelid margin itching may suggest the presence Demodex blepharitis [313].

Chronic pain disorders (the body's heightened sensitivity to non-noxious stimuli) can present with ocular discomfort symptoms similar to those of dry eye.; fibromyalgia, chronic pelvic pain, irritable bowel syndrome are among the most established risk factors [227]. Development of neuropathic pain within the trigeminal somatosensory system has been proposed to underlie this link [227]. A study of twins in the United Kingdom found evidence for shared genetic factors underlying these disorders and DED, suggesting a common chronic pain predisposition that accounts for clustering of pain disorders [299]. Other pain conditions that have been linked to DED are migraine and other headache types, osteoarthritis, back pain, and temporomandibular joint disorder [214,227,228,314].

There is also strong evidence for an association between DED and depression, anxiety, burnout, and stress (including post-traumatic stress disorder) [212,214,227,231-233]. A meta-analysis of studies in patients with DED found an overall prevalence of 40% for

depression (OR of 1.8), and of 39% for anxiety (OR of 2.3) [231]. These relationships are likely bidirectional, and studies show an association with psychiatric disorders with DED symptoms that is greater than with signs. Proposed mechanisms that link psychiatric disease to DED include altered pain perception, somatization, and increased serum inflammatory markers [227]. Clinicians may find it beneficial to enquire about mental health history in patients with DED, as appropriate, and consider referrals to mental health professionals when indicated.

Obstructive sleep apnea has been linked to DED [212,235]. It increases the risk of floppy eyelid syndrome, which may lead to exposure keratopathy at night and to increased conjunctival inflammation and MGD [315]. Leakage of air from a continuous positive airway pressure (CPAP) mask may also lead to irritation, conjunctivitis and DED [316,317]. Insomnia and poor sleep quality may be a consequence of DED, but could also result in DED [141,227,235]. For example, obstructive sleep apnea is associated with a persistent low-intensity inflammatory state which may be pertinent to DED etiology [318].

Other disorders that have been convincingly linked with DED are osteoporosis (which may reflect common underlying mechanisms such as sex hormone or vitamin D deficiency, or a role of purinergic signalling) [319], autism (which may reflect hypersensitivity to external stimuli) [214], Parkinson's disease (which may be linked to decreased blink rates and lower tear secretion) [236,320] and sinusitis (which may sometimes involve the tear ducts, or reflect common mechanisms like allergy, or may represent referred pain) [214].

6.6.2 Ophthalmic disorders

MGD may be the most important factor associated with DED [237]. Its prevalence varies widely between studies, but has an estimated global prevalence of 35.8%, and the majority of patients with DED have underlying MGD that is considered to be a major cause of evaporative DED [237,238,321]. Signs of MGD occur earlier in the natural history of DED progression (typically between 24-29 years) than other clinical signs such as tear film instability, hyperosmolarity (31-38 years) and ocular surface staining (46-52 years) [322].

Anterior blepharitis is a common chronic inflammatory disorder of the eyelid margin located anterior to the gray line (see section 3.5.3.3.1). This area includes the eyelid skin and the eyelashes and its follicles. Anterior blepharitis is traditionally subcategorized based on its etiology: bacterial, seborrheic, fungal or parasitic. Anterior blepharitis is a risk factor for other forms of ocular surface disease including DED, MGD, chalazion and keratitis [239,323].

Other disorders of the ocular surface have also been associated with a disruption in the tear film or inflammation at the ocular surface, potentially leading to DED (see Section X). Conditions that have been most associated with increased risk of DED are pterygium [160,244,324], pingueculum [245-247], conjunctivochalasis [248-250], and allergic conjunctivitis (see section 3.1.2.1.1).

Patients with glaucoma are also at increased risk for DED, most often associated with use of antihypertensive eye drops. Both active ingredients [269] and preservatives can affect the ocular surface. In addition, glaucoma surgery can affect the ocular surface. [212,325,326]

Facial nerve paralysis, including Bell's palsy, can lead to exposure keratopathy through incomplete blinking, and has also been associated with impaired lacrimation and MGD [214,242,243]. Facial nerve dysfunction may lead to lagophthalmos and reduced blink force, compromising tear film distribution and postulated to compromise meibomian gland expression [327].

6.6.3 Surgery and other procedures

Eye surgery has consistently been linked to an increased risk of DED. The greatest amount of evidence is available for refractive surgery, [256-258,328], cataract surgery [255,259] and intravitreal injections, [260] but it is likely that all eye surgery comes with an increased risk of DED. [212,214,261,262] Mechanisms that lead to DED after eye surgery include incisional procedures leading to transectional nerve damage, surgical trauma to the ocular surface during surgery, phototoxicity, toxicity from eyedrops (e.g. from povidone iodine, anaesthetics, or preservatives), and damage and stress from repeated drying, irrigation and exposure to surgical illumination. [261,329,330]

Eyelid and periorbital surgery, such as blepharoplasty, ptosis surgery and brow surgery, have also been associated with an increased risk of DED. [261] Mechanisms include lacrimal gland injury and post-operative incomplete eyelid closure and incomplete blinking. Similarly, periocular botulinum toxin injections can lead to lagophthalmos. The TFOS Lifestyle Report discusses the impact of eye, eyelid and periocular procedures on the ocular surface in further detail. [261,263]

Allogeneic hematopoietic stem cell transplantation, mostly used in the treatment of various haematological diseases, can lead to graft-versus-host disease. In this disease the graft's immune cells attack the host's body cells, and this can become chronic. It can affect multiple organs including the eyes and principally involves the ocular surface which can lead to severe DED. More than 50% of recipients develop DED, mostly 6 to 24 months after transplantation [264].

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6.6.4 Medication use (See Section 10)

A large population-based study in the Netherlands found that 52 of the 99 most commonly used medications were associated with an increased risk of DED. [266] Although not all these medications may be causally linked to DED, and some of the associations may reflect an association with underlying disease (severity), it highlights the importance of assessing medication status in patients with DED. This section discusses various medications that have been most frequently associated with DED, but it is recognised that medication-associated DED is not limited to these medications only.

Medications with anticholinergic (side-)effects have been consistently and causally linked to DED, such as antihistamine, antiarrhythmic, bronchodilator, some antidepressant, anti-Parkinson's, and antispasmodic medications. [261,262,266-268] These drugs affect the muscarinic receptors of the lacrimal glands and conjunctival goblet cells, causing decreased aqueous and mucous secretion.

Several medication-related pathophysiological mechanisms have been demonstrated: allergic, toxic, immune-inflammatory effects; chemical interaction with the tear film leading to a disrupted lipid layer; reduced aqueous secretion; damage to goblet cells and epithelium of the cornea and conjunctiva; and neurotoxic effects on the corneal nerves and eyelids including meibomian glands. [261,262,274,325,326]

Systemic chemotherapeutic and other anticancer agents may have cytotoxic effects at the ocular surface and affect tear film quality and reflex tear secretion. [262,270-274,331] Examples include alkylating agents such as cyclophosphamide, antimetabolites such as 5-fluorouracil and methotrexate, monoclonal antibodies such as rituximab and the aromatase inhibitors.

Vitamin A derivatives or retinoic acids, including isotretinoin, are used for acne vulgaris and in anti-ageing regimes, and may be administered topically or orally. They are secreted into the tear film by the lacrimal gland [332], and are associated with tear film instability, lower Schirmer test scores, and atrophy of meibomian glands, leading to DED. [263,275-277,306,333]

Interleukin (IL)3 and 4 receptor antagonists (such as dupilumab) are increasingly used in patients with atopic dermatitis with good effect but may cause several adverse effects at the ocular surface including a mild to sometimes severe conjunctivitis, punctate keratitis, blepharitis, and loss of meibomian glands. [278-280]

Hormone replacement therapy has been associated with increased prevalence of DED. [213,218,262]

Proton pump inhibitors, antacids, and psychostimulant agents used for attention-deficit/hyperactivity disorder have also been linked to a highly increased odds of having DED, but biological pathways are currently unclear. [266,267]

6.6.5 Environmental factors

Climatic risk factors may play an important role in the etiology of DED. The TFOS Lifestyle Environmental Conditions report has recently and comprehensively summarized evidence for environmental risk factors for DED. [283] Both low and high temperatures have been associated with DED, as has low humidity, in several experimental and population-based studies. Air-conditioning and wind are well-known risk factors that patients often report as triggering symptoms, [334] although limited evidence is available from research studies. Air pollution such as from NO_2 and CO was found to be probably associated with DED, and soil pollution from chromium is likely associated with DED and Sjögren disease. Evidence for a risk of other air, soil, and water pollutants was not conclusive. [283]

Contact lens wear has been reported as a factor associated with DED in many cross-sectional population-based studies, but prospective studies are lacking. It has been postulated that mechanisms leading to DED may include thinning of the tear film after insertion of the contact lenses, increased friction between the lens and the ocular surface, leading to meibomian gland dropout, decreased tear film stability, ocular surface staining and lid wiper epitheliopathy. [238] It is important to recognise that contact lens discomfort symptoms overlap with DED, and symptoms do not necessarily reflect underlying DED. [212,262] (See Section 10 Contact lenses and DED)

Screen or visual display terminal use has been consistently linked to DED. [285-287,335] As few as 1 to 2 hours of screen use per day may even be associated with adverse ocular surface effects. [285] Important mechanisms include decreased blink rate and incomplete blinking that leads to increased evaporation and decreased lipid release from meibomian glands. [336] It is also important to consider digital eye strain, independently of DED, in the differential diagnosis of persons who are heavily exposed to a digital environment. [286] This requires a full refractive correction to be determined for the distances required and a binocular vision assessment to ensure suboptimal visual input is not the cause.

Cosmetic products may be associated with adverse effects at the ocular surface and may aggravate or initiate DED symptoms. The TFOS Lifestyle cosmetics report identified 10 ingredients that are commonly present in cosmetics that particularly have significant adverse effects: benzalkonium chloride, chlorphenesin, formaldehyde-releasing compounds, parabens, phenoxyethanol, phthalates, prostaglandin analogues, retinoids, salicylic acid, and tea tree oil. [263] Finally, recent reviews have found impacts

on the ocular surface from: environmental endocrine disruptors in foods, packaging and pesticides. [337] No clear or consistent link between smoking/vaping and DED has been identified, although effects on the tear film are evident. [227,338-341] While a direct causal relationship between smoking and DED may not be certain, it is reasonable to suggest that smoking cessation should be encouraged as part of a comprehensive approach to promoting ocular surface and holistic health in patients with DED.

6.6.6 Demographic factors

Female sex and gender are strong risk factors for DED (see Section 6). As discussed in Section 5, sex-related differences in prevalence are underpinned by differences in the ocular surface and adnexa including anatomy and immunity, a higher prevalence of autoimmune disorders and psychiatric disease (and their related medications), higher general pain sensitivity in women and reduced androgen levels (already lower than in men) after menopause. [218,289,342]

Increasing age is also a well-established risk factor for DED, but DED is prevalent in both the young and the old. Systematic reviews have found that signs show a stronger relationship with age than symptoms. MGD, also, increases in prevalence with age. [288,321,343-346]

Prevalence values for DED and MGD are higher in Asian countries than in Western countries [289] and a pooled analysis of interethnic differences suggested differences in disease type and age at which signs manifest in Asian and Caucasian populations. [291] There is also evidence that white races are less affected than non-white races in Western studies. [212] It is difficult to unravel precise causes for these differences, which likely reflect socio-economic, cultural, genetic, anatomical, lifestyle and environmental differences between groups.

6.6.7 Other factors

Nutrition may play a role in the development of DED, but there is no strong evidence to support recommendation of an optimal diet. Poor nutrition that drives systemic disorders may be associated with risk factors for DED. Limited but increasing evidence links alterations in the gut microbiome to ocular surface health, possibly by altering the immune system. [292,296,297,347-350] The TFOS Lifestyle Nutrition report concluded that there is evidence that deficiencies of vitamins A and C and omega 3 fatty acids are risk factors for DED. Moderate evidence was also found for vitamin B12 and D deficiencies as risk factors.[292] There is no clear evidence that alcohol use, [227,292], a Mediterranean diet [351] and water intake [292] represent risk or protective factors for DED.

Genetic factors likely contribute moderately to DED. A large twin study in the United Kingdom established heritability accounted for 29% for DED symptoms, and 41% for DED diagnosis. The remaining 60-70% was attributed to unique environmental factors. [136] A recent genome-wide association study in Taiwan with over 14,000 DED cases and almost 26,000 controls found eleven independent risk loci, including *MUC16,* which encodes for a mucin protein that is expressed at the ocular surface. A polygenic risk score including 932 loci was able to detect individuals with a high-risk of DED. [298] A limitation of this study was the use of self-reported DED only and that the findings were not replicated in an independent cohort. This lack of replication is also a major limitation for several, mostly small, candidate gene studies that have found a link between genetic variations and DED. [352-358] Further studies are warranted before genetic testing becomes clinically useful in DED.

6.6.8 Risk factors for MGD

There is debate about the impact of MGD on risk factors and prevalence of DED. It is recognised as a risk factor for the disease and recent estimates of DED prevalence suggest that MGD may be present in 50-70% of cases of DED. [359,360] Nonetheless, there have been attempts to establish demographic risk factors for MGD which may or may not be independent of those for DED more broadly. Age is frequently reported as a risk factor in MGD, (Arita et al., 2019) particularly in elderly populations [188] and where MGD is diagnosed using meibography to determine meibomian gland loss or drop out. [11,361] Ethnic background (Asian compared with other backgrounds) is consistently reported as a risk factor. The impact of sex in MGD is however equivocal, in contrast to its impact in DED more broadly where female sex predominates.

In examining population-based studies, there is a higher rate of non-obvious MGD in Caucasian males [175] and a higher risk of MGD in males based on gland plugging and telangiectasis in the Singapore Malays study.(Siak et al., 2012) In a smaller population-based study from Japan where symptomatic disease was diagnosed using lid margin irregularities and gland plugging, males similarly had a higher risk of MGD compared with females. (Arita et al., 2019) Male sex was not an independent risk factor for MGD in a population-based study of adults in Iran.(Hashemi et al., 2017) There is wide variability in the impact of sex on MGD depending on how the disease is diagnosed using individual or combined lid signs (gland drop-out, orifice plugging, altered meibum secretion, degree of gland expressibility, lid telangiectasia, either presence/absence or presence above a certain level), whether symptoms are present, study design and whether the study is based on a clinical sample, specific disease group or is population-based.

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For the meta-analysis reported above (Section 6.3.7), large population-based studies only have been included and two analyses report either all MGD, irrespective of definition and clinically significant MGD signs with symptoms, broadly based on the TFOS MGD Report. For population-based studies reporting any lid changes with or without symptoms (Any MGD – Figure 3, F), MGD is more prevalent in older men (above 70 years of age) than older women but there is no difference between sexes in the rates of clinically significant MGD (Figure 3, G).

6.7 Morbidity and impact

DED severely affects the lives of sufferers. It negatively impacts quality of life (QoL) including physical, psychological and emotional well-being, social functioning, daily living activities and independence. General and mental health, social functioning, physical, emotional states, bodily pain and vitality are significantly poorer in those with DED compared with those without [362-365] and that health status worsens in those with more severe disease. [364,365] In individuals with mild and moderate disease (n =217) blurred vision, productivity loss, and visits to eye care practitioners were increased compared with age-matched normal individuals (n=67). [366]

Adverse QoL effects appear to be consistent over time, irrespective of geography and with variations in ethnic background [367,368] and will likely increase with population aging.

The economic cost of DED can be measured in direct resource utilisation (service provision, medication costs), out-of-pocket costs, cost of lost productivity and reduced QoL (quantified as a utility). There are limited studies which include an associated utility algorithm or that meaningfully evaluate productivity impacts of DED.[369-371]

6.8 Summary and outstanding questions

This update has considered the prevalence of DED in studies that have shown disease rates by age and sex. Eight major diagnostic groups were identified and meta-analyses reported here include DED diagnosed using the Womens Health Study criteria, symptom report; claims data from health or insurance databases; signs and symptoms; diagnosis according to the TFOS DEWS II criteria; clinical diagnosis; any MGD or clinically significant MGD (grade 2 or above). Prevalence varied with diagnostic criteria where not all disease increased with age or showed a female predominance. Broadly, using the Womens Health Study criteria, DED increased with age and was more common in women. A clinical diagnosis of DED showed a female preponderance but not an age-related effect. Symptoms and signs were more common in women with higher rates in younger and older adults. Any or severe MGD was age-related, with males more likely to show any MGD. Some studies included in the ‘Any MGD’ analysis reported non-obvious MGD and the age and sex effects here have not been stratified.

These findings are perhaps not unexpected given the multifactorial nature of DED, the specificity of ocular symptom measurements and differences in the etiology of different subtypes of DED [1]. Prevalence data in some recent studies may be confounded by the impact of the COVID-19 pandemic. Previous meta-analysis showed that both the pandemic and mitigating factors (mask and screen use) were associated with greater ocular symptoms and signs of ocular surface disease. [26,145,191,197]

Studies reporting rates of DED in those under 20 are limited. Rates are lower than adults for clinically diagnosed DED, DED with signs and symptoms (under 10 years) and any or significant MGD. High rates of symptom-reporting are evident in those under 20 however, although it is recognised that symptom report alone is not specific for DED and childhood anterior or posterior blepharitis, Demodex and allergy may be common co-morbidities or contributors to ocular symptoms. Most studies reporting symptoms did not report signs although they may be present.

Risk factors are reported as consistent or probable, and potentially modifiable or non-modifiable. New risk factors related to environment, climate and lifestyle are included. There is some evidence for an increase in prevalence in DED over time which may conceivably be due to the impact of new risk factors including changes in the digital environment. Given the differences in prevalence and age/sex associations with different diagnostic criteria for DED, it may be important to disaggregate risk factors particularly for DED and MGD where possible. Given the high prevalence of symptom-reporting in childhood, appropriate triaging for other conditions and hypothesis-driven and appropriately powered studies to explore risk factors in children would be valuable.

Outstanding questions include:

1. Disease severity. A limited number of studies explored the prevalence, risk factors or natural history by disease severity, which could help to triage and manage those more likely to experience more severe DED.
2. Geographical mapping was not considered as part of this update.
3. The generalizability of prevalence measures for DED in children and adults under 40. These mostly originate from studies in Asia and there are limited studies in other regions.
4. The need for appropriately powered studies to determine risk factors in those under 40.

7. Pathophysiology

7.1 Introduction

The consensus view of the 2017 TFOS DEWS II report envisioned a tear film centric model of the pathophysiology of DED, [6] broadly classified by compromised quantity; Aqueous-Deficient Dry Eye (ADDE) or quality; Evaporative Dry Eye (EDE) of the tear film. In DED, tear hyperosmolarity is considered to set up a cascade of signaling events within surface epithelial cells, that leads to the release of inflammatory mediators and proteases. Such mediators, together with the tear hyperosmolarity itself, are conceived to cause goblet cell and epithelial cell loss and damage to the epithelial glycocalyx. Damage is reinforced by inflammatory mediators from activated T-cells, recruited to the ocular surface. The net result is the characteristic punctate epitheliopathy of DED and a tear film instability which leads at some point to early tear film break-up. This break-up exacerbates and amplifies tear hyperosmolarity and completes the vicious circle events that ultimately lead to ocular surface damage and self-perpetuation of the disease. Epithelial injury and a defective glycocalyx, loss of tear volume and of goblet cell mucin, lead to increased frictional damage and friction-related symptoms. The tear hyperosmolarity and epithelial injury caused by DED, stimulates corneal nerve endings, leading to symptoms of discomfort, increased blink rate and potentially, a compensatory, reflex increase in lacrimal tear secretion." The TFOS DEWS II report also highlighted causes of ADDED that included lacrimal gland infiltration and dysfunction, neurosecretory or reflex blocks, androgen deficiency or aging-related downregulation of secretion, obstruction of the lacrimal ducts in cicatricial disease, and iatrogenic causes such as prescription medication and surgical damage to trigeminal nerves. For EDED, conditions that affected the ocular surface included MGD, anterior blepharitis, xerophthalmia, ocular allergy, androgen deficiency as well as iatrogenic causes such as topical preservative use, contact lens wear and certain anti-glaucoma drugs.

Subsequent research has led to a more nuanced understanding of the disease. Given the inconsistent relationship between symptoms of DED and signs such as measurable inflammation, hyperosmolarity, ocular surface staining, low tear volume, low TBUT, or MGD, [360,372] the data suggest that the associated signs help identify subtypes of the disease, with symptoms providing no predictive ability as to which etiology is active in that patient. [360] The implications of the lack of correlation of signs and symptoms in DED are particularly important in relation to pathophysiology, because while most historical literature assumes that inflammation and inflammatory pathways are the common effector of DED, evaporative subsets show a muted if any increase in inflammatory mediators in the tear film, [372-376] and thus inflammation cannot be assumed to be active in every patient with DED. For example, one study based on mass spectrometry found that evaporative patients exhibited no increase in proteins associated with the inflammatory response compared to normal controls, while 51% of differentially upregulated proteins in the aqueous cohort were associated with inflammation. [372] Similar data showed that while aqueous- deficient subjects (Schirmer value ≤ 5 mm)

had elevated lipid peroxides in tears compared to healthy controls, subjects with 6–10 mm wetting were not significantly different from normal controls. [373] Another study based on multiplex bead analysis found that evaporative patients tend to exhibit much higher levels of epidermal growth factor, but no difference in the IL-6 and IL-8 cytokines compared to healthy controls and essentially no detectable TNF-α in tears, [375] reinforcing the idea that not all forms of DED are driven by the same underlying pathways. In mouse models, bilateral lacrimal gland excision showed dramatic increases in IL-1ß, IL-6 and TNF-α protein concentration in the tear film over four weeks, while topical benzalkonium chloride and environmental chamber-induced DED showed no significant difference from controls in those tear film cytokine proteins, despite concurrent increases in tear osmolarity and cytokines measured from corneal mRNA transcripts. [377] Immune-mediated etiologies, represented by immunological diseases that include Sjögren disease, Stevens-Johnson's syndrome, and ocular graft versus host disease (GVHD), are widely regarded as being more severe than evaporative counterparts. [378] At the extreme end, severe forms of dry eye found in ocular GVHD tend to be accompanied by fibrotic processes, [379,380] and Sjögren-related DED is associated with lymphocytic infiltration of the lacrimal gland, neither of which is evident in common evaporative DED [380] In a rabbit model, cauterization of the meibomian glands results in only moderate increases in hyperosmolarity and loss of goblet cells as compared to models in which lacrimal excretory ducts and accessory glands of rabbits are sealed. [381] These data may help explain why anti-inflammatory medications have a dichotomous effect, with some patients reporting improvement, [382,383] while others report high failure rates of cyclosporine and lifitegrast in the general population. [384-387] As many patients are still poorly served by the available therapeutic options, it may be that our understanding of the pathophysiology of DED is incomplete.

Note that a variety of the pathophysiological aspects of the disease are discussed in more detail elsewhere. For example, androgen deficiency is discussed in the Sex, Gender, and Hormones section, neuropathic damage is covered in the Pain and Sensation section, the tear proteome is outlined in the Tear Film section and iatrogenic causes are described in detail in the TFOS Lifestyle Report (Gomes et al., 2023) and in the Iatrogenic section of the digest report.

7.2 Initiating triggers of disease

Except in certain situations of injury, [176] surgery [256,328] and therapy [388], establishing causality is difficult when diagnosing DED. Metabolic disease induces mitochondrial stress [389-394] and advanced glycation end products in the lacrimal gland [395,396], hormonal changes that alter glandular production and fatty acid metabolism [3,397-400], and biophysical stresses of friction [401-405], hyperosmolarity [406-414], and swelling pressure [415]. These effects are compounded by the biological activity of dysregulated or self-reactive

immune cells [416-419], cellular stresses from cytokines [377,420-426], proteases [427-429], reactive aldehyde species[430], extracellular deoxyribonucleic acid (DNA) and neutrophil extracellular traps [431-433], exogenous toxins [390,411], damage or danger-associated molecular patterns [408], gut dysbiosis [434,435], as well as neurogenic inflammation. [436-441] Typical pathways of initiation of evaporative processes include phenotypic alterations in corneal epithelial cells that lead to a compromised glycocalyx [442], keratinization of the meibomian gland that alters the lipid profile of the tear film, [443,444] and incomplete blinking or reduced blink rate during screen use that exposes the ocular surface to desiccating stress. [445] On the aqueous deficiency side, inflammatory ingress into the lacrimal gland can be driven by androgen deficiency or autoimmunity, causing a cascade of protease release, cytokine expression, inflammatory cell recruitment, dendritic cell maturation, and an adaptive T-cell mediated response. [3,418,446] As disease severity increases, the evidence suggests there is a progressive accumulation of these mechanisms. [227,418,447,448]

7.3 Hyperosmolarity

When exposed to hyperosmolar conditions, epithelial cells begin to change their morphology and lose their microplicae. [449] Increasing levels of hyperosmolarity (excess salt) on the ocular surface cause an increase in epithelial apoptosis, [450] transglutaminase mediated cornification of epithelial cells, [451] and increased secretion of IL-1ß, IL-6, IL-8, TNF-α, and MMP-9 from ocular surface epithelial cells. [452-455] Hyperosmolarity-induced epithelial cell stress results in desquamation, revealing the immature glycocalyx and microplicae-free cells beneath. [449] These immature cells are hydrophobic and contribute directly to tear film instability. When patches of hydrophobic cells are adjacent to normal hydrophilic cells, the differential surface tension causes the tear film to breakup and evaporate within a few seconds, dramatically increasing the local osmolarity and exposing cells to a toxic environment. [456,457]

In addition to the hypothesis that inflammatory cells infiltrating either the lacrimal gland or conjunctiva are a source of oxidative damage at the ocular surface, [458,459] *In vitro* evidence suggests that hyperosmolarity directly induces reactive oxygen species (hydroxyl and peroxyl activity) in human corneal epithelial cells, along with reductions in the anti-oxidant superoxide dismutase-2, vitamin D, Notch ligands Dll3 and Jag 1, and a tripling of the pro-apoptotic Bax/Bcl2 ratio. [460] Reactive oxygen species undergo lipid peroxidation and cause reactive aldehyde species to be produced, causing a cascade of protease release, cytokine expression, inflammatory cell recruitment, dendritic cell maturation, and an adaptive T-cell mediated response [430,461]. *In vitro* data have shown that exposure to hyperosmolarity causes mitochondrial DNSA to leak into the cytoplasm of human corneal epithelial cells, activating the cGAS-STING pathway and increasing cytokines such as CXCL10 and IFN-β, [462] while human conjunctival impression cytology

samples from evaporative, short TBUT patients (2.9 s) confirmed an increase in STING proteins compared to more normal controls. [463] Evaporative hyperosmolar patients (327 mOsm/L, 7.0 s TBUT, 10.4 mm average Schirmer value) exhibit specific upregulation of IFN-γ, without concomitant increases in IL-2, IL-6, IL-10, TNF-a or IL-17A. [464] IFN- γ, which is strongly associated with dose dependent CD40, MICA and MHC II expression along with esithelial cytotoxicity in corneal epithelial cells, [465-467] increases NLRP3 oxidative stress and contributes to pyroptosis in human corneal epithelial cells under hypertonic conditions. [468] These data mirror earlier findings that hyperosmolarity disrupts the balance of oxygenases and antioxidant enzymes such as SOD1 and PRDX4, stimulates lipid peroxidation (e.g., 4–hydroxynonenal & malondialdehyde), increases COX2, and damages corneal mitochondrial DNA. [469,470] Hyperosmolarity also potentiates the negative effects of the oxidizing blue light in corneal and conjunctival epithelial cells, increasing H_2O_2 production, phototoxicity and altering the mitochondrial membrane potential in these cells compared to those in normal media. [471] Downstream of hyperosmolarity, a TLR-4 dependent upregulation of Dual oxidase 2, a member of the NADPH oxidase family that regulates the production of intracellular reactive oxygen species, increases alongside high mobility group box 1 production in human corneal epithelial cells. [408,472] This result was consistent with data that MyD88-/- mice lacking TLR-4 signaling exhibited significantly less corneal fluorescein staining, cytokine and protease expression following 5 days of hyperosmolar stress and scopolamine administration, [473] implicating TLRs and damage or danger-associated molecular patterns (e.g., HSPs, HMGB1, S100A, tenascin-C) as essential agents in evaporative, hyperosmolar DED pathophysiology. [473-475] Although inflammatory ingress may be an eventual result of hyperosmolar exposure, [476] clinically, it is likely a matter of time and severity that governs which patients progress to the inflammatory phenotype as a result of hyperosmolarity. [381] For example, at low levels of hyperosmolarity, protective responses such as the increase in IGF binding protein-3 are observed, but at increasing levels of osmolarity and length of exposure, cell viability is challenged as respiration and glycolysis are decreased alongside endoplasmic reticulum stress and caspase-3 activation, leading to unchecked mitophagy and eventual cell death. [393,477,478] Similarly, short term exposure to low humidity environments showed an increase in damage or danger-associated molecular patterns such as HSP-60 and a small 2.4-fold increase in MMP-9 mRNA expression in patients with evaporative DED, but no statistically significant increase in IL-6 or IL-8 expression. [474] These data suggest that the severity of tear hyperosmolarity should help inform the underlying pathophysiology; wherein a strongly elevated osmolarity will lead to oxidative stress and inflammation, while a middling or low osmolarity is suggestive of a subclinical inflammatory state [479] and may mitigate against prescription of anti-inflammatory therapy.

7.4 Proteases

Another aspect of the pathophysiology of aqueous deficient disease that separates it from evaporative disease is the disrupted balance of protease and anti-protease activity on the ocular surface. The normal tear film exhibits an equilibrium of protease and protease inhibitors. [427] Excess protease is observed in Sjögren disease, where cathepsin S catabolizes anti-proteases such as cystatin C, [428] MMP-9 is highly expressed while thrombospondin-1 lags behind, [480] and plasmin activity is increased ten-fold in Sjögren compared to normal tears. [481] In non-autoimmune DED, protease activity is stratified by disease severity and is strongly associated with aqueous deficiency. For example, mild evaporative-type subjects with low TBUT (5.0–7.2 seconds) and normal Schirmer value (13–19 mm) exhibited only slightly higher tear MMP-9 activity (35-66 ng/mL), compared to the tears of more severe aqueous-deficient cohorts, (Sjögren disease, Stevens-Johnson, 6 mm Schirmer value) with levels in the 101–381 ng/mL range. [378] These data are consistent with observations that 11-14% of early stage, patients with EDE (4.9 second TBUT, 312 mOsm/L, 14.0mm Schirmer value) presented with ELISA-validated MMP-9 positivity. [482] Increases in neutrophil elastase, MMPs [483] and neutrophil extracellular traps are seen in severe aqueous-deficient subjects (chronic ocular GVHD, Sjögren disease and ocular cicatricial pemphigoid) that are not commonly observed in non-autoimmune DED and healthy controls. [431] Similarly, the anti-protease Cystatin S was found to have a significant, inverse relationship with aqueous severity in non-autoimmune DED subjects, falling from about 2,000 ng/mL in controls (16 mm Schirmer value) to 400 ng/mL in moderate aqueous-deficient DED (4 mm Schirmer value), while MMP-9 was only mildly upregulated to the 20–40 ng/mL range in those subjects. [484] The increase in cathepsin S, a serine protease that is known to degrade a variety of essential components of the normal tear film and glycocalyx (lactoferrin, sIgA, proteoglycan 4), [428,485] is also stratified by inverse tear volume; increasing monotonically as tear volume decreases. [376] Of note, although cathepsin S is dramatically enhanced in Sjögren disease compared to non-autoimmune subjects on average, non-autoimmune subjects with 0–5 mm wetting on the Schirmer strip exhibited equal or higher levels of cathepsin S than patients with Sjögren disease with > 15 mm of wetting. [376]

These data invoke the question of causality, whether aqueous deficiency leads to inflammation and protease release, or vice versa. In an animal model of severe aqueous deficiency, excision of the lacrimal gland resulted in a significant upregulation of serine proteases in corneal tissue including tryptase, urokinase plasminogen activator receptor and protease activated receptor 2 expression, which upon activation, is able to induce MAPK / ERK-1 & 2 signalling with downstream NF-κβ, ICAM-1 & cytokine expression, [429] suggesting that aqueous deficiency can precede a measurable protease burden at the ocular surface. In contrast, MMP-9 positivity was found in 84% of GVHD as compared to 33% of non-autoimmune DED patients, even as the GVHD patients

trended towards increased Schirmer test values, lower OSDI and higher TBUT than the DED controls, [486] suggesting that immune infiltration into the lacrimal gland precedes the protease release in GVHD.

Ultimately, in non-autoimmune DED, the transition from low levels of protease release to inflammatory DED with excessive protease concentrations is likely related to progressive alterations in the lacrimal gland and ocular surface. These might result from androgen deficiency, [3,6] accumulation of advanced glycation end-products in the lacrimal gland, [395] diabetic keratopathy and nerve degeneration, [487] or other factors that lead to significantly less aqueous flow, thereby tipping the homeostasis towards a severe hyperosmolar state and associated excessive, pathogenic protease release. It is recognised that in marked EDE or lipid deficiency, tear and ocular surface homeostasis may be sufficiently disrupted to cause inflammation and that in advanced disease, features of both are displayed.

7.5 Sub-functional or absent glycocalyx

The glycocalyx lubricates, [488] retains water at the ocular surface, [489] provides wetting to the ocular surface, [490] acts as a barrier and helps remove debris from the ocular surface. [491] The hydrophilic epithelial surface provides 82.5 sec/cm of specific resistance compared to 12.9 sec/cm for the lipid layer, [489] establishing the glycocalyx as one of the components responsible for preventing evaporation. Electron micrographs of intracellular, pre-expression glycocalyx showed that "the increase in numbers of subsurface vesicles that occur in some external eye diseases may reflect an attempt to increase the binding of the mucus to the eye surface… In late keratoconjunctivitis sicca, however, the subsurface vesicles are absent. [492]" These foundational observations identified the difference between evaporative DED and other ocular surface conditions, where exogenous stresses lead to a compromised epithelial glycocalyx – the proximal cause for an increase in evaporation, hyperosmolarity and its downstream complications in EDE.

Evidence that the glycocalyx plays a central role in disease pathogenesis has begun to accumulate; for example, knockout of MUC4, the most abundant membrane associated mucin in the conjunctiva with an ectodomain predicted to extend > 2 µM above the apical cell surface, resulted in significantly reduced microplicae, tear film disruption, and increased rose bengal dye penetrance into deeper layers of the ocular surface. [493] *In vitro* evidence from corneal and conjunctival co-culture has shown that hyperosmolarity (caused by a decreased blink rate) directly downregulated components of the epithelial glycocalyx, increased TLR-4 expression, initiated release of cytokines and proteases including MMP-9, and eventually, cellular apoptosis. [445] The hyperosmolarity compromised glycocalyx led to significantly lowered TBUT, lower Schirmer value, and

persistently elevated fluorescein staining, whereas replenishment of the glycocalyx using human recombinant lubricin (recombinant human proteoglycan 4) was able to reverse these clinical indicators, as well as inhibit epithelial NF-kB translocation and normalize IL-8, TNF-a, IL-1ß, TLR-4 and MMP-9 expression within one day of supplementation. [445] A striking aspect of this study is that the common clinical expressions of DED were recapitulated without immune cell involvement, establishing the hyperosmolarity-compromised glycocalyx as a causal, initiating event in evaporative disease. A related study found that intraperitoneal injections of streptozotocin, used to induce metabolic hyperglycemia similar to type 1 diabetes in mice, caused dramatic reductions in the extent of the corneal glycocalyx, with associated reductions in tear film volume, number of goblet cells and upregulation of TLR-4, MAPK, IL-1, IL-6 and IFN-γ genes within as little as one week following injection. [494] In a C57BL/6 mouse model, scopolamine and desiccating stress induced an almost complete abrogation of secreted (MUC5AC, MUC2) and transmembrane mucins (MUC1, MUC4, MUC15 & MUC16) along with significant increases in conjunctival IFN-γ, leading to persistent epithelial defects. [495] Importantly, an *in vitro* study evaluating the impact of applied hyperglycemia in media (15 & 30 mM glucose vs. 5 mM control) showed that elevated glucose exposure had no effect on the amount or distribution of membrane associated mucins in either corneal or conjunctival epithelial cells, [496] suggesting that the pathophysiology of streptozotocin-induced glycocalyx loss is more likely due to global metabolic disease than exposure to hyperglycemia alone. In support of these data, a study of vitreo-retinal surgery found that conjunctival MUC4 & MUC16 gene expression increased post-surgery and tear osmolarity was reduced in normal subjects, likely as a protective response, but older diabetic subjects exhibited a lower goblet cell density, lower MUC5AC and increased cytokine response by comparison, once again linking systemic metabolic disease to ocular surface glycocalyx impairment. [497] In a mouse model of GVHD, allogeneic transplantation of a mixture of spleen and bone marrow cells resulted in a significant reduction in the area and thickness of the corneal glycocalyx, reductions in MUC4 and MUC5AC and a coincident reduction of tear film volume and an increase in fluorescein staining, which were partially abrogated after application of topical rebamipide. [498]

More recent data have shown that components of the glycocalyx also regulate immune cells[499], transduce extracellular environments to intracellular signaling pathways[500], and actively inhibit proteases such as MMP-9 [501], which would further implicate the catabolism or downregulation of the glycocalyx as an initiating event in DED pathogenesis. As the glycocalyx is altered in ocular surface pathology, large decreases in sialic acid [502] and increases in galectin-3 are observed in the tear film that strongly correlate with disease severity. [503,504] When released into the tear film, galectin-3 seems to amplify the IL-1ß mediated inflammatory response, [505] which is similar to how

degraded low molecular weight hyaluronic acid becomes pro-inflammatory in other tissue systems. [506,507]

In summary, hyperosmolarity, metabolic disease and inflammation impair the ocular surface glycocalyx and initiate the characteristic clinical signs of DED. The degree of impairment of the glycocalyx seems to be a fulcrum for transitioning from common evaporative DED to an inflammatory, aqueous-deficient state, as the balance of fluid output overwhelms the fluid input.

7.6 Meibomian gland dysfunction

Meibomian gland dysfunction (MGD) is a chronic, diffuse abnormality of the meibomian glands, commonly characterized by terminal duct obstruction and/or qualitative and quantitative changes in the glandular secretion. [237] Although it is a disease distinct from DED, when the severity of MGD is of a sufficient degree, it may give rise to EDE. [323] While a widely cited study is often misquoted to suggest that 86% of all DED subjects have MGD, the denominator of that estimate excluded patients with DED that could not be classified as either aqueous or evaporative, and when included, found that 60.7% of those with DED had evidence of MGD. [508] Larger, more recent estimates of the prevalence of MGD within those with DED estimate a number, from 51.3% to 70.3%. [359,360] Those with pure MGD exhibited the lowest severity of signs of ocular surface damage, [359] consistent with the assumption that evaporative DED is often a subclinical inflammatory state. [372] Indeed, in patients with MGD, significant negative correlations were found between inflammatory mediators such as IL-8, C5a and Schirmer value results. (Mahajan et al., 2021) One study however has suggested there may be inflammatory mediators and cytokines present in meibum in patients with MGD [509]. As a causative pathophysiology of DED, it is currently estimated that between 33% to 50% of DED patients have sufficiently impactful MGD to initiate a dry eye state. [360]

By leveraging novel models of primary human meibomian gland epithelial cells, ductal from human tissue, [510,511] 3D cultures and organotypic cultures, [462,512-517] researchers have further investigated the pathophysiology of MGD. PPARγ agonists, such as rosiglitazone, may significantly promote cell differentiation, lipogenesis and anti-inflammation in human meibomian gland epithelial cells. [168,518-522]

The meibomian gland is located in a physiologically hypoxic environment [523], that is impaired in MGD. [524] An increased expression of hypoxia-inducible factor 1α (HIF1α) plays a role in the beneficial effect of hypoxia on the meibomian glands, [525] and as such, a low oxygen environment may be important in imitating the *in vivo* condition in culture. Animal models include injection of complete Freund's adjuvant in rabbits [526] and mice, [527] electrocauterization of meibomian gland orifices in rats, [528,529] transitory alkali

exposure of the rat eyelid margin, [530] Soat1-null mice, [531,532] APOE KO mice, [533] and Elovl1-deficient mice. [534]

Recent studies have indicated that it may be possible to restore gland structure after atrophy. In mice, fibroblast growth factor receptor 2 gene (FGFR2) knockout could lead to significant gland acinar atrophy, [535] but this change is reversible if the knockout condition is removed. Moreover, this recovery relies on the extent of ductal atrophy, which indicates that ductal epithelia may serve as a reservoir for meibomian gland progenitor cells for regeneration. [536] The importance of FGFR2 and other FGFRs were also reported in humans, in which FGFR inhibiting anti-cancer drugs could induce significant gland atrophy, and patients who use these drugs may develop MGD. [537]

The relationship between systemic lipids and MGD has been a focus of several recent studies. Patients with elevated serum total cholesterol, low-density lipoprotein and triglyceride levels exhibited significantly higher levels of meibomian gland loss compared to healthy controls. [538] Mice fed with a high-fat diet developed hypertrophic meibomian glands, [539] decreased PPAR-γ expression, increased meibomian gland acini cell apoptosis and mitochondrial damage, and activation of MAPK and NF-κB signaling within the gland. [540] A high fat diet significantly altered the rhythmicity of meibomian gland, which may offer new insights into the regulation of the glands by dietary lipids. [541] Patients who have long term dyslipidemia also showed significant meibomian gland atrophy and changes in meibum quality, even while undergoing statin treatment. [542] Dietary cholesterol has a direct impact on meibum components and meibomian gland pathophysiology. [543-545] In contrast, dyslipidemia and increased triglyceride levels were found to be protective factors for meibomian gland atrophy in an elderly female population. [546] A systematic review has suggested there is moderate evidence for a beneficial effect of omega-3 supplements on MGD. [547]

Meibum proteins also play an important role in ocular surface homeostasis. A discussion of the role of meibum proteins in MGD is beyond the scope of this review but these have been described more fully in Jeyalatha et al., 2017. [548] Ectodysplasin A protein secreted from meibomian glands plays a significant role in regulation the proliferation of corneal epithelial cells through the epidermal growth factor receptor signaling pathway. This discovery connects meibomian gland function with corneal epithelial homeostasis, [549] and suggests future research pathways surrounding the feedback between the eyelid and ocular surface.

7.7 Inflammatory cell recruitment

By leveraging novel single-cell RNA sequencing (scRNA-seq) methods, CD45+ lymphocyte populations were quantified in mouse corneas before and after short term oral scopolamine hydrobromide and desiccating stress. T cells (18.6%), resident

macrophages (18.2%), B cells (12.8%), type 2 conventional dendritic cells (cDC2)/macrophage (9.9%), NK cells (9.3%), monocytes (8.9%) and neutrophils (7.9%) were the most prominent cell types in normal corneas, whereas a striking shift towards resident macrophages (55.2%), MMP12 and MMP13 high macrophages (11.5%) and type 2 conventional dendritic cells (cDC2)/macrophage (8.5%) were observed following the induced hyperosmolar ADDE eye. [550] In the mouse conjunctiva, pharmacologic suppression of tear secretion followed by desiccating stress recruited monocytes from the blood and promoted maturation of Ly6ChiMHClo monocytes to Ly6CloMHCIIhi macrophages. [551] In a hyperosmolar model that exposed mice to a low humidity environment without the scopolamine administration, scRNA-seq revealed that conjunctival CD4+ T cells, Mo/Mφs, and DCs were amplified during DED progression, CD8+ T cells gradually decreased, and epithelial cells exhibited EMT-like characteristics that created a positive feedback loop of pro-inflammatory MφC3 activation. [552] Importantly, although mouse models tend to echo human DED in terms of innate or myeloid immune cells (macrophages, dendritic cells, neutrophils), lymphoid cells such as NK, NKT, or γδ, or adaptive CD4+, CD8+ or Th17 lymphocytes do not necessarily follow the same trajectories between species. [553] For example, brush cytology of human conjunctival tissues revealed an increase in IL-17A producing γδ cells from ≈ 4% in mild evaporative disease (3.3s TBUT, 26 OSDI) to 7–10% of the CD45+ cells in more symptomatic human samples (2.0s TBUT, 49 OSDI), which was far lower than the increase from 25% to 33% of CD45+ cells in normal versus induced DED mouse conjunctivas. [554] Of note, conditional knock down of lymphangiogenesis in a scopolamine, dessicating stress murine model resulted in the significant reduction of TNF-α, IL-1ß, IFN-γ and IL-8 within corneal tissues but did not lessen the release of those cytokines in the lacrimal gland, highlighting the importance of angiogenic immune cell trafficking in the progression of aqueous DED. [555]

7.8 Immune cell dysfunction

Dysfunctional regulatory T cells (Tregs) exhibiting reduced Foxp3, CD24 and CTLA-4 expression lose the ability to suppress the chronic Th17 mediated inflammation in an IL-6 dependent manner. [556] Blocking the substance P / neurokinin-1 receptor interaction, [439] or the TLR4/MyD88 pathway [557], both of which are upstream of NF-κB, were shown to normalize the Treg/Th17 balance. In mice colonized with intestinal microbiota from patients with Sjögren disease, a reduced frequency of CD4+Foxp3+ T regulatory cells in cervical lymph nodes was observed [558], while in aged mice, CD4+CD25+Foxp3+ T cells were dysfunctional, lost suppressive ability, and produced significant amounts of inflammatory cytokines IL-17 and IFN-γ. [559]

7.9 Future directions

Recent studies exploring the pathophysiology of DED have continued to define the etiologically distinct subpopulations within the disease, recognizing limited evidence for inflammation in evaporative DED, and expanding roles for metabolic, hormonal, physical and molecular impacts on the ocular surface. Neural regulation of the epithelium is taking on increasing importance, as are factors from the ocular surface that damage nerves, such as hyperosmolarity, reactive oxygen species, and other potentially excitotoxic stressors. Yet major questions remain about how to best classify the subsets of disease within a single patient and to what extent observed pathways are causal to symptomatology. The natural history of the disease is unclear, when or if inflammation becomes involved, or when acute inflammation transitions into a chronic state. While animal models of the pathophysiology have provided valuable mechanistic insights, they often fail to fully replicate human disease progression. If, as the existing data show, inflammation is most strongly associated with aqueous deficiency, whether this due to a relative increase in hyperosmolarity compared to that seen in evaporative disease, or whether inflammatory mediators and proteases from the diseased lacrimal gland driving downstream changes in the epithelium underpin these effects. In part due to these questions about pathophysiology, it is currently not possible to predict which patients will respond to anti-inflammatory or other types of therapy and which ones will show positive responses to lubricants. While yet to be established, there may be value in evaluating human immune cell dynamics and cell populations using advanced imaging methods such as functional in vivo confocal microscopy. [560] As the diversity of drugs expands, it will be increasingly necessary to align etiology with mechanism of action. Hundreds of millions of dollars of development rests on predicting responders from non-responders. Thus, it is becoming ever clearer that classical indices of DED such as corneal staining and TBUT are insufficient to support modern drug development efforts, and that more informative methods of determining disease subtype are critical for future research.

8. Tear Film

8.1 Introduction

The Tear Film Report of TFOS DEWS II described the biophysical and biochemical aspects of tears and how these change in DED. [561] The report noted that DED is characterized by loss of tear volume, more rapid breakup of the tear film and increased evaporation of tears from the ocular surface. The tear film lipids, proteins, mucins and electrolytes contribute to the integrity of the tear film but exactly how they interact was an area of active research. Tear film osmolarity, proteins and mucins could potentially be used as biomarkers for DED. Some of these themes have been explored by research since the publication of the TFOS DEWS II report, and these are highlighted in the current report.

For the current report, the authors undertook searches in Pubmed, Scopus and Web of Science databases for keywords ‘tear film’ and ‘dry eye’ since the publication of the TFOS DEWS II Tear Film Report in 2017. After reviewing the titles and abstracts of documents obtained, the authors have concentrated on those areas that have produced the greatest number of articles. This report focusses on concepts and more detail on specific components can be found within the cited references.

8.2 Clinical measurements of the tear film

The reproducibility of non-invasive TBUT (measured without fluorescein) obtained with combinations of several commercially available instruments have been reported. Instruments generally had good agreement, [562-566] but measurements from different devices are not interchangeable. Reproducibility tended to worsen for those with DED, [564] Another study showed that NIBUT, whether measured by small-cone (E300) or large-bowl (K5M), had poor agreement with FBUT, even when precise timing of blinking was taken into consideration, and Bland-Altman analysis showed limits of agreement that spanned the entire dynamic range of the instruments (31.4 seconds) with an average difference of 3.9 seconds between the methodologies, making the techniques neither interchangeable nor accurate compared to FBUT, and not interchangeable with each other. [567] One of the challenges in interpreting the reported correlations of signs of DED is the influence of subjectivity and low inter-rater reproducibility.

Simple correlations between symptoms and tear film or ocular surface characteristics were sought. Whilst correlations do not imply causation, they can point the way to a greater understanding of the mechanisms involved in signs and symptoms of DED and identify areas for further investigation, recognising that there may be confounders in individual studies and populations may differ. There was no correlation between tear evaporation and symptoms (OSDI score), nor between tear evaporation and tear production. [568] However, one study found significant positive correlations between OSDI and tear evaporation rate scores in subjects with refractive errors. Both OSDI and tear evaporation rate were lower in those without refractive errors. [569] These findings may have been confounded by the inclusion of participants with uncorrected refractive errors, smokers or differences in other demographic factors, but it may be important to adjust for refractive error in studies of tear film and symptoms.

OSDI score has been negatively correlated with NITBUT, [563,570-574] tear meniscus height (TMH) [570,571] and meibomian gland area, [574] and positively correlated with meibomian gland loss. [573,574] NITBUT was positively correlated with TMH, meibomian gland dropout grade [570,571] and corneal or lid margin staining. [563] Females with refractive errors had significantly lower tear film lipid layer (TFLL) thickness, TMH, and NITBUT scores than those without refractive errors.[570] A strong correlation was found between TMH and

NITBUT, but no correlations between these parameters and OSDI scores.[575] These differences may be due to difference between populations and between techniques. TBUT has been correlated with meibomian gland irregularity determined from meibography images taken with a keratograph.[576]

Tear meniscus height was significantly correlated with the Schirmer value and tear film breakup time and negatively correlated with ocular surface staining score, but there were no correlations with any MGD indicator. [571]

A study of normal subjects and those with MGD found a positive correlation between precorneal tear film thickness and thinning rate and negative correlations between precorneal tear film thickness and TFLL thickness and between precorneal tear film thinning rate and TFLL thickness. [577]. There were no associations between TFLL thinning rate and any of precorneal tear film thinning rate, precorneal tear film thickness or TFLL thickness.[577] TFLL thickness has been positively correlated with age, meibomian gland expressibility, and negatively correlated with meibomian gland dropout [572,578] and OSDI score.[572]

Several studies have been conducted to determine whether tear film osmolarity correlated with clinical signs and symptoms of DED. Across population-based studies, tear osmolarity is generally not correlated with tear film breakup time, corneal fluorescein staining score, lid hyperemia, tear production, blink interval, Ocular Protection Index, Schirmer I test, meibum expressibility, meibum quality or MGD. [579-583] However, this may depend on how DED was classified, as other studies found correlations between tear osmolarity and OSDI discomfort subscore, corneal and conjunctival staining scores [584] OSDI score, ocular surface staining and Schirmer I.[585] Data from the DREAM (Dry Eye Assessment and Management) study, whilst demonstrating some correlations between tear osmolarity and signs and symptoms, were interpreted as being not indicative of causation as changes in tear osmolarity were not associated with changes in signs and symptoms of DED. [586] Other recent longitudinal studies, however, have suggested a relationship between change in tear osmolarity with treatment and iatrogenic effects changes, suggesting a link in certain DED subtypes. [587-592] The Diagnostic Methodology Subcommittee report of TFOS DEWS III [1] gives more background on correlations and the limitations of this approach, and the hyperosmolarity section of the Pathophysiology section of the TFOS DEWS III Digest report (Section 7) discusses the role of osmolarity in more detail.

8.3 The tear lipids – composition and function

Several studies have examined the tear film lipid layer (TFLL). These have examined changes to the lipids in DED and/or MGD patients compared to controls, as well as which lipids contribute to tear film stability and reduced tear evaporation.

Broadly, meibum, combinations of meibum lipids and a thicker TFLL were associated with a significantly slower evaporation flux, [593-596] and significantly slower tear breakup. [597] Tear film proteins can also reduce evaporation of water, and this is enhanced when some of the proteins (human serum albumin and lactoferrin) but not others (human lysozyme and bovine mucin) are used in conjunction with mixture of polar and non-polar lipids found in meibum. [598] Lysozyme can reduce the surface tension of lipids on water, by disrupting the order of lipid molecules. [599]

Patients with MGD have significantly lower peak height ratios of the CH3/CH2 bands in NMR studies than normal subjects, and this was hypothesized to be due to changes in branched hydrocarbon chains, which contain fewer CH2 moieties, and straight chain hydrocarbons, which contain more CH3 moieties. [600]The authors further hypothesized that, as van der Waals interactions between CH2 moieties were responsible for lipid ordering, the more CH2 in meibum from MGD patients could contribute to a more ordered TFLL. A more ordered TFLL can contribute to a more patchy layer resulting in a deterioration in spreading and decreased elasticity. [600] Another factor that is involved in TFLL stability is saturation of lipids, with increased saturation resulting in stiffer, thicker, and more elastic films at high surface pressures. [601] Lipid saturation is a major factor that contributes to lipid disorder and the phase transition temperature of lipid layers. [602] Meibum of MGD patients has lower surface pressures in Langmuir trough experiments compared to meibum in those without MGD, and formed more brittle unstable patchy layers. [603]

The cholesteryl ester to wax ester ratio decreases in patients with MGD. [604-606] An optimal mixture of wax and cholesterol esters in the TFLL may be necessary to disrupt the ordered packing of pure lipid species, leading to better lipid spreading, [607] a more stable tear film, [608] and thinner lipid layer patterns. [609] In a model system, mixtures of cholesteryl oleate (CE) and behenyl oleate (a wax ester mimic) plus phosphatidylcholine increased the surface pressure of films on phosphate buffered saline. [610] A layer of behenyl oleate can form a crystalline state on water and reduce its evaporation [611] Iso-branched wax esters help to reduce evaporation and increase surface pressure. [612] On the other hand, addition of cholesterol to cholesteryl esters increases film rigidity. [613] People with thin TFLLs (who tended to be those with EDE) and people with irregular TFLL patterns (who tended to be those with ADDE), had lower levels of cholesteryl esters and lysophospholipids, but higher levels of glycerolipids and phospholipids in their tears than those with normal TFLL patterns. [614]

Other meibum and tear film lipids that have been examined are the O-acyl-ω-hydroxy fatty acids (OAHFAs). Ultra-long OAHFAs found in TLLF and meibum are capable of integrating other species in the polar lipid layer thereby molding its properties and/or providing a base for the creation of hierarchical structures within the TFLL. [615] *In vitro*,

OAHFAs can spread well on the surface of phosphate buffered saline to form a solid monolayer with a crystalline structure. [616] Longer chain OAHFAs prevented evaporation of water at a mean molecular area of approximately 18 Å2/molecule. [616]

Mixtures of OAHFAs and wax esters may produce even more evaporation resistant films. [616] Indeed, wax esters existing in the solid state under physiological conditions were capable of forming a mixed condensed monolayer with OAHFAs, and such a monolayer exhibited very high evaporation resistance *in vitro*. [617] Whilst OAHFAs can induce stable multilamellar cholesterol ester films on water or phosphate buffered saline, [618] OAHFAs alone reduced evaporation, with cholesterol esters having no effect. [618] An additional double bond in the hydroxy fatty acid chain of OAHFAs was accompanied by more disordered molecular organization, which led to a loss of the evaporation resistance.[619] The ultralong chain lengths observed in naturally occurring OAHFAs may require the presence of double bonds to achieve an appropriate balance between spreadability and evaporation resistance.[619] The observations of ordered lipids resulting in deterioration of spreading and reduced elasticity, whereas a more disordered molecular organisation leads to a loss of evaporation resistance require further exploration of how the ordering of different types of or combinations of fatty acids can optimise the evaporative resistance, spreadability and elasticity of the film.

Individuals with symptomatic MGD or symptomatic mixed MGD/ADDE had reduced abundance of several OAHFA species in tears, and to a lesser extent in meibum, compared to normal subjects. [620] Other meibum-derived OAHFAs had negative correlations with precorneal tear film thinning rate, and one meibum-derived OAHFA had a positive correlation with precorneal tear film thinning rate.[621] Conversely, tear film-derived OAHFAs had no association with the precorneal tear thinning rate.[621]

There have also been several studies examining the contributions of sphingolipids to the TFLL and meibum. Meibomian gland loss has been associated with loss of some sphingosines from the TFLL. [609] Individuals with poor meibum quality had changes to their sphingolipids in tears and meibum.[622,623] The presence of both sphingomyelin and ceramide increases surface tension due to the change their position in the TFLL under lateral pressure.[624] Sphingomyelin at the interface of the TFLL with the aqueous phase has a role in capturing the protein lysozyme and entrapping it in the TFLL.[625]

Finally, several experiments have shed important light on how meibum lipids are synthesized and the consequences of disturbance in their synthesis. Using mice with specific genes knocked out has shown that fatty acid w-hydroxylase Cyp4f39, acyl-CoA reductase FAR2, the elongases of very long chain fatty acids- 1i or -3, alcohol acyltransferases Awat1 and Awat2, and sterol O-acyltransferase 1 are all involved in various stages of the production of meibum lipids such as OAHFAs, wax esters and

cholesterol esters. [626-633] Some of the major findings when genes for these enzymes were knocked out was shortening of tear film break-up time,[626] plugged meibomian gland orifices, tear film instability and increased tear evaporation, [627,631] increases in blink frequency and evaporation from the ocular surface, [628], shorter chain, branched and unsaturated cholesterol esters [629], and changes in the melting temperature of their meibum.[629,630]

8.4 The tear proteome

Recent findings continue to emphasize the importance of the tear proteome as a non-invasive tool for trying to discriminate between types of DED [634,635] and also for disease monitoring, for instance revealing changes in the levels of proteoglycan 4, in tears of those with Sjögren disease. [636]

One interesting study explored the potential of tear MUC5AC and IL-8 levels to distinguish between Sjögren and non-Sjögren DED, indicating that lower MUC5AC and higher IL-8 levels could serve as biomarkers for Sjögren syndrome, aiding in its diagnosis.[637] Similarly, the tear thrombospondin-1/matrix metalloproteinase-9 ratio is significantly reduced in Sjögren disease compared to non-Sjögren DED, offering another potential biomarker. [480] Additionally, the MMP-9/lactoferrin ratio was positively correlated with ocular inflammation and tear film stability in stable controlled patients with Sjögren disease. [638]

Tear biomarkers related to ubiquitination (LMO7 and HUWE1) and the regulation of intracellular vesicle dynamics (TPD52) are significantly elevated in patients with Sjögren disease, suggesting a role for these pathways in the underlying mechanisms of the disease. [639] Other robust protein biomarkers have been identified using LCMS/MS in multiple cohorts, highlighting several proteases and protease inhibitors plus noting the relevance of oxidoreductase proteins in Sjögren disease. [640] Oxidative stress biomarkers serve as indicators for assessing the extent of ocular surface damage, and research using both laboratory models and individuals with DED underscores redox imbalance as one of the many pathophysiological factors driving the disease. [641] Sjögren disease may also be characterised by reduced or absent levels of complement regulators (CD59, CD55, and CD46), alongside upregulation of C3. This imbalance may drive excessive complement activation in ocular tissues and may suggest a possible therapeutic pathway. [642]

Ongoing research continues to shed light on the role of tear cytokines in the context of DED. A recent meta-analysis of tear film cytokines in Sjögren disease revealed elevated levels of inflammatory cytokines, including IFN-γ, TNF-α, IL-1α, IL-1Ra, IL-4, IL-6, IL-8, IL-10, IL-17, IL-21, and IL-22. Interestingly, IL-23 levels were significantly lower compared to healthy controls. [643] Additionally, reduced levels of epidermal fatty acid-

binding protein (E-FABP), which regulates inflammatory pathways on the ocular surface, further suggest its potential as a biomarker for epithelial damage and altered lipid metabolism in the disease. [644] However, despite these advances, a persistent challenge in the field is the lack of standardization in sample collection and analytical techniques, [645] which hinders the translation of these findings into clinical practice.

Tear proteomic profiling has also revealed altered biological pathways related to corneal sensitivity and nerve parameters in DED and neuropathic corneal pain. [646,647] (See section 8.5)

One emerging area of interest in this field involves examining the complex interplay between the tear proteome and the ocular surface microbiome, and its potential contribution to the onset of dry eye. Recent data indicate that the interaction between these two systems is beneficial to combat pathogens and maintain ocular health through modulation of the inflammatory response. [648] Moreover, the application of whole-metagenome sequencing has enabled researchers to pinpoint unique microbial compositions in individuals with DED, with the relative abundances of certain bacteria being correlated with specific tear proteins in the tear fluid. [649] Understanding the crosstalk between these two components of the tear fluid may provide opportunities for the development of personalized therapeutics tailored to individual patient profiles.

8.5 Lipid-protein-mucin interactions

Interactions between the various components of the tear film are important for maintaining its stability and protecting the ocular surface and the expression of these components is interconnected. For instance, there is a compensatory mechanism where the absence of lipid secretion, caused by stearoyl-CoA desaturase-1 deficiency, leads to increased tear volume and enhanced expression of mucins, along with changes in the expression of lipid metabolism genes. [632] The knowledge gained from understanding the interactions between tear film components is also being used to develop dry eye treatment strategies. Two treatment studies suggest that mucomimetic polymers can improve the structure and functionality of the lipid layer. [650,651] Cross-linked hyaluronic acid is particularly effective in promoting tear ferning and the spreading of meibum *in vitro*, suggesting therapeutic potential of these polymers for ocular surface health. [652]

8.6 Mucins

The realm of ocular mucins has witnessed several scientific discoveries, shedding additional light on their important role in maintaining ocular surface health. Membrane-associated mucins act as regulators of transcellular barrier function, tear film stability and apical epithelial cell architecture. [493,653] Membrane-associated mucins are distinctly distributed along the conjunctiva of those with and without DED, emphasizing the vital

role of these mucins in minimizing eyelid friction during blinking. [654] Advanced modelling approaches have enabled researchers to establish a relationship between the loss of membrane-associated mucins and the premature rupture of the tear film, predicting quantitatively the shortening of NIBUT observed in DED. [655]

Through comprehensive genomics analyses, researchers have identified mucin variants lacking sialylation due to a point mutation in a sialyltransferase gene, St6galnac1, which plays an important role in synthesizing sialyl-Tn and serves to protect the conjunctival mucosa against foreign particles. [656] Other investigations have highlighted the rapid turnover of gel-forming mucins in the healthy ocular surface [657] and the potential benefits of thymosin β4, resolvin D2 and microRNA inhibition in their ability to stimulate the secretion of gel-forming mucins. [658-660] A randomized clinical trial comparing thymosin β4 to placebo in a 28 day study showed no difference to placebo in corneal fluorescein staining and symptoms. [661]

8.7 MicroRNAs (miRNAs)

The altered expression of miRNAs in tear fluid has positioned them as promising candidates for non-invasive biomarkers in DED. To date, approximately 300 distinct miRNAs have been identified in tear fluid, many of which are recently discovered and are poorly understood in terms of their regulatory functions. [662] RNA sequencing experiments has revealed that extracellular vesicles in the tear film in both non-Sjögren DED and healthy controls carry distinct miRNA profiles, with 126 differentially expressed miRNAs between the groups. [663] Among these, nine miRNAs (miR-127-5p, miR-1273h-3p, miR-1288-5p, miR-130b-5p, miR-139-3p, miR-1910-5p, miR-203b-5p, miR-22-5p, and miR-4632-3p) were significantly upregulated in DED and were associated with inflammation, indicating a potential role in disease pathogenesis. Other studies, on the other hand, have shown an inverse relationship between miRNAs in tears and inflammation. For instance, miR-223 inhibits hyperosmolarity-induced inflammation through downregulating NLRP3 activation in human corneal epithelial cells and those with DED. [412] Comparison of the expression of 43 miRNAs in the tears of those with Sjögren disease and healthy controls have revealed 14 significantly differentially expressed miRNAs that may be involved in the pathogenesis of Sjögren disease, though none were correlated with ocular staining scores. [664] As the precise molecular mechanisms by which these miRNAs contribute to DED remain largely unclear, and the lack of standardized methods hampers comparison across studies, further research is essential to clarify the specific roles of miRNAs within the ocular environment.

8.8 Translational dry eye models of tear film

8.8.1 *In vitro* models

Translational models of the tear film aim to bridge the gap between basic research and clinical applications. Significant advances in recreating the ocular surface *in vitro* have emerged using organs-on-chips and stem cell-derived organoids. Organ-on-chip technology employs microfluidic devices to replicate the ocular surface environment, allowing for detailed study of tear film components such as mucins and inflammatory cytokines. By utilizing microfluidic platforms with segmented channels, researchers can simulate dry eye conditions by exposing cultured human corneal epithelial cells to an air-liquid interface within a chip. [665] Additionally, induced pluripotent stem cells have facilitated the creation of organoids, offering a new platform to study the molecular mechanisms involved in DED. These approaches have allowed the generation of functional conjunctival epithelial lineage cells, including goblet cells. [445,666,667]

Biomimetic models that rely on the enzymatic removal of mucins in cell culture have emerged to more accurately recapitulate the pathological changes in lubrication, adhesion, and barrier function often observed in mucin-deficient DED. [668] In addition, hydrophilic and hydrophobic glass surfaces have been used to model the interactions of the tear film with a healthy cornea or a hydrophobic cornea in the absence of a glycocalyx. [669] Also noteworthy is the development of miniaturized analogs of a blinking human eye, which utilize a dome-shaped 3D cell culture scaffold to mimic the dynamic interface between the ocular surface and the external environment. [445] Other investigators have used molecular modeling of the tear film and machine learning models to better understand the contribution of proteins to tear film stability and proteomic changes in MGD, respectively. [670,671]

8.8.2 *In vivo* models

Compared to *in vitro* models, there have been relatively fewer innovations in the development of animal models of DED in recent years. The established models rely on surgical procedures, such as removal of the exocrine glands, exposure of the eyes to drugs like benzalkonium chloride, and placement in a dry environment, with or without scopolamine (See Section 5, TFOS DEWS II Tear Film Report). [561] Additionally, there are numerous genetically modified models targeting genes involved in tear production, inflammation, or autoimmune responses. These models continue to generate important results in the development and approval of therapeutic agents for DED. [672] Almost all current models mimic a severe aqueous-deficient, inflammatory form of DED (see Section 7) which is not reflective of EDE, and there is a need to develop alternative models for designing and testing therapies that better address different subtypes of DED. More recently, there has been a renewed emphasis on the use of canine models of evaporative and aqueous-deficient DED to better align with human disease. [673]

8.9 Summary and future directions

This review has identified several areas for future research that may help in the understanding of the pathogenesis and subclassification of DED, as well as identifying better biomarkers to help clinicians classify and monitor DED. Researchers should use lipids that more closely align with those in meibum, rather than mimics of meibum lipids for future *in vitro* experiments. Further research is needed to explore the relationship between disordered lipids that result in spreading and increased elasticity, compared to ordered lipids that lead to improved resistance to evaporation. Additionally, incorporating more accurate models of the muco-aqueous layer in laboratory experiments may help elucidate the roles of lipids, mucins, proteins, and other components in stability and evaporation. There is also a need for a more detailed understanding of whether tear biomarkers can be used to differentiate subtypes of DED as described in the TFOS DEWS III Diagnostic Methodology report. [1] Analyses of microbiome changes across individuals of different ethnicities and countries of residence may provide further insights into its potential role in DED pathogenesis or as a marker for the disease. Understanding the potential role of different microRNAs in DED pathogenesis, DED subtype or as biomarkers could be a highly promising area for future investigation.

9. Pain and Sensation

9.1 Introduction

As described in the TFOS DEWS II Pain and Sensation Report, [40] the ocular surface, particularly the cornea, is densely innervated with sensory fibers that have important functions in the maintenance of ocular surface health. The dynamic nature of ocular surface nerves is often underappreciated, as this system continuously undergoes remodeling in adults, particularly nerve terminals in the corneal epithelium. Many common insults can amplify this remodeling, including accidental or surgical injury or disease processes that lead to chronic inflammation (e.g., Sjögren disease, Steven-Johnson disease, herpes keratitis). Damage to ocular sensory nerves can result in corneal nerve loss, a change in nerve architecture, and altered sensitivity to stimulation. Although nerve regeneration can occur after such damage, it is typically gradual and incomplete. As a result, it often fails to fully restore the original density, architecture, and function of the corneal innervation, leading to persistent changes in neural excitability. This update will consider new evidence in corneal nerve remodeling during normal physiology and following trauma, surgery or inflammation and recent evidence for the anatomical and functional status of the corneal nerves in diagnosing and managing DED.

9.2 Corneal nerve remodeling in adults

The development of corneal nerves begins in the fifth gestational month with the formation of sensory axons around the cornea, followed by their radial extension into the corneal tissue. Although the specific molecular signals controlling corneal nerve growth are still unknown, molecules such as nerve growth factor (NGF) and various neurotrophic factors released by corneal cells contribute to the development and survival of corneal nerves. The growth of the corneal nerves does not stop once development is complete but continues to occur continuously. In adults, corneal subbasal nerves and their terminals undergo continuous morphological rearrangements throughout life while the stromal nerves present few morphological changes. [674-676]

Observations from living human eyes using *in vivo* confocal microscopy reveal that subbasal nerves move centripetally at rates of 10–20 μm per day. These nerves elongate by adding new material near the site of nerve penetration to the epithelium from the Bowman's layer. Distal nerve segments eventually degenerate or slough into the tear film due to the turnover of the corneal epithelium. [40,677] Additionally, intraepithelial nerve terminals undergo spontaneous morphological changes through long-term reconfigurations and short-term reorganization in response to outward migrations of differentiating epithelial cells. Notably, corneal nerve remodeling is more prominent in the central regions of the cornea than in the periphery. Research in living transgenic and knock-in mice reveals that, over time, there are noticeable changes in the subbasal nerve fibers and intraepithelial nerve terminals. The presence of continuous remodeling is supported by the expression of growth-associated protein 43sub in the epithelial nerves of intact corneas. [675,677]

Continuous remodeling is insufficient to maintain lifelong corneal innervation. As mammals age, the density of corneal nerve terminals decreases, leading to reduced corneal sensitivity and changes in tearing regulation. In older individuals, there is a noticeable reduction in subbasal and intraepithelial nerve density and increased tortuosity and disorientation of subbasal nerves. These changes affect all corneal sensory nerves, whether nociceptive or cold thermosensitive, and result in an abundance of simple nerve terminals and a scarcity of complex nerve terminals in aging individuals. The reduced density and dysfunction of corneal nerves, especially cold thermoreceptors, seems to be responsible for the altered tearing and sensitivity in older population. [678]

9.3 Corneal nerve regeneration after surgery

Corneal nerves may be severed during corneal and anterior segment surgery, such as photorefractive keratectomy (PRK), laser-assisted *in situ* keratomileusis (LASIK), cataract surgery, iridectomy, trabeculectomy, and corneal transplantation. The survival of corneal nerves relies on transporting essential substances from their parent nerve

cells in the trigeminal ganglion. Therefore, surgical procedures that disrupt corneal nerve fibers can lead to rapid degeneration of the distal axons, reduced corneal sensitivity, and impaired functional integrity of the ocular surface.

Although corneal nerves can regenerate, this process is slow and imperfect. After most corneal surgeries, the regeneration of nerves is characterized by decreased nerve density, changes in nerve structure, and diminished corneal sensitivity. Regeneration is more delayed and incomplete when nerves are cut closer to their origin. [679] Consequently, surgical disruption of the subbasal and subepithelial nerve plexuses typically results in less severe and short-term damage to corneal innervation than deep or penetrating incisions affecting major stromal nerve bundles.

Although corneal sensitivity typically returns to preoperative levels after LASIK within 6-12 months, [680] a significant proportion of patients may experience long-term dry eye symptoms due to impaired nerve regeneration. [681,682] Nerve damage in small incision lenticule extraction surgery is less marked than in LASIK, resulting in a faster nerve regeneration three months post-surgery. However, no significant difference is observed at six months. [257] A meta-analysis of corneal sensitivity recovery has shown a more rapid early recovery of corneal sensitivity with small incision lenticule surgery compared with femtosecond LASIK. [683](See also TFOS Lifestyle report [684])

In PRK, where the corneal epithelium is removed, and the corneal stroma is reshaped with an excimer laser without creating a flap, nerve regeneration and recovery of corneal sensitivity occur more rapidly than in LASIK. [685] Despite this, subbasal nerve density, architecture, and corneal sensitivity may remain reduced for up to 1-2 years post-PRK. [686]

In cataract surgery, small, perilimbal incisions have minimized the risk of significant injury to corneal innervation. However, in corneal transplantation procedures such as penetrating keratoplasty, a full-thickness incision cuts all corneal nerves, resulting in complete denervation of the transplanted cornea. Nerve regeneration following penetrating keratoplasty is slow, and even years later, the innervation density of the transplanted tissue remains lower than that of the host peripheral cornea. [687] Stromal nerves regenerate poorly, which may be attributed to the misalignment of Schwann cell channels in the donor cornea with the stromal nerve stumps in the host cornea. This contrasts with the perilimbal incisions used in cataract surgery, where stromal nerves on opposing sides of the incision remain closely aligned. Following penetrating keratoplasty, limited nerve regeneration occurs. A few subbasal nerve fibers elongate through the epithelium at the graft margin to enter the donor basal epithelium. Regenerated subbasal nerves may exhibit atypical orientations and morphologies. [687]

The corneal nerve density and corneal sensitivity remain significantly reduced compared to healthy corneas even decades after surgery. [688]

Although theoretically expected not to alter corneal innervation, other corneal transplant techniques, such as Descemet membrane endothelial keratoplasty, show a temporary decrease in nerve density early after transplantation. However, complete recovery of corneal nerve density and function to preoperative values typically occurs within 6-10 months post-surgery. [689] Studies have also reported similar results with Descemet's stripping automated endothelial keratoplasty, where the corneal sensations were noted to be normal within 6 months following surgery. [690] Another study comparing corneal sensation between Descemet membrane endothelial keratoplasty and penetrating keratoplasty showed that corneal sensation improved significantly following Descemet membrane endothelial keratoplasty but was slightly but not significantly decreased after penetrating keratoplasty. [691] These findings suggest better preserved corneal sensations following endothelial transplants compared to PK.

9.4 Nerve regeneration in pathological conditions

The peripheral nervous system has impressive regenerative capabilities following injury. However, injuries to the afferent axons of trigeminal neurons can lead to significant morphological and functional changes, which depend on the magnitude and location of the damage. [679,692] The regeneration of neurons after injury is influenced by a supportive environment for axon growth and the involvement of non-neuronal cells like Schwann cells. [693]

As described in the TFOS DEWS II Pain and Sensation Report, [40] mechanical trauma or inflammatory damage to the peripheral axons of corneal trigeminal neurons causes a complex cellular response and changes to their spontaneous and stimulus evoked firing rates. The expression, distribution and activation thresholds of the transduction ion channels changes. These disturbances lead to increased responsiveness to normal stimuli (allodynia), spontaneous firing without intended stimulation and increased abnormal or unpleasant sensations from a stimulus that would normally elicit a response (hyperalgesia).

The mechanisms that stimulate and direct neurite outgrowth from injured and intact areas of corneal innervation following local nerve injury still need to be fully understood. Corneal epithelial cells release several growth factors following an injury, which may play essential roles. Nerve growth factor is upregulated after corneal epithelial wounding and topical recombinant human nerve growth factor has been used to stimulate corneal nerve regeneration and recovery of corneal sensitivity [694] in the treatment of neurotrophic keratitis, where sensitivity is reduced. [695] The role of nerve growth factor in

the mechanisms resulting in allodynia or hyperalgesia, however, has not been established, although restoration of nerve function may be advantageous.

Ocular and systemic diseases, including herpes virus keratitis, diabetes, and ADDE, can negatively impact corneal nerves. [696] Diabetes significantly alters corneal nerve morphology and function, reducing nerve density and sensitivity. In diabetic patients, the appearance of subbasal nerves resemble intraepidermal small fiber neuropathy, making *in vivo* confocal microscopy of corneal nerves a valuable biomarker for monitoring diabetic neuropathy. [697]

In summary, in healthy adult corneas, the subbasal nerve fibers and intraepithelial nerve endings undergo continuous remodelling and regenerate rapidly following damage, whereas stromal nerves maintain their structure over time. The dynamic nature of intraepithelial nerve endings accounts for their rapid regeneration after injury, while subbasal nerve fibers regenerate more slowly, and stromal nerve trunks may not fully regenerate. Corneal nerve morphology is affected by trauma, ocular surgery, infections, chronic tear deficiency, and various systemic diseases. Although damaged corneal innervation can regenerate like other peripheral nerves, their morphology and function are often incompletely restored, leading to reduced sensitivity, abnormal sensations, and pain.

9.5 Nerve abnormalities and DED

Many manifestations placed under the heading of “dry eye” occur because of morphological and functional changes in ocular innervation. Changes in ocular sensory nerves induce symptoms such as unpleasant sensations of different intensities, ranging from dryness or ocular discomfort to lacerating and burning pain. [698] Conversely, abnormal nerve functioning can lead to alterations in tissue trophism and in the regulation of tear production and blinking, which in turn contributes to both symptoms and signs.[699] In other words, sensory innervation can be altered by chronic eye dryness and can also contribute to DED pathogenesis.

Reduction of the ocular surface moistness either by reduced tearing or increased tear evaporation is a stressful situation for the corneal epithelium exposed to a hyperosmotic tear film and the environment (See section 7). Tear hyperosmolarity can independently affect subbasal corneal nerves in an animal model. [700] As a response to this hyperosmolar and desiccating stress, corneal epithelial cells and immune resident cells produce local inflammatory mediators, primarily IL-1 and TNF- α, which stimulate the production of matrix metalloproteases, activate dendritic cells and local inflammation occurs, [458] leading to sensitization of nociceptive nerve terminals and development of discomfort and pain sensations. When dryness becomes chronic, it also leads to corneal nerve damage and, consequently, the morpho-functional changes of the

sensory nerves during chronic eye dryness resemble both those produced during inflammation and those produced by nerve injury. The eyelid movement causes mechanical friction at the ocular surface when the tear film is thin and does not lubricate well. Together with the chronic inflammation, this movement damages the epithelium and the intraepithelial nerve terminals, and eventually the subbasal nerve fibers, triggering the mechanisms of nerve degeneration and regeneration. [701]

In clinical practice, these degeneration and regeneration processes are evidenced by signs such as the reduction in the density and branching of the subbasal plexus nerve fibers, and the increase in their tortuosity when explored by *in vivo* confocal microscopy, as well as by a reduction in corneal sensitivity to stimulation (shown by increased sensation thresholds) that often occurs simultaneously with hyperalgesia and spontaneous pain sensations and is aggravated by tear film instability. [702] A recent *in vivo* confocal microscopy of corneal nerves confirmed these nerve and branch density, and length reductions in 23 participants with DED and showed that osmolarity exhibited a weak negative correlation to these nerve parameters, whereas other ocular surface signs were not associated with any nerve parameters. [703] Despite the nerve changes reported, it does appear however that not all sensory modalities are equally affected and corneal hypersensitivity to cold stimuli has been reported in DED, [704] in computer vision syndrome [705] and contact lens discomfort. [706] Repeated evaporative cooling at the ocular surface may change the excitability of corneal receptors and their subsequent responsiveness. [40,702,707] Tear film instability can cause changes in suprathreshold scaling by both cold thermoreceptors and polymodal nociceptors. [708]

Most of the alterations of ocular surface sensitivity in chronic eye dryness are due to functional changes in corneal nerves that resemble those seen in injured nerves. There is an increased excitability of corneal nerves consecutive to the increased activity and expression of sodium channels,[709-711] resulting in an increased spontaneous firing of cold thermoreceptors that not only leads to dryness sensations but also to dysregulation of protective mechanisms driven by thermal sensory input such as tearing and blinking, contributing to increase and perpetuate the ocular surface disturbances.

In contrast to the pathophysiology of neuropathic pain, defined by pain or altered sensation due to nerve damage, nociplastic pain occurs in the context of no discernible tissue or nerve damage. Instead, nociplastic pain is believed to be related to dysfunctional central sensory processing of pain through a process called sensory sensitization. Either neuropathic or nociplastic pain can cause chronic ocular pain and and are particularly associated with a subset of DED patients who report pain of a magnitude that is out of proportion to ocular surface findings (Table X).c[712] Quantitative sensory testing and functional magnetic resonance imaging studies remain the primary

methods of assessing nociplastic pain, but the underlying pathophysiology remains poorly understood.

Table 4. Mechanistic characterization of pain, reproduced from De Lott et al., 2025. [712]

Type	Neurobiology	Clinical signs & symptoms	Treatment	Non-ocular and ocular-related examples
Nociceptive	Actual or threatened tissue damage via activation of nociceptors	Localized to the site of injury and surrounding area	• Remove source of nociception • Promote tissue healing • Topical medications • Short term systemic medications: NSAIDs, acetaminophen, anti-inflammatories, opioids	Non-ocular: burn Ocular: corneal abrasion
Neuropathic	Damaged somatosensory system, either peripheral or central	Localized to a dermatome (peripheral) or site of injury (e.g., CNS demyelinating lesion in the spinal cord causing leg paresthesias and pain)	• Peripheral neuropathic: nerve blocks, topical medications (e.g., capsaicin) • Systemic medications: gabapentinoids, tricyclic antidepressants, SNRIs, sodium channel blockers	Non-ocular: diabetic neuropathy Ocular: Post-herpetic neuralgia
Nociplastic	Altered pain perception and processing in the central nervous system	Regional and diffuse pain Multisensory sensitivity COPCs	• Lifestyle: education, sleep, exercise • Behavioral/Integrative: self management, psychotherapy, etc. • Medications: SNRIs	Non-ocular: fibromyalgia Ocular: referred pain from chronic tension-type headache

9.5.1 Impact of DED on nerve structure and function in animal models

Animal models have been developed to replicate human correlates of DED. While initial focus examined immune abnormalities, recent studies have found that various insults that create tear and ocular surface abnormalities also impact upon corneal nerves.

Several mouse models replicate DED in Sjögren disease, including an IL-2 receptor α-chain knockout model. [713] In one study, intraepithelial corneal nerve density of knockout was compared to wildtype mice using confocal microscopy at 4 weeks, 6 weeks, 8 weeks, and 10 to 11 weeks after birth. After combining data from each time point, overall density was significantly decreased in the knockout compared to wildtype. These

results suggest that corneal nerve alterations coincide with the onset of DED, with reduced nerve fiber density commonly noted.

Nerve alterations have also been examined in a GVHD murine model, [714] using an allogeneic bone marrow transplant (case) group (strain B10.D2 to strain BALB/c) and a syngeneic bone marrow transplant group (strain BALB/c to strain BALB/c, control group) at 1-, 2-, 3-, and 4-weeks post-transplant. Corneal nerve tortuosity and branching were increased in the allogeneic bone marrow transplant group at 4-weeks post-transplant compared to 1-week post-transplant. Conversely, no significant differences were observed in branching or tortuosity in the syngeneic group at any time point. These findings support the induction of morphologic changes in corneal nerves associated with the progression of GVHD-associated DED.

Other protocols have been developed to model DED following ocular surgery, such as photorefractive keratectomy (PRK). [715] PRK (-9D) was performed monocularly in New Zealand rabbits while the contralateral cornea was used as a control. Corneal nerve fiber density was evaluated in post-PRK corneas with acetylcholinesterase histochemistry staining at 1 day (n = 3), 1 month (n = 3), 2 months (n = 3), 3 months (n = 3), and 6 months (n = 4) after surgery. Compared to controls (n = 3), nerve density was significantly decreased in post-PRK corneas after 1 day, 1 month, and 2 months, but was not different at 3- and 6-months post-surgery. Nerve morphology alterations persisted however, (increased tortuosity and aberrant innervation) until the final time point at 6 months. Findings that surgery may result in corneal nerve alterations that do not fully return to baseline by 6 months post-surgery, consistent with the effects of refractive surgery in human. [680]

Other studies focused on DED prompted by iatrogenic challenges. One murine study used confocal microscopy with staining using an anti-β3 tubulin antibody to examine corneal nerve fiber density after repeated topical administration of benzalkonium chloride for 14 days. [716] There was a significant reduction in density in benzalkonium chloride-treated mice compared to controls.

Partial or complete lacrimal gland excision has also been used to model DED. [436] In one study, the number of corneal nerve terminals in the surgical group was assessed qualitatively by confocal microscopy and compared to a sham-surgery and control group. The surgical group demonstrated fewer corneal nerve terminals compared to both sham-surgery and control groups. This study also examined nerve function with electrophysiological studies and found that post-surgical animals developed time-dependent corneal mechanical hypersensitivity accompanied by increased spontaneous ciliary nerve fiber electrical activity. Other investigators have found similar functional changes after gland excision, with increased activity and expression of sodium channels

that lead to increased excitability of corneal nerves. This results in a greater spontaneous firing of corneal cold thermoreceptors, causing sensations of dryness, and an enhanced excitability of corneal nociceptors that results in low frequency spontaneous firing and increased response to stimulation, resulting in sustained discomfort and hyperalgesia. [710,717,718] These findings suggest that, beyond nerve density and morphology, corneal nerve function can be impacted differentially in DED subtypes.

Desiccating stress is often induced using a combination of scopolamine and low humidity conditions (e.g., humidity: 25%, constant airflow). In one study, dessication reduced corneal sensitivity over time, [719] linking dry eye onset with corneal functional abnormalities. In summary, animal models can be used to study ocular surface nerve anatomy and sensitivity (and their upstream connection) as they relate to various insults that lead to chronic dry eye. Changes in morphology and sensitivity have been noted across many models, both reduced abilities to detect stimuli (further aggravating ocular surface disturbances) and the development of spontaneous discomfort and pain, that primarily stem from functional alterations in corneal nerves, resembling those observed in peripheral nerves after injury. These data highlight the intricate relationship between corneal nerves and dry eye.

9.5.2 Structural and functional nerve alterations in DED in human

In DED, alteration of corneal nerve structure and function is frequently implicated in the onset, progression, and severity of symptoms. Disruption of the tear film in DED can result in nerve fiber damage which stimulates regenerative processes that are often improper or incomplete. The TFOS DEWS II Pain and Sensation Report in 2017 [720] determined that while IVCM is a reliable method for detecting abnormal corneal nerve morphology, nerve density was a somewhat unreliable marker for nerve fiber damage, but that other morphological parameters were more useful including increased tortuosity, reflectivity and increased beading. More recently a higher frequency of microneuromas has been documented. [702] These structural aberrations, in turn, may further impair tear film production and the blinking reflex resulting in additional nerve damage and conceivably neuropathic dysfunction. [702] Typical manifestations of neuropathic dysfunction include ocular pain, loss of sensation, and tear hyperosmolarity. Damage to nerve fibers in DED can also result in physiologic disruption of the corneal nerve's usual neurotrophic functions, including regulation of nerve growth factor and substance P, that help mediate epithelial growth, local immunoregulation, and nerve fiber regeneration. [702,721] Together, these structural and functional changes likely play a significant role in the pathophysiology of DED.

Interestingly, some individuals with DED report symptoms out of proportion to ocular surface alterations and others are found to have signs that are more severe than their symptoms suggest. [158] One possible explanation for the symptom>sign discordance is the presence of central abnormalities, with inappropriate processing of somatosensory signals due to changes in the central nervous system referred to as central sensitization. Central sensitization is associated with generalized somatosensory dysfunction and cutaneous hypersensitivity in addition to excessive ocular pain.

9.5.2.1 Corneal nerve anatomy in DED

Different manifestations of corneal nerve fiber abnormalities likely have clinical significance for patients with various subtypes of DED but are not yet fully understood. In recent years, several studies have improved the understanding of IVCM in DED. Table 5 summarises the key findings from studies exploring corneal nerve tortuosity.

Nerve tortuosity appears to higher in Sjögren DED compared with non-Sjögren DED [722] [723] based on the Oliveira-Soto and Efron (OSE) grading scale, [724]. However, greater tortuosity was seen in corneal neuropathic pain associated with autoimmune disease compared with corneal neuropathic disease in non-autoimmune DED, MGD or MGD without pain and controls. [725] [726]

Table 5. Corneal nerve tortuosity in DED subtypes

Author(s)/year	Location	Sample Size	Methodology	Findings
[727]	Japan	GVHD (n = 12) Controls (n = 10)	Imaging: multi-image (3), central cornea + inferior limbal epithelia Analysis: ImageJ/NeuronJ, OSE grading scale	↑ tortuosity in GVHD compared to controls.
[722]	United States	SDED (n = 22) NSDED (n = 12) Controls (n = 5)	Imaging: multi-image (3), central cornea Analysis: ImageJ/NeuronJ, OSE grading scale	↑ tortuosity in SDED and NSDED compared to controls.
[725]	France	AIDED-NCP (n = 7)	Imaging: single	↑ tortuosity in

		MGD-NCP (n = 11) AIDED (n = 8) MGD (n = 8) Controls (n = 10)	image, central cornea Analysis: ImageJ/NeuronJ, study-designed tortuosity scale	AIDED-NCP compared to controls. Tortuosity not different in MGD-NCP, painless AIDED, or painless MGD compared to controls.
[723]	China	SDED (n = 22) NSDED (n = 20)	Imaging: multi-image (5), central cornea Analysis: ImageJ/NeuronJ, OSE grading	↑ tortuosity in SDED compared to NSDED.
[726]	France	SDED (n = 71) MGD (n = 20) Control (n = 20)	Imaging: multi-image (5), central cornea Analysis: ImageJ/NeuronJ, OSE grading	↑ tortuosity in SDED compared to MGD and controls.

GVHD=graft-versus-host disease; DED=Dry eye disease; SDED=Sjögren DED; NSDED= non-Sjögren DED; AIDED= autoimmune DED; MGD= meibomian gland dysfunction; NCP= neuropathic corneal pain.

Corneal nerve fiber density (CNFD) has also been studied in various DED subtypes (Table 6) and in most studies appears to be significantly decreased in Sjögren DED, GVHD, non-Sjögren DED and neuropathic corneal pain compared to controls, [722,728 729,730 731] although within studies density is similar between DED types. [732] One small study has evaluated MGD and found no reduction compared with controls, [725] however a larger study found a reduction in both ADDE and EDE compared with controls. [733] The reduction in density is greater with increased symptoms. Those with normal-to-mild symptoms have similar CNFD to controls. [734] These studies reinforce that CNFD values are generally lower in some DED subtypes, most notably in auto-immune associated DED and that CNFD may be affected by degree of symptoms severity/ocular pain.

Table 6. Corneal nerve fiber density in DED subtypes

Author/year	Location	Sample Size	Methodology	Findings
[734]	India	Normal-mild EDE (n = 29)	Imaging: single image, central cornea	↓ CNFD in EDE with moderate-to-severe symptoms compared

		Moderate-severe EDE (n = 23) Control (n=43)	Analysis: ACCMetrics	to controls. CNFD not different in EDE with normal-to-mild symptoms compared to controls.
[727]	Japan	GVHD (n = 12) Control (n = 10)	Imaging: multi-image (3), central cornea + inferior limbal epithelia Analysis: ImageJ/NeuronJ	CNFD not different in GVHD compared to controls.
[722]	United States	SDED (n = 22) NSDED (n = 12) Control (n = 5)	Imaging: multi-image (3), central cornea Analysis: ImageJ/NeuronJ	↓ CNFD in SDED and NSDED compared to controls.
[730]	Japan	NSDED (n = 25) Control (n = 25)	Imaging: multi-image (5), central cornea Analysis: ImageJ/NeuronJ	↓ CNFD in NSDED compared to controls.
[735]	France	DED (n = 32) Control (n = 15)	Imaging: multi-image (5), central cornea Analysis: ImageJ/NeuronJ	↓ CNFD in DED compared to controls.
[732]	Italy	SDED (n = 20) GVHD (n = 19) Control (n = 30)	Imaging: multi-image (3), central cornea	↓ CNFD in SDED and GVHD compared to controls.

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			Analysis: ACCMetrics	
[736]	Turkey	GVHD (n = 22) Control (n = 28)	Imaging: multi-image (3), central cornea Analysis: ImageJ/NeuronJ	↓ CNFD decreased in GVHD compared to controls.
[729]	United States	NCP (n = 25) DED (n = 30) Control (n = 16)	Imaging: multi-image (3), central cornea Analysis: ImageJ/NeuronJ	↓ CNFD in the NCP and DED compared to controls.
[731]	United States	ADDE (n = 24) EDE (n = 46) Control (n = 45)	Imaging: multi-image (3), central cornea Analysis: ImageJ/NeuronJ	↓ CNFD in ADDE and EDE groups to controls.
[725]	France	AIDED-NCP (n = 7) MGD-NCP (n = 11) AIDED (n = 8) MGD (n = 8) Control (n = 10)	Imaging: single image, central cornea Analysis: ImageJ/NeuronJ	↓ CNFD in AIDED and MGD-NCP compared to controls. CNFD not different in AIDED-NCP or MGD compared to controls.
[723]	China	SDED (n = 22) Control (n = 20)	Imaging: multi-image (5), central cornea Analysis: ImageJ/NeuronJ,	↓ CNFD decreased in SDED compared to NSDED.
[737]	India	ADDE/EDE + symptoms (n = 10) Control (n = 15)	Imaging: single image, central cornea	↓ CNFD in those with DE symptoms compared to controls.

		ADDE +/- symptoms (n = 57) EDE +/- symptoms (n = 16) Control (n = 15)	Analysis: ACCMetrics	CNFD not different in ADDE (+/- symptoms) or EDE (+/- symptoms) compared to controls.
[738]	China	DED (n = 155) Control (n = 20)	Imaging: multi-image (≥10), central Analysis: CS-Net	↓ CNFD in DED compared to controls.
[726]	France	SDED (n = 71) MGD (n = 20) Control (n = 20)	Imaging: multi-image (5), central cornea Analysis: ImageJ/NeuronJ	↓ CNFD in SDED compared to controls. CNFD not different in SDED compared to MGD.
[739]	China	DED (n = 25) Control (n = 20)	Imaging: single image, central cornea Analysis: ImageJ/NeuronJ, ACCMetrics	↓ CNFD in DED compared to controls.
[733]	Turkey	ADDE (n = 22) + EDE (n = 21) Control (n = 20)	Imaging: multi-image (3), central cornea Analysis: ImageJ/NeuronJ	↓ CNFD in ADDE and EDE compared to controls.

DED= dry eye disease; EDE= evaporative dry eye; GVHD= graft-versus-host disease; SDED= Sjögren dry eye disease; NSDED= non-Sjögren dry eye disease; NCP= neuropathic corneal pain; ADDE= aqueous-deficient dry eye; CNFD= corneal nerve fiber density; MGD = meibomian gland dysfunction.

Newer nerve parameters, such as microneuromas, defined as irregular expansions of subbasal nerve endings that suggest nerve damage underlying ocular symptoms, have

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been included in some studies (Table 7). Conceivably, either due to the subjective determination of microneuromas, or the difficulty of discriminating a turn in the nerve from an ending, there are inconsistencies in the literature. [740,741] Some studies found an increase in microneuromas among DED subtypes compared to controls, most notably in autoimmune DED and MGD, [725,742] with a greater increase where there was co-existing neuropathic corneal pain. [725] While these studies suggest that microneuroma frequency or number is increased among several DED subtypes compared to controls, other studies have not replicated these findings in those with symptoms of DED. [743] These findings point to the need for additional studies that standardize microneuroma determination and further explore their utility as a biomarker for DED.

Table 7. Corneal nerve microneuromas in DED subtypes.

Author(s)/year	Location	Sample Size	Methodology	Findings
[729]	United States	NCP (n=25) DED (n=30) Control (n = 16)	Imaging: multi-image (3), central cornea Analysis: ImageJ/NeuronJ	MN frequency ↑ in NCP compared to controls. MN frequency not different in DED compared to controls.
[725]	France	AIDED-NCP (n = 7) MGD-NCP (n = 11) AIDED (n = 8) MGD (n = 8) Control (n = 10)	Imaging: single image, central cornea Analysis: ImageJ/NeuronJ	MN frequency ↑ in AIDE, AIDE-NCP, MGD, and MGD-NCP compared to controls.
[743]	United States	DED (n = 119) DED + refractive surgery (n = 19) Control (n = 18)	Imaging: multi-image (3), central cornea Analysis: ACCMetrics	MN frequency not different in DED or DED + refractive surgery compared to controls.
[742]	India	ADDE/EDE + symptoms (n = 14) Control (n = 27)	Imaging: single image, central cornea	MN frequency ↑ in those with DE symptoms compared to controls.

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		ADDE +/- symptoms (n = 24) EDE +/- symptoms (n = 65) Control (n = 27)	Analysis: ACCMetrics	MN frequency not different in ADDE (+/- symptoms) or EDE (+/- symptoms) compared to controls.
[726]	France	SDED (n = 71) MGD (n = 20) Control (n = 20)	Imaging: multi-image (5), central cornea Analysis: ImageJ/NeuronJ	MN frequency ↑ in SDED compared to controls but not compared to MGD.[726]

DED= dry eye disease; NCP= neuropathic corneal pain; AIDED= autoimmune dry eye disease; MGD= meibomian gland dysfunction; ADDE= aqueous-deficient dry eye; EDE= evaporative dry eye; SDED= Sjögren dry eye disease; MN= microneuroma.

Recent work has explored the use of AI technology in nerve analysis, particularly the use of deep learning AI models to automatically segment corneal nerves.[744]. Using the TFOS DEWS II diagnostic criteria, [131] individuals with DED had reduced CNFD compared with controls and values were similar to that of manual annotation.[738] Applying new technologies to IVCM will allow for more consistent and faster quantification of images that can applied across centers, facilitating comparisons across different populations.

9.5.2.2 Corneal sensitivity in DED

Corneal sensitivity testing is an important diagnostic tool that can provide valuable insights into underlying somatosensory abnormalities in DED. The TFOS DEWS II report highlights that corneal sensitivity to mechanical stimuli tends to be reduced in patients with ADDE.[720] However, studies using air/gas esthesiometers have shown equivocal findings, with increased [745,746], decreased [747,748], and similar [749] corneal sensitivity in various DED subtypes. This could be due to differences in stimulus parameters (e.g., cold, chemical stimulus included in some testing paradigms) or due to differences in neural responses across various DED subtypes (e.g., ADDE vs MGD).[750] Thus, consideration of the type of stimulus, the subtype of DED and patient factors is critical to interpreting corneal sensitivity results. Table 7 summarizes studies evaluating sensitivity or pain responses in DED.

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In a study of corneal mechanical sensitivity in those with DED symptoms tested using an air jet aesthesiometer, 13% showed hypersensitivity and 11% had hyposensitivity. [749] When grouped by DED subtype, there was an association between ADDE and reduced sensitivity.[726] There is growing awareness that some individuals with symptoms of DED likely have neuropathic contributors to symptoms and this group, as a whole, display corneal hypersensitivity. Individuals with hypersensitivity had more severe ocular pain complaints while those with hyposensitivity had more severe epithelial disruption.[751] These data highlight that differences in corneal sensation and symptom reporting may align with differences in DED profiles.[751] In a study using the Cochet-Bonnet aesthesiometer, mechanical corneal and pain sensitivity were assessed in individuals with short tear film breakup DED, defined as TBUT<5secs and Schirmer value >5mm, and controls (n=46). Pain sensitivity threshold but not mechanical sensitivity threshold was higher in the DED group. [752] Corneal hyperalgesia was present in 37% of the DED group and there was a strong relationship between pain sensitivity and subjective pain scores. [752] Cold sensitivity measured using cooling scores was assessed in a DED and control group and was compared with mechanical sensitivity threshold using a Cochet-Bonnet esthesiometer. Treatment with a lubricant containing TRPM8 agonist (0.01% menthol, a compound that activates cold thermoreceptors) or lubricant alone was applied bilaterally in a cross-over design. While there was no difference in mechanical sensitivity, the DED group reported a higher cooling response and score compared to controls, suggesting mechanical and cold sensitivity may be differentially impacted in DED. [704] The duration of disease was associated with a greater cooling response, suggesting that nociceptor activity may change over disease duration. Taken together, these results suggest abnormal pain and conceivably cold stimulus processing in different DED subtypes and that different qualities of stimulus processing and other ocular surface sites, may help to understand neurogenic changes in different subtypes of DED.

Increased sensitivity (reduced mechanical threshold) has been reported after treatment of DED or GVHD with autologous serum tears, [753] Sjögren DED with cyclosporine, [754] and short TBUT DED with diquafosol (Table 8).[755] This underscores the dynamic nature of corneal sensitivity, highlighting changes in the short term, with disease, and with treatment, the implications of which need further study.

The relationship between corneal nerve structure and function is equivocal. Nerve structure was not consistently associated with gross corneal sensitivity using a cotton wisp test, where a significant relationship was only noted between corneal sensitivity and corneal nerve fiber area.[756] In a diabetic population, there was a stronger relationship between anatomy and function. [757] These data highlight that relationships between corneal nerve anatomy and function may vary with disease type and that relying on structural findings alone may not suffice in evaluating nerve functional health.

Given the importance of understanding nerve function in DED, there have been efforts to develop new esthesiometers aimed at improving precision, range, stimulus type and portability, to facilitate routine clinical use. These are reviewed in the TFOS DEWS III Diagnostic Methodology Report. [1] Briefly these have included non-contact airjet esthesiometers, [758,759] liquid jet esthesiometers, [705,760,761] and a single-use filament mechanical esthesiometer (Kerasense, Dompè farmaceutici SPA, Italy). [759,762] While the initial reports on these esthesiometers are encouraging [705], further research is necessary to explore their clinical utility, particularly in individuals with DED.

Table 8. Corneal sensitivity in DED

Author/year	Location	Sample population	Methodology	Findings
Variation in testing paradigms and DED subpopulations				
[751]	USA	DE symptoms (none-severe) (n=129)	Corneal detection and pain thresholds and relationship with symptoms and signs of DED (Belmonte esthesiometer)	Those with hypersensitivity had more severe ocular pain complaints while those with hyposensitivity had more severe epithelial disruption
[704]	USA	DED (TBUT≤5 secs, and corneal staining≥4; n=33) Control (n=15)	Cold sensitivity score by comparing responses to lubricants with or without a cold receptor stimulator (TRPM8 agonist: 0.01% menthol) (Cochet-Bonnet esthesiometer)	Mechanical sensitivity was similar between groups. Cold sensitivity reflected by cooling scores was greater in DED than controls (24.6 vs 12.1, p=0.0005) and shorter disease duration (<10 years) had higher cooling scores than longer disease duration (>10 years)
[752]	Japan	Short TBUT DED (TBUT<5secs, Schirmer value>5mm; (n= 60)) Control (n=46)	Mechanical corneal sensitivity and pain sensitivity (fiber length that elicited pain) thresholds (Cochet Bonnet esthesiometer)	Detection thresholds were similar between the groups Short TBUT DED had higher pain thresholds, (26.3±23.1 vs.

				6.9±16.4 mm, p<0.01) which correlated with subjective pain scores (scale 0-4; r=0.24, p<0.05)
[749]	USA	DED (DEQ ≥6) N= 403	Corneal detection thresholds in DED (Belmonte esthesiometer)	Mean corneal detection threshold of 87 ± 46 mL/min. 13% of participants with hypersensitivity and 11% with hyposensitivity
Alterations in corneal sensitivity in response to therapy:				
[754]	France	SSDED (primary/ secondary; n=30) Control (n=15)	Corneal sensitivity tested after 6 months of 0.05% cyclosporine use (Cochet-Bonnet)	Increased corneal sensitivity over 6 months of treatment (5.1 to 5.6mm, p=0.03)
[755]	Japan	Short- TBUT DED (TBUT <5secs) DQS group (n=12) AT (n=15)	Corneal detection and pain thresholds measured after 5 weeks of DQS or AT therapy. (Cochet-Bonnet)	DQS treatment significantly lowered pain sensitivity compared to AT after 5 weeks of therapy. Neither lowered detection thresholds
[753]	USA	DED (Ocular GVHD; n=20)	Corneal sensitivity tested after 12 weeks of 20% AST use (Cochet-Bonnet)	Mean corneal sensitivity increased at 12 weeks of 20% AST in 17 patients with ocular GVHD (31.1±23.8 to 51.6±12.6mm, p=0.001)

DED= dry eye disease; GVHD = graft versus host disease; TBUT = tear film breakup time; NCP= neuropathic corneal pain; AT = artificial tears; AST = autologous serum tears

In conclusion, sensitivity measurements can examine certain functional qualities of corneal nerves in DED. There is heterogeneity across and within DED subtypes, with conditions such as ADDE generally showing reduced corneal sensitivity, and with symptom severity overall relating to increased sensitivity. Variability in results can also be attributed to the type of nociceptors stimulated, nature of the stimulus, threshold vs pain sensitivity, esthesiometer type, and factors such as disease duration and severity, underscoring the complexity of DED. As research progresses, the integration of

sensitivity assessment into routine clinical practice could enhance the management and therapeutic outcomes for patients with DED**.** Most studies have focused on corneal sensitivity only and there is limited evidence for other ocular surface regions including the bulbar and palpebral conjunctiva and lid margin, which may also contribute to ocular symptoms. A commercial, quantitative and suitably sensitive, non-contact esthesiometer is urgently needed along with evidence-based normative and reference values for application in different subtypes of DED.

9.6 Anesthetic challenge in DED

The TFOS DEWS II report emphasized the role of neurosensory abnormalities in the etiology of DED and the need to evaluate somatosensory functions in its assessment. [720] The topical anesthetic challenge has emerged as an important tool for assessing neurosensory abnormalities. This simple and rapid test can differentiate between pain originating from peripheral nociceptor activation versus central (or non-ocular surface) mediated pain arising from proximal sensory pathways or the central nervous system. [720] The test involves applying a drop of topical anesthetic (e.g., 0.5% proparacaine hydrochloride) to the ocular surface and evaluating pain relief. Pain relief typically indicates peripheral neuropathic or nociceptive causes of pain, as proparacaine stabilizes nociceptor neuronal membranes, impeding initiation and conduction of nerve impulses. [763] In contrast, the absence of relief suggests a central or non-ocular surface etiology of pain, while partial improvement may indicate a mixed component to pain. [764] However, it is important to note that in patients with complete pain relief, the test cannot differentiate between pain due to peripheral nerve abnormalities or nociceptive causes like DED. [763] It is further not informative if no pain is present prior to anesthetic placement (in individuals with pain that waxes and wanes).

Persistent pain after anesthesia can indicate central abnormalities in pain processing pathways. [765] Specifically, individuals with persistent pain after topical anesthesia often exhibit other features of somatosensory dysfunction. A study of veterans with DED symptoms were categorized based on anesthetic challenge response. Those with persistent pain had greater discordance between signs and symptoms and lower non-ocular cutaneous cold and hot thresholds (see Section 9.7) compared with those where ocular pain was reduced following topical anesthesia. [766] These data suggest increased cutaneous sensitivity both at a site innervated by the trigeminal nerve (forehead) and at a distant site (forearm), supporting the contribution of central mechanisms in individuals with persistent ocular pain following anesthesia. Similarly, individuals with DED who experience with central-dominant pain had higher ocular and non-ocular pain scores compared with those who had greater reduction in pain with topical anesthesia (peripheral-dominant).[767] These findings have therapeutic implications as treatments for peripheral vs centrally mediated pain vary. For instance, patients with peripheral

dominant pain may benefit more from anti-inflammatory and topical neuromodulation, while those with central pain may benefit from systemic neuromodulation.[763] Further studies are needed, however, that examine whether results of diagnostic tests can be used to predict therapeutic responses.

The anesthetic challenge is a simple and accessible test that can aid in the evaluation of somatosensory (dys)function in DED. When used appropriately, it can help identify the location of pain generation which may assist in formulating a personalized treatment plan for individuals with ocular pain.

9.7 Quantitative sensory testing in DED

Somatosensory function, including central abnormalities, can be investigated using quantitative sensory testing, a method of assessing an individual's response to various stimuli (thermal, mechanical, vibratory). Similar to ocular sensory testing, non-ocular function can be evaluated using detection and pain thresholds, measuring degree of pain to a fixed stimulus, temporal summation (a phenomenon where repeated stimuli of the same intensity cause a gradual increase in pain intensity), and aftersensations (a sensation that persists after the external stimulus that caused it has stopped). Abnormalities in the latter two tests (temporal summation and aftersensations) have been linked to central abnormalities. Several studies have applied these protocols to the study of DED (Table 9).

Table 9: Quantitative Sensory Testing in DED

Author/year	Location	Sample and Size	QST Metrics	Results
[768]	United States	DED (n=118)	Detection threshold for vibration, cool, warm, cold pain, and hot pain at the forearm Pain, aftersensation and temporal summation of cold pain and hot pain stimuli at the forearm	HTPS positively correlated with burning pain, wind sensitivity, and ocular pain.
[769]	United States	Concordant DED (n=25) Discordant	Intensity rating of pressure pain at the thumbnail	Intensity rating for pressure pain and auditory tones were not statistically different in

		DED (n=23) Total DED (n =48) Controls (n= 26)	Intensity rating of auditory tones in both ears	discordant DED. Pressure pain and auditory sensitivity were not correlated with DED symptoms.
[770]	United States	DED (n=326)	Detection threshold for vibration, cool, warm, cold pain, and hot pain at the forearm Pain, aftersensation and temporal summation of cold pain and hot pain stimuli at the forearm	Aftersensations of cold pain and hot pain at the forehead and forearm positively correlated with DED discordance scores. Intensity ratings at threshold of cold pain and hot pain positively correlated with DED discordance scores.
[771]	United States	LASIK (n=43)	MPT at the forehead and forearm Temporal summation and aftersensation intensity of mechanical pain at the forearm Conditioned pain modulation at the forearm	MPT at the forehead negatively correlated with baseline ocular pain. MPTS positively cordrelated with baseline DE symptoms. Aftersensations were associated with increased baseline DED symptoms and ocular pain. Presence of aftersensations pre-LASIK was a predictor for chronic DE symptoms at 6 months post-LASIK.
[772]	United States	DED (n=235)	Aftersensation intensity of cold pain	Intensity of ocular pain due to light was a

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	and hot pain at the forearm	predictor for the presence of aftersensations.

DED = dry eye disease; HTPS = hot pain temporal stimulation; MPT = mechanical pain threshold; MPTS = mechanical pain temporal summation

Hot and cold thermal stimuli (hot and cold) are frequently used to evaluate somatosensory (dys)function. Despite variation in quantitative sensory testing protocols, increased responsiveness to hot pain has been noted in individuals with DED where there are increased symptoms disproportionate to the signs (measured as a discordance score). [768] DED discordance score positively correlated with hot pain aftersensation intensity at the forehead. [770] These studies support the finding that abnormal perception of hot pain at sites near and distant to the eye correlate with DED symptoms in the absence of ocular surface signs and that the altered perception may arise from central abnormalities.

Mechanical pain can also be used as a stimulus to evaluate somatosensory (dys)function. Prior to LASIK surgery, mechanical cutaneous pain thresholds at the forehead negatively correlated with ocular pain intensity, indicating higher sensitivity in individuals with more severe pain. The presence of aftersensations to a cutaneous mechanical stimulus prior to surgery predicted symptoms of DED six month after surgery. [771] Using a thumbnail pressure test, no significant differences were found in mechanical pain testing or auditory testing in discordant DED compared with concordant DED or controls, although the sample size was relatively small. [769] These discrepancies highlight the need for more standardization in protocols when applied to the study of DED and perhaps better definition of DED subtypes.

9.8 Brain imaging in DED

Advanced neuroimaging techniques, such as using functional magnetic resonance imaging (fMRI), have been applied to the study of individuals with chronic ocular pain. [773,774] fMRI is a neuroimaging technique that uses a light stimulus and measures regional changes in brain metabolism over time and has been extensively utilized in various studies. [775] In a small study of individuals with and without ocular pain, the chronic ocular pain group exhibited greater activation in brain regions like the primary somatosensory, insular, and anterior mid-cingulate cortices compare with controls. Instillation of topical anesthetic reduced activation in the primary somatosensory and anterior mid-cingulate cortices. These results suggest different activations patterns to light in individuals with chronic pain and photophobia, supporting the contribution of central mechanisms in driving pain in this group. [773]

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fMRI studies have also studied the neural impact of various therapies used to treat ocular pain. In a study examining the impact of botulinum toxin A administered to the forehead, frontalis, procerus and corrugators, showed that in those individuals who reported reduced unpleasantness scores when viewing the light stimulus during the post-injection scan, exhibited activation in the spinal trigeminal nucleus in response to light stimuli prior to injections and this was not evident in those who did not show an improved response post-injection. [776] This suggests that photophobia may be driven by different neural pathways, and that individuals with activity within the spinal trigeminal nucleus during light stimulation may be the ones more likely to response to botulinum toxin. Use of the FL-41 spectacle tint to reduce photophobia led to significant reductions in light-evoked blood oxygen level dependent signals (used as an indirect measure of neural activity) in the bilateral primary and secondary somatosensory, bilateral insular, right temporal pole, precuneus, anterior cingulate cortex and paracingulate cortices as well as bilateral cerebellar hemispheric lobule VI, although the responses in regions associated with pain processing were not eliminated completely. [777] These findings indicate that fMRI may play a role in predicting and monitoring responses to therapy and suggest that more studies are needed to examine central mechanisms to pain in various DED subtypes.

9.9 Future directions and conclusions

Ocular pain and sensation are intimately associated with the structural anatomy of corneal nerves. This structure can be altered by trauma, surgery, DED, other systemic diseases, natural aging, and incomplete or improper neural regeneration. These alterations are pleomorphic but most commonly manifest as reduced nerve fiber density and increased nerve tortuosity. Aberrant corneal nerve anatomy generally correlates with altered sensation, increased ocular pain, and symptoms typical of DED. Functional abnormalities of corneal nerves are also implicated in altered ocular sensation and include changes in growth factor activity and increased expression of sodium channels. Ocular pain, in some individuals can be driven by generalized dysfunction of peripheral and central nervous systems, with studies supporting activation of the primary somatosensory, insular, and anterior mid-cingulate cortices in individuals with chronic ocular pain, which may be mitigated with certain therapies.

Research into this field is ongoing and future developments may include the elucidation of specific pathways controlling corneal nerve development and regeneration, exploration of neuroimmune crosstalk in DED, greater implementation of artificial intelligence networks in processing large bodies of data, more sophisticated assessments of peripheral and central nerve function, and targeted treatments to address dysfunction in an individual patient.

10. Iatrogenic

10.1 Introduction

Iatrogenic disease is an adverse clinical condition resulting from diagnosis or medical treatment performed by a health professional. It affects many patients worldwide in all fields of medicine, including ophthalmology. [778] In the eye, the ocular surface and the tear film are probably the most affected. This assumption is related to the fact that this anterior interface represents the first target for the effect or penetration of the first line of treatment for most eye diseases and is involved directly or indirectly in the most common surgical and non-surgical procedures for the eye. Among all ocular surface disorders, DED is the most prevalent. [779] In addition to topical medications and surgical and non-surgical procedures, systemic drugs and the use of contact lenses are major causes of iatrogenic DED (Table 9).

The TFOS DEWS II recognized the importance of this topic, and a specific report about Iatrogenic DED was included.[778] The impact of elective medications and procedures was also highlighted in the "Lifestyle Epidemic: Ocular Surface Disease TFOS Workshop".[684] This review presents an update since these reports of the most common iatrogenic causes of DED, including a summary of pathophysiology and recommendations for management.

Table 10. Classification of iatrogenic DED (Reproduced from [778])

I. Drug-induced
A. Topical
B. Systemic
II. Contact lens-induced
III. Ophthalmic surgery*
A. Refractive surgery
B. Keratoplasty (Penetrating, lamellar and endothelial)
C. Cataract surgery D. Lid surgery E. Other surgeries
1. Conjunctival surgery 2. Glaucoma surgery 3. Vitreoretinal surgery 4. Strabismus surgery 5. Intrastromal corneal ring segment implantation

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6. Others

IV. Non-surgical ophthalmic procedures

A. Botulinum toxin
B. Crosslinking (CXL)
C. Cosmetic procedures
D. Others

V. Non-ophthalmic conditions

A. Graft-versus-host disease (GVHD)
B. Others

*Ophthalmic surgeries were extensively reviewed previously and are not included in this update. [684,778]

10.2 Topical drug-induced DED

10.2.1 Prevalence

As mentioned in TFOS DEWS II, evaluating DED caused by topical medications is challenging, as clinical trials often exclude patients with ocular surface diseases, which may lead to an underestimation of symptoms of DED.[778] Most epidemiological studies evaluate the effect of anti-glaucoma topical medications on the ocular surface. Given glaucoma and DED are two common conditions that can occur concurrently in the same individual, these results may be confounded. Furthermore, patients undergoing treatment for glaucoma frequently show signs of deterioration in the ocular surface.[780]

In a multicenter study conducted across four European countries with 9,658 patients, over 40% of individuals treated for glaucoma reported symptoms of DED, such as pain or discomfort during the application of eye drops, foreign body sensation, dry eye, and burning.[781]. Furthermore, more than 20% of patients exhibited signs of blepharitis, conjunctival hyperemia, or keratitis. These findings were more prevalent in patients treated with preservative-containing eye drops. Similarly, a recent cross-sectional comparative study with 320 patients concluded that glaucoma patients are more affected by DED than non-glaucoma patients, showing a lower TBUT and greater corneal staining in eyes with glaucoma using multiple eyedrops and daily doses.[782] Another cross-sectional study with 101 patients undergoing anti-glaucoma treatment detected signs and symptoms of DED in over 50% of the patients and advanced changes in the ocular surface in 27% of the individuals examined.[783] Reduced values on the Schirmer test were observed in 61%, decreased TBUT in 78%, and staining of the ocular surface in 22% of the subjects.

In a German study involving 20,506 glaucoma patients from 900 centers, the prevalence of DED was higher in women (56.9%) than in men (45.7%), and this difference was more pronounced in patients over 50 years old. The prevalence increases with age and duration of glaucoma, and it occurs more frequently when three or more medications are used.[784] In another epidemiological study including 4,107 glaucomatous patients, the most frequently reported dry eye symptoms were: excessive discomfort after applying anti-glaucoma eye drops (43%), pressure behind the eyelids (40%); foreign body sensation (31%); dry eye sensation (23%); excessive reflex tearing (21%); and eyelid itching (18%) of cases. These symptoms were significantly more prevalent when using preservative-containing than preservative-free eye drops.[785,786]

10.2.2 Topical drugs contributing to DED

According to TFOS DEWS II, various topical medications and excipients have been implicated in DED. [778] (Table 11) Of particular concern is the use of chronic medications, such as anti-glaucoma, antiallergic, and anti-inflammatory eye drops. However, there needs to be more specific data on active compounds, as ophthalmic formulations are often evaluated with preservatives, making it difficult to isolate the impact of the medications, preservatives, and excipients. Also, common excipients in ophthalmic preparations, such as solutions, ointments, suspensions, and emulsions, may cause dry eye symptoms. [778] The chemical properties of the formulation, including isotonicity/hypotonicity and pH, can influence the tear film and local tolerance after application. [778]

Topical drug-induced DED has mainly been studied in individuals in long-term treatment for glaucoma and ocular hypertension. [778] Other than the presence of benzalkonium chloride, the main risk factors for iatrogenic DED in anti-hypertensive topical drug users are the treatment duration, higher IOP, and glaucoma severity. [787,788] The frequency of mild or greater DED symptoms tended to increase with an increasing number of anti-glaucoma medications. [788] Interestingly, patients on brimonidine performed the worst. Patients on timolol reported pain induced by light, and those on latanoprost complained of stinging. [788]

Table 11. Topical drugs implicated in DED [778]

Agents used to treat glaucoma
Betablocking agents (eg: Betaxolol, Timolol)
Adrenergic agonist drugs (eg: Apraclonidine, Brimonidine)
Carbonic anhydrase inhibitors (eg: Brinzolamide, Dorzolamide)
Cholinergic agents (eg: Pilocarpine)

Prostaglandin analogs (eg: Bimatoprost, Latanoprost, Travoprost)
Agents used to treat allergies (eg: Emedastine Olopatadine)
Antiviral agents (eg: Aciclovir, Idoxuridine, Trifluridine)
Decongestants (eg: Naphazoline, Tetryzoline)
Miotics (eg: Dapiprazol, Pilocarpine)
Mydriatics and cyclopegics (eg: Cyclopentolate Tropicamide)
Preservatives (eg: Benzalkonium chloride)
Topical and local anesthetics (eg: Proxymetacaine, Tetracaine)
Topical ocular NSAIDs (eg: Bromfenac, Diclofenac, Ketorolac, Nepafenac)

10.2.3 Mechanism

Topical medications can affect the ocular surface through various mechanisms, causing allergic, toxic, and/or immune-inflammatory effects or through chemical interactions with different components of the ocular surface. [778] These effects can result from disruption of the lipid layer of the tear film due to the detergent properties of the compounds, reduced aqueous secretion, damage to the ocular surface epithelium, neurotoxic effects on the corneal nerves, and injury to the eyelids, including the skin or meibomian glands. [789-792]

Indirect effects may also arise if chronic inflammation induced by the topical medication stimulates the precursors of the keratinized envelope, leading to the entrapment of mucous cell contents and squamous metaplasia. [790] Keratinization of the eyelid margins can further worsen MGD. [789] Additionally, destruction and/or dysfunction of goblet cells, which are increasingly recognized as having an important role in immunomodulation of the ocular surface, can exacerbate chronic inflammation. [793]

As reported in TFOS DEWS II, distinguishing between spontaneous changes in ocular surface disease and medication-induced effects presents a clinical challenge. [778] Conjunctival allergic reactions caused by eye drop use may be indicative, but both conjunctival congestion and papillary conjunctivitis can occur with or without atopy. Additionally, delayed allergic reactions may arise, often mimicking blepharitis with mild inflammation. [778] Similarly, determining whether corneal staining is due to pre-existing or induced DED, or caused by toxic epithelial damage and corneal melting, such as that induced by overuse of anesthetics or non-steroidal anti-inflammatory agents, can be complicated. With many factors to consider, the relationship between eye drop use and ocular inflammation, tear film instability, or ocular surface staining is often difficult to

establish, especially when treatment is essential for severe or vision-threatening conditions. This is particularly true in glaucoma, where treatments are usually prolonged.

10.2.4 Role of preservatives and excipients

Preservatives used in topical ophthalmic medications are derived from different chemical families, such as mercury derivatives, alcohols, parabens, ethylenediaminetetraacetic acid, chlorhexidine, and quaternary ammonium. They act as a surfactant to solubilize ionic components, facilitating stabilization of medications, and inhibit microbial activity. [792] Benzalkonium chloride is a quaternary ammonium compound widely used as a preservative in eye drops that has its toxicity to the ocular surface well-documented. Benzalkonium chloride concentrations as low as 0.0001% can cause damage to corneal epithelial cells, with more severe toxic effects observed at higher concentrations, such as 0.01% and 0.2%. [792,794]

The impact of benzalkonium chloride on the ocular surface involves multiple mechanisms of action. One main effect is through mitochondrial dysfunction, which leads to dysfunction and subsequent cell apoptosis. [390,790] Additionally, exposure increases the production of reactive oxygen species, inducing oxidative stress in corneal epithelial cells. [390,790] This stress directly contributes to the apoptosis and damage to ocular surface and trabecular cells.

Benzalkonium chloride can activate various inflammatory pathways, increasing the production of the inflammation marker HLA-DR and the expression of pro-inflammatory cytokines such as IL-6, IL-8, and CCL2, resulting in inflammation of the ocular surface and worsening of DED symptoms. [790-792] Furthermore, benzalkonium chloride causes DNA damage, contributing to cytotoxicity and reduced cell viability. It also affects gene expression in trabecular and ocular surface cells, impacting genes related to apoptosis and inflammation, such as Fas and caspase-3. [795,796]

Corneal neurotoxicity is another harmful effect of benzalkonium chloride, which decreases corneal nerve fiber density and reduces tear production while also causing nerve inflammation and degeneration. [790,797] These combined effects result in significant damage to ocular surface cells, exacerbating conditions such as DED. [798] Recent research suggests that benzalkonium chloride may affect both the mucin and lipid layers of the tear film, as damage to goblet cells and Meibomian glands has already been documented, leading to tear film instability. [326,786,799]

Increased tear osmolarity has also been observed in patients using preserved eye drops compared to preservative-free topical medications. [262] Once the tear film loses its protective properties, the compromised tear film not only leads to symptoms of DED and corneal damage but can also spread cytotoxic inflammatory mediators across the ocular

surface. Consequently, increased corneal epithelial permeability has been observed in DED with more significant deterioration when using artificial tears containing benzalkonium chloride compared to preservative-free drops. [800] Changes in the tear film may, therefore, initiate a series of biological alterations on the ocular surface, leading to subsequent neurogenic inflammation and further tear film impairment, creating a vicious cycle. New preservatives, such as Polyquad, Purite, and SofZia, have significantly reduced cytotoxic effects compared to benzalkonium chloride. However, their impact on DED patients needs to be further investigated. [262,684]

Preservative-free eye drops are associated with better ocular surface health and higher tolerability. [800] A systematic review of randomized clinical trials comparing beta-blockers with and without preservatives in patients with glaucoma or ocular hypertension found that, although the difference in intraocular pressure reduction was not clinically relevant, preservative-free eye drops performed better in tear film breakup time and Schirmer test. [800] Additionally, in patients with DED, preservative-free eye drops demonstrated a significant reduction in ocular inflammation symptoms and increased antioxidants in the tear film compared to preserved eye drops. [801] Another observational study in patients with DED who switched from preserved drops to preservative-free drops containing hyaluronate showed significant improvement in the OSDI and reduced frequency of superficial punctate keratitis. [802]

The use of preservative-free prostaglandins has also shown significant benefits for ocular surface health compared to preservative-containing versions. The PRAMOS study demonstrated that patients using preservative-free prostaglandin analogs had a lower prevalence of conjunctival hyperemia and corneal staining compared to those using preserved eye drops. [803] A prospective study showed that switching from preservative-containing prostaglandin-timolol fixed combinations to a preservative-free formulation resulted in a significant improvement in ocular surface disease symptoms, and a reduction in conjunctival hyperemia. [804] Another randomized controlled trial revealed that transitioning from preservative-containing glaucoma therapies to preservative-free formulations improved both ocular surface health and intraocular pressure (IOP) control. [589] These findings are supported by studies indicating that the absence of preservatives, such as benzalkonium chloride, prevents chronic ocular surface toxicity, leading to improved treatment adherence and long-term efficacy. [805,806]

10.2.5 Recommendations for management

The first step is to investigate which medication is causing DED and try to stop its use. This subtraction can be challenging when discontinuing the treatment, which presents a risk to the eye's health. Sometimes, multiple drugs and components are involved, or adverse effects appear long after treatment initiation, making identification of which is

causing DED even more difficult. [778] In some cases, eye drops may be necessary to treat dry eye symptoms, but adding preserved drops to eyes already suffering from dryness caused by other drops may be ineffective and worsen the condition. Once the responsible drug is identified, efforts should be made to discontinue using the preservative or medication. Considering that toxicity is dose-dependent, reducing the number of preserved eye drops can minimize adverse. [778] In glaucoma patients, more invasive and definitive options such as laser trabeculoplasty or surgery may replace or diminish the use of topical medication when the ocular surface and QoL are very compromised. [778]

Conceivably the adverse effects of toxic preservatives should reduce with time. There is a significant shift in first-line primary open angle glaucoma therapy in most countries, from topical therapies towards laser therapies. [807] Preservative-free therapies are increasingly available and there is good evidence for their ocular surface benefits. A significant issue related to preservative-free eye drops is their cost, as they are generally more expensive than preserved drops. This is primarily due to higher production costs and patent fees associated with preservative-free drops. Allocation of healthcare and out-of-pocket costs for patients vary with region and insurers or health system payers, and preservative-free alternatives may not be available in certain jurisdictions.

10.3 Systemic drug-induced DED

Systemic medications can contribute to DED through different mechanisms, including reduction of tear production, disruption of nerve input and reflex secretion, inducing inflammatory responses in secretory glands, or directly irritating the ocular surface through their presence in tears.[262,808] Some systemic medications can exacerbate immune responses, causing further harm to the ocular surface and its innervation, decreasing sensitivity or increasing pain, which worsens DED symptoms.[262,808] This section reviews how some of the most common systemic medications can negatively impact the ocular surface.

10.3.1 Prevalence

At least 1/5 of the best-selling systemic drugs in the US have been associated with the development of DED. In the elderly population, 62% of DED cases are related to systemic medications, including nonsteroidal anti- inflammatory drugs (NSAIDs), diuretics, vasodilators, analgesics/antipyretics, antiulcer agents, sulfonylureas, cardiac glycosides, anxiolytics/benzodiazepines, anti-infectives, antidepressants/antipsychotics, hypotensive agents, and antihistamines. [809]

In a secondary analysis of the Dry Eye Assessment and Management Study to evaluate whether systemic medication use is associated with DED severity, the authors found that 160 (30%) of the 535 participants used medications for hypertension, 129 (24%) used statins, 118 (22%) used antidepressant medications, 117 (22%) used antihistamines, and 16 (3%) used systemic corticosteroids. [810] A multivariable analysis demonstrated that antihistamines and corticosteroids were associated with the highest OSDI score among these systemic medications. Users of seizure medications had a higher composite signs severity score compared to non-users. Compared to non-users, antihistamines, aspirin, and vitamin D3 users had significantly worse average TBUT. Unexpectedly, MGD scores were worse in users of vitamin D3, although this could be confounded if those users of vitamin D3 had vitamin D deficiency. Users of diuretics had significantly better scores for MGD compared to non-users. [810]

In a Military Health System (MHS) beneficiaries database, newly diagnosed DED (≅ 40 000 beneficiaries) and prevalent DED (> 285 000 beneficiaries) patients were compared with matched non-DED patients. [811] In both the newly diagnosed and prevalent DED patients, comorbidities were significantly higher in the DED vs non-DED groups. Systemic medication use was also significantly higher in the DED than in the non-DED groups. The most prescribed medications were narcotic analgesics/strong pain killers, decongestants/vasoconstrictors, b-blockers, antidepressants, diuretics, and anxiolytics. [811] Risk factors are described in the Epidemiology Section 9 above.

10.3.2 Medications and mechanisms

10.3.2.1 Tamsulosin

Tamsulosin, a medication that blocks alpha-1 receptors, is widely prescribed for treating benign prostatic hyperplasia, but is also utilized in managing ureteral stones, prostatitis, and female voiding dysfunction. Epidemiological research suggests a link between DED and benign prostatic hyperplasia itself, as well as the medications used to treat it, including tamsulosin (See Section 5.5.1).

10.3.2.2 Antihistamines/anticholinergic drugs

A large class of systemic medications that lead to the signs and symptoms of DED are those with anticholinergic activity. The anticholinergic class covers a wide range of therapeutic drug categories, including antidepressants, antipsychotics or neuroleptics, antiparkinsonians, H1 antihistamines, decongestants, and antispasmodics.[812]

Anticholinergic medications, such as those used for overactive bladder, have been linked to a reduction in tear film breakup time and an increase in DED symptoms, including burning and foreign body sensation. Studies have shown that medications like

solifenacin can significantly worsen dry eye symptoms and signs compared to placebo.[813,814]

Oral antihistamines and anticholinergic drugs are commonly used to treat allergies and rhinitis. These medications, especially in combinations like pseudoephedrine and cetirizine, can increase the sensation of dryness in the eyes and mouth.[815]

Results of the comparison of DED symptoms, measured by the OSDI, between users and non-users of medications showed that patients who used antihistamines had worse TBUT and OSDI scores compared to non-users. Anticholinergic drugs, such as antihistamines, contribute to DED by acting on the G-protein coupled muscarinic receptors of the lacrimal gland acini and mucus-producing conjunctival goblet cells, affecting the production of the aqueous and mucous components of the tear film. Furthermore, functional cholinergic receptors have been identified in the epithelial cells of the meibomian glands, leading to decreased tear film stability. Thus, despite being commonly used to treat allergy-related eye symptoms, antihistamines confer a high anticholinergic burden, contributing to the development and worsening of DED signs and symptoms.[268,810,816]

10.3.2.3 Isotretinoin

13-cis-Retinoic acid or isotretinoin, is a vitamin A derivative widely used in treating moderate-severe acne due to its atrophic effects on the sebaceous glands. This medication acts as a pro-drug that, upon conversion, induces apoptosis in various cell types, including those of the sebaceous and meibomian glands. Toxicity may manifest as blepharoconjunctivitis, presenting symptoms such as crusting at the eyelid margins and conjunctival redness, and is often associated with dry eyes, light sensitivity, and contact lens intolerance.[261,271,817]

Recent studies utilizing standardized tools, such as the OSDI, indicate that DED symptoms are more commonly found in patients treated with Isotretinoin. The drug can lead to glandular atrophy, ductal keratinization, and small fiber neuropathy, which may reduce corneal sensitivity.[818,819] Other studies have reported severe DED symptoms and decreased tear film stability among patients who start using Isotretinoin, associated with reduced TBUT, lower Schirmer test value, and decreased central corneal thickness.[275,820] Additionally, a few studies show that even after discontinuation of the medication, these signs and symptoms can persist for months post-treatment.[275,820] The adverse effect of Istotretinoin on meibomian gland lipid production is likely to be through suppression of the peroxisome proliferator-activated receptor γ pathway which inhibits meibocyte differentiation and meibum characteristics. [821]Topical isotretinoin is also recognised as a risk factor for iatrogenic DED (see Section 6.6.4). [822]

10.3.2.4 Chloroquine / Hydroxychloroquine

Hydroxychloroquine is a medication commonly used to treat Sjögren disease and other rheumatologic diseases, primarily aimed at relieving symptoms related to joint pain and fatigue, with a well-recognized anti-inflammatory effect. Research on the ophthalmological effects of hydroxychloroquine is highly controversial, as studies from the early 21st century have shown that the substance can be secreted into the tear film, potentially worsening DED symptoms. [823] On the other hand, other studies indicate that patients with Sjögren disease did not experience significant clinical improvement in xerophthalmia. [824,825] Likewise, other studies have shown no improvement in symptoms, tear production, corneal staining, or inflammatory markers after up to 24 weeks of treatment, with patients not experiencing any symptom relief. [824,826] The divergent results regarding its efficacy may stem from the common use of hydroxychloroquine in the treatment of autoimmune disorders, which are often associated with DED, highlighting the need for further studies on this medication.

10.3.2.5 Corticosteroids and non-steroidal anti-inflammatories

Research on the impact of non-steroidal anti-inflammatory drugs on the ocular surface is less extensive, and more recent studies should be conducted. However, selective cyclo-oxygenase-2 inhibitors, such as celecoxib and rofecoxib, have been associated with ocular side effects, primarily conjunctivitis and blurred vision. These effects disappear within 72 hours of discontinuation. Although other ocular side effects are less common and lack clear evidence of a direct link, the consistent findings of blurred vision and conjunctivitis suggest a potential connection. The occurrence of these ocular symptoms should lead physicians to consider discontinuing the medication, as this action generally resolves the symptoms without long-term effects. [827]

Aspirin, a non-selective and irreversible inhibitor of cyclo-oxygenase-1 and -2, is expected to reduce the inflammatory component of DED, leading to an improvement in signs and symptoms. [828] However, the findings of previous studies on the effects of aspirin on DED are contradictory. For instance, a 2018 prospective cross-sectional study involving 106 individuals without a prior diagnosis of DED showed that aspirin users had lower tear osmolarity, increased TBUT, and lower OSDI scores. [828] However, no significant difference in corneal staining was found despite greater symptomatic relief. In line with these results, other older studies were unable to find an association between aspirin and a higher prevalence of DED. [829] This underscores the need for future studies to provide a clearer understanding of aspirin's effects on DED.

The use of corticosteroids is a common practice in medicine for treating inflammatory disorders and has been associated with worsening DED severity, affecting both signs and symptoms.[810] However, it is important to consider the route of administration, as it

can significantly impact the efficacy and potential side effects of these drugs. For instance, when administered topically, corticosteroids are effective in treating DED. [830] Additionally, many users of systemic corticosteroids have underlying autoimmune conditions that are independently associated with DED, making it challenging to isolate whether the worsening of DED signs and symptoms is due to medication use or the progression of the underlying disease. [96] More prospective studies are needed to determine the effects of systemic corticosteroid use on DED. [831]

10.3.2.6 Antibiotics

Antibiotics are commonly used to treat various infections, but few studies have been conducted regarding their impact on the ocular surface. Antimicrobials, particularly tetracyclines and macrolides, are generally used in the treatment of MGD and have shown promising results. [832,833]

The correlation between DED and antimicrobial use is sometimes indirect, as antibiotics can cause Stevens-Johnson syndrome. Some studies have shown that antibiotics are notable risk factors for this condition, potentially leading to severe damage to the ocular surface. [834,835]

10.3.2.7 Antidepressants / anxiolytics / mood stabilizers

The use of antidepressants and antipsychotics is strongly associated with an increased risk of DED. A systematic literature review suggests that both depression and antidepressant use independently contribute to the development of DED. Various clinical and population-based studies have linked the diagnosis and severity of depression to DED and its symptoms. Additionally, other extensive population studies have also found an association between antidepressant use and DED. [836]

Possible mechanisms for the effects of antidepressants on DED might be related to their anticholinergic properties. Another plausible theory is that antidepressants disrupt the corneal epithelial barrier and promote an inflammatory response on the ocular surface by increasing serotonin levels in the tears, which in turn induces an inflammatory response and apoptosis of corneal epithelial cells by activating NF-κB signalling. [837,838]

The prevalence of DED in patients using antipsychotics compared to the general population, with higher prevalence also noted in those on multiple medications compared to those on a single medication. Patients treated with clozapine (monotherapy) and those on a combination of clozapine and quetiapine (polytherapy) showed the highest prevalence of DED. [839]

Selective serotonin reuptake inhibitors can impact the ocular surface by impairing tear film stability. [840] Conversely, serotonin-norepinephrine reuptake inhibitors, effective in treating chronic pain syndromes, may help alleviate DED symptoms. Patients using these inhibitors often have lower OSDI scores, suggesting they may be safer for the ocular surface compared to other antidepressants. Thus, while DED is common among antidepressant users, it may be better managed with serotonin-norepinephrine reuptake inhibitors. [841]

Lithium, widely used for the maintenance and acute treatment of bipolar disorders, has a narrow therapeutic index and various side effects, making its management challenging. Although the adverse effects of lithium are well-documented, its ocular impacts are less understood, which may affect patient compliance. Ocular side effects associated with lithium use include exophthalmos, abnormal eye movements, ocular myasthenia gravis, papilledema, photophobia, and alterations in the tear film, which can lead to DED. [842]

Given the complexity of these interactions, co-management of individuals with DED and medicated for mental health conditions is advisable.

10.3.2.8 Hormone replacement therapy

DED is common among postmenopausal women, with estrogen hormone imbalance being identified as a potential causative factor (see Section 6.6). [310,843] Hormone replacement therapy may be used to relieve menopausal symptoms, using either estrogen alone or in combination with progesterone or progestin. However, estrogen replacement therapy after menopause has been associated with an increased prevalence of DED (See Section 6.6). [129] Each additional three years of hormone replacement therapy resulted in a significant 15% increase in the occurrence of dry eye.[843] The same study showed that the onset of DED was linked to initiating estrogen therapy.[843] Conversely, the use of androgens, both topically and systemically, have been reported to improve the signs and symptoms of MGD and DED. [3,48,844,845]

10.4 Contact lenses and DED

The relationship between contact lens wear and DED is complex. Contact lens wear is recognised as a consistent risk factor for DED (Section 6.6). Ocular symptoms occur more commonly in contact lens wearers than non-wearers and many large epidemiological studies have used ocular symptom reporting as a surrogate for DED. [149] Signs such as corneal staining and tear film instability, due to the partitioning of the tear film during wear occur. [846] Signs of meibomian gland alterations are also more common in contact lens wearers than in age-matched non-wearers. [847] Contact lens discomfort is characterized by episodic ocular symptoms of discomfort and dryness which resolve

when the contact lens is removed, as distinct from those in DED, [284] however this distinction is not consistently considered when reporting symptoms in contact lens wearers. A further complexity is that other complications of contact lens wear may also lead to very similar ocular symptom-reporting. [847] Contact lens wearers may also have existing or subsequently develop, DED. This has led to the use of series of descriptors which are often used interchangeably, including contact lens discomfort, contact lens-induced dry eye and contact lens-associated dry eye. With these caveats, this section will endeavour to summarise the evidence for prevalence of, mechanisms underlying and remedial strategies for DED associated with contact lens wear.

10.4.1 Prevalence of DED in contact lens wear

The prevalence of symptoms of DED in soft contact lens wearers in a series of recent prospective and retrospective studies in select population groups ranged between 31% to 77%. [848-850] A clinical diagnosis of DED by an eyecare professional was made in 24% of office workers in a Chinese sample [850] and in 14% of University students wearing contact lenses in Thailand. [851]. The frequency of DED in scleral lens wearers has been more difficult to determine. These are mostly used for medical and therapeutic indications, and an estimate of 56% of scleral lens wearers had severe DED symptoms based on OSDI. [852,853] There is a lack of large-scale epidemiological studies on the prevalence of DED associated with orthokeratology lenses. In a small retrospective study of contact lens complications conducted at a tertiary hospital in China, 24% of orthokeratology lens wearers presented with DED. [854] Understanding the pathogenesis, and provision of mechanism-oriented treatment of DED in contact lens wearers is important.

10.4.2 Mechanism

Contact lens use causes a variety of biophysical and biochemical changes in the tear film, as well as alterations in the ocular surface structure and function which are summarized in Table 12. These mechanisms collectively lead to reduced tear film stability, increased tear evaporation, and ocular surface damage, [846,847] and likely contribute to DED in contact lens wearers.

Rigid and soft contact lens wear increase tear film evaporation,[855] although they interact differently with the ocular surface. Mechanical interactions are different between rigid and soft contact lenses due to material and fitting characteristics. [847,856,857] Upper lid wiper epitheliopathy is more common in rigid than soft lens wear. [858] There is a higher frequency of incomplete blinking, [859] leading to uneven tear distribution and exacerbation of dry eye symptoms. Corneal staining in the 3 to 9 o'clock positions is common among rigid lens wearers,[860,861] attributed to thinning of the tear film [862]

adjacent to the lens edge, whereas soft contact lens wearers more often exhibit inferior arcuate cornea staining.[861]

It is conceivable that mechanisms in contemporary scleral lens use might resemble those observed with rigid corneal contact lenses, including friction between scleral lenses and ocular tissues, [856] increased mechanical stimulation of the ocular surface, [855] and tear film instability.[863]

Orthokeratology lenses share some mechanisms for DED with soft and rigid contact lenses, but the unique mechanism of these lenses is related to their reverse geometry design, [854] which not only alters corneal shape but also significantly affects the density and distribution of corneal nerve fibers, [864,865] which may not fully recover shortly after cessation of wear. [866,867]

The evidence for changes in corneal sensitivity underpinning sensation report in contact lens wear is limited. Contemporary rigid corneal and soft lenses do not appear to alter corneal sensitivity, although a reduction in central corneal mechanical sensitivity is observed in orthokeratology lens use [866,868] Reduced corneal sensitivity may reduce tear secretion and blink frequency, thereby exacerbating symptoms of DED. [867] There is a substantial augmentation in tear inflammatory mediators in orthokeratology lens wear, such as IL-17A, IL-6, and prostaglandin E2. [864] The TFOS Contact Lens Discomfort neurobiology report described the multifactorial effects of contact lens wear, include mechanical, cooling, drying, change in osmolarity and chemical impacts on ocular surface neurosensory processes. [869] Altered sensory processing of cooling stimuli delivered to the cornea has recently been reported in symptomatic soft contact lens wearers suggesting nerve sensitisation or maladaptation in the absence of sensitivity changes, [870] which is intriguing but causality for DED has not been demonstrated.

Subclinical inflammation at the ocular surface is reported in several modalities of contact lens wear, evidenced by recruitment and activation of inflammatory cells to the cornea, conjunctiva and lid margin and the presence of mediators in tears [846] for review). In naïve wearers, epithelial immune cells are recruited to the central and peripheral cornea within 2 hours of soft contact lens wear and morphological changes suggest enhanced antigen capture capacity. [871] Increased density of inflammatory cells is seen at the lid margin in reusable hydrogel and silicone hydrogel wearers but not in daily disposable wear. [872] The link between subclinical inflammation and DED in contact lens wearers, however, has not been confirmed.

Table 12. Effects of contact lens wear on the tear film and ocular surface

Biophysical tear film alterations	Effect
Division of the tear film into pre- and post-lens layers	

Tear meniscus volume	Reduced
Lipid layer spreading	Reduced
Tear stability	Reduced
Structural ocular surface alterations	
Conjunctival goblet cell density↓	Reduced
MG expressibility ↓,	Reduced
MG obstruction and atrophy	Increased
Basal corneal nerve density↓, nerve tortuosity↑:	Reduced
Ocular surface sensitivity	Corneal sensitivity unchanged in SCL use; altered sensory processing; conjunctival sensitivity increased Corneal sensitivity reduced in OK use
Ocular surface friction and lid wiper epitheliopathy	Increased
Biochemical tear film alterations	
Tear cholesterol	Increased
Malondialdehyde & 4-hydroxy 2-nonenal	Increased
Beta-2 microglobulin	Reduced
Proline rich protein 4	Reduced
Lacritin	Reduced
Lipocalin 1D1	Reduced
Secretory IgA	Effects equivocal, may depend on duration of wear and wear modality
Albumin	Increased
Deleted in Malignant Brain Tumors-1	Increased
Prolactin inducible protein	Increased
MUC5AC	Reduced
Inflammatory tear film/ocular surface changes	
Epithelial immune cells	Increased at ocular surface in SCL wear, increased antigen capture capacity. Density reduced at central cornea in OK
Tear cytokines IL-7,8,13,15	Increased

SCL = soft contact lens; OK = orthokeratology

10.4.3 Recommendations for management of DED in contact lens wearers

Therapeutic and device-related management of DED in contact lens wearers broadly follows that described in the TFOS DEWS III Management and Therapy report, (Jones et al., 2025) with caveats about the use of preserved medications during contact lens wear and noting that many products are not specifically registered for use in contact lens wearers. Given the potential interdependencies between DED, MGD and contact lens discomfort, non-specific strategies might include switching to daily disposable contact lenses, attention to the lens fitting relationship, avoiding preserved care systems with reusable contact lenses, improving blink completeness, attention to environmental triggers, ruling out ocular or systemic co-morbidities and recommending the use of

unpreserved lubricants, with or without lipid additives. [847 873 874] Discontinuing from lens wear or reducing wear time may be successful [855] unless there is underlying DED.

In contact lens wearers with MGD and consequential evaporative DED, there is evidence for the benefits of treatments including IPL [875], microblepharon exfoliation [876] or thermal pulsation, [877] including improved TBUT, and symptoms of DED. Low level light therapy or photobiomodulation therapy has been used in the treatment of MGD and evaporative DED [2], but there is no high-level evidence for benefit in contact lens wearers, although an observational study without a control group exists in wearers with discomfort. [878] While there are no pharmacological treatments approved for MGD in contact lens wearers, a phase 1 [879] and phase 2 [880] clinical trial both demonstrated benefit in improving comfortable wear time and meibomian gland signs in a population of symptomatic wearers with MGD, using a topical selenium sulphide treatment.

10.5 Procedures

10.5.1 Botulinum toxin

Paradoxically, botulinum toxin A has been used in treating DED. [881] In the literature, BTX-A was injected into the periorbital area including the medial portion of the orbicularis muscle and not solely into the lateral canthal region, for the treatment of existing DED. The mechanism of action of botulinum toxin on lacrimal drainage in such instances has been suggested to be due to a paralysis of the orbicularis oculi muscle around the canaliculi with a decreased compression as well as weakness of apposition of the puncta during blinking. [882,883]

10.5.1.1 Mechanism

Dry eye due to botulinum toxin type-A injection for treatment of blepharospasm or after blepharoplasty and peri-orbital surgery has been previously reported in literature. The mechanisms by which DE is induced are multifactorial:

10.5.1.2 Direct inhibition of tear secretion:

Botulinum toxin blocks acetylcholine release within the lacrimal gland, disrupts neuromuscular junctions, and inhibits parasympathetic nerves, thereby suppressing both basal and reflex tear secretion. [884] This leads to a significant decrease in TBUT and Schirmer test results. [882,883,885]

10.5.1.3 Impact on meibomian gland function:

Botulinum toxin can also diminish lipid production by affecting neuro-muscular transmission around the muscles of the meibomian glands. [886] Additionally, botulinum

toxin-induced paralysis of the preseptal orbicularis oculi and Riolan's muscle weakens eyelid closure and reduce meibum secretion, thereby decreasing lipid layer thickness and tear film stability. [887] However, further studies are needed to examine the long-term effects on the meibomian glands.

10.5.1.4 Regulation of inflammatory responses:

Botulinum toxin injections may trigger inflammatory responses at the ocular surface and in the lacrimal glands, such as increased release of cytokines (e.g., TNF-α and IL-1β), [888] which may further impact tear stability and tear secretion.

10.5.1.5 Chemodenervation of orbicularis oculi:

Botulinum neurotoxin induces chemodenervation of the orbicularis oculi muscle which can lead to poor blinking, lagophthalmos, and ectropion that may result in corneal dryness with symptoms of irritation, foreign body sensation, and epiphora. Epiphora is likely due to a combination of factors including DED causing reflex tearing, hypotonicity of the medial pretarsal fibers causing decreased outflow of tears, and malposition of the eyelids, causing impaired retention of tears.[884]

10.5.1.6 Recommendations for management

1. For patients experiencing DED due to botulinum toxin treatment for other conditions, individualized management plans should be developed based on specific symptoms and tear volume.[884]
2. Patients undergoing botulinum toxin treatment should be regularly monitored for tear secretion levels, ocular surface condition, inflammatory responses, and changes in orbicularis muscle function. Patients should be educated about symptoms of DED and management, including maintaining good eye hygiene practices and avoiding excessive use of electronic screens.
3. Grading eye lid muscle tone and eye lid laxity and documenting any eye symptoms or findings prior to injection is of importance. It is recommended that patients who are to undergo botulinum toxin injections should undergo the snap-back test which measures muscle tone and the lower lid distraction test that measures lid laxity resulting primarily from the stretching of canthal ligaments. Patients with severe test results are more prone to dry eye development after injection of the toxin into the lateral canthal region. Patient should be asked about any early symptoms including eye irritation, foreign body sensation, or tearing that were not present previously and/or by noticing any change in snapback and distraction tests during the treatment course. In presence of positive test findings and dry eye symptomatology, one should temporarily discontinue repeated BTX-A injections into the lateral canthal rhytids. [889]

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10.5.2 Corneal collagen crosslinking

10.5.2.1 Mechanism

Corneal collagen cross-linking (CXL) is a treatment that uses ultraviolet A light and riboflavin (vitamin B2) as a photosensitizer to strengthen the cornea. This technique reinforces the chemical bonds within the cornea, thereby halting or delaying the progression of corneal ectatic diseases such as keratoconus.[890,891] Both epithelium-on and epithelium-off CXL have been reported to have a positive effect on DED, and may improve tear film homeostasis and reduce DED symptoms in patients with keratoconus.[892]

Paradoxically, CXL can also induce DED symptoms through multiple mechanisms. Firstly, epithelial removal and delayed healing may induce DED.[893-895] This delay can affect the uniform distribution of tears and the corneal defences,[893,896,897] by compromising tear film stability[708,898] and making the ocular surface more prone to DED symptoms.

CXL may also reduce corneal nerve density, [899] impairing corneal sensory nerve function and decreasing corneal sensitivity, which in turn affects the tear secretion. [899] However, studies indicate that in most patients subbasal nerve regeneration occurs within 2-3 months after CXL,[899] reaching preoperative levels in 6-12 months. [900,901] Additionally, although CXL does not directly alter meibomian gland morphology, [902] the effects on corneal sensory nerves [902], the use of eyelid speculums during surgery, post-CXL inflammatory responses with elevation of IL-6, and medications used post-surgery [902] can indirectly affect gland function and secretion quality.

10.5.2.2 Recommendations for management

To effectively manage DED that may arise after CXL, a comprehensive approach is essential. Firstly, bandage contact lenses, due to their material and characteristics, help protect exposed nerve endings and the corneal epithelium, reducing frictional damage from blinking and thereby alleviating dry eye symptoms. [903]

Secondly, during the postoperative phase, the use of corneal protectants, tear substitutes, and tear secretagogues is important. These medications should be selected based on their ability to replace the mucin glycocalyx lost from the corneal epithelial cells, promote the restoration of the mucin gel on the epithelial cell surface, enhance the adhesion of the tear film to the corneal epithelium, and aid in the recovery of the lipid layer of the precorneal tear film. [903,904]

Lastly, advancements in modern CXL technology also help reduce the risk of postoperative complications. These improvements include transepithelial cross-linking, localized and personalized techniques, and accelerated procedures, which aim to enhance surgical outcomes and minimize corneal manipulation time. [898,905-907]

10.5.3 Other procedures

10.5.3.1 Eye cosmetics and beauty treatments

The use of eye cosmetic products and procedures represent a lifestyle challenge that may exacerbate or promote the development of ocular surface and adnexal disease. This topic has been recently explored in detail. [263]

10.6 Non-ophthalmic conditions

10.6.1 Radiotherapy

10.6.1.1 Mechanism

DED is associated with radiation therapy for head and neck cancer, [908,909] temporal tumors, breast cancer, and Graves' eye disease. This condition often results from direct damage to tissues around the eye.[909-911] The incidence of radiation-related dry eye depends on the type of radiation therapy, tumor location [912,913] and radiation dose.[909,914,915]

Mechanisms of Radiation-Induced Dry Eye are multifactoral:
(1) Damage to the Lacrimal Glands and Tear Drainage System: Radiation therapy can cause inflammation of the tear ducts or obstruction of tear duct openings,[916,917] significantly reducing lacrimal gland function and leading to DED symptoms. Ultrastructural analysis of post-radiation lacrimal glands reveals extensive tissue damage, including membrane rupture, loose cell junctions, and nuclear fragmentation.[918] Additionally, the expression of markers such as lysozyme, S-100, and CD117 is reduced in the lacrimal tissues.[918]
(2) Corneal and Conjunctival Damage: Radiation therapy can cause direct damage to corneal epithelial cells and loss of conjunctival goblet cells,[914] which can destabilize the tear film. Additionally, inadequate conjunctival closure during brachytherapy can lead to conjunctival and scleral necrosis, further affecting ocular surface health and tear film stability.[919]
(3) Eyelid Damage: The meibomian glands exhibit acute inflammatory responses to radiation therapy characterized by cystic dilatation of ducts containing keratin, and atrophic decrease of the glands and ducts or their complete loss.[920]

10.6.1.2 Recommendations for management

The management of radiation-related DED can be divided into three key approaches:

1. Preventive measures to minimize the risk such as limiting the radiation dose to the lacrimal glands, ideally keeping the dose below 30 Gy.[914,915,921,922]
2. Proper patient positioning, eye shielding, and adequate patient immobilization during radiotherapy sessions to reduce radiation- related toxicity.[912]
3. Therapeutic interventions for DED. [2]

Long-term management is essential for maintaining good ocular health and QoL. Regular ophthalmologic examinations and assessments to monitor changes in DED allow for timely adjustments in treatment plans and interventions. A multidisciplinary team, including ophthalmologists and oncologists, can provide comprehensive care and personalized treatment plans, thereby maximizing QoL.

10.6.2 Bariatric surgery

This topic has been reviewed in detail in two recent reports. [292,684]

10.6.3 Stem cell or bone marrow transplant

10.6.3.1 Mechanism

GVHD is a common complication following human leukocyte antigen (HLA)-matched allogeneic hematopoietic stem cell transplantation. GVHD can affect various organs, including the skin, gastrointestinal tract, liver, lungs, and eyes, with dry eye being the most common manifestation of ocular GVHD. [923,924] Ocular GVHD is seen in 60-90% of transplant recipients, primarily manifesting as secondary inflammation and fibrosis of the lacrimal and meibomian glands. [925,926] The mechanisms of DED in GVHD include the following aspects:

1. Lacrimal Gland Fibrosis: Fibrosis of the lacrimal gland is a major cause of GVHD-related dry eye. Studies have shown the presence of activated CD34+ fibroblasts in the lacrimal glands of GVHD patients, with T-cell infiltration around the ducts, thickening of blood vessels and duct basement membranes correlating with the severity of fibrosis. [927,928]
2. Renin-Angiotensin-Aldosterone: Angiotensin 1 type 1 receptor antagonists, such as valsartan, can reduce fibrosis, decrease inflammatory cell density, and increase tear secretion. [929] Angiotensin-mediated fibroblast activation and upregulation of TGF-β expression via the angiotensin 1 type 1 receptor are the primary mechanisms of fibrosis. [930]

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3. Immune-Mediated Inflammation: In GVHD patients, increased density of T cells and dendritic cells [931] in the lacrimal glands, conjunctiva, and cornea contribute to inflammation and fibrotic responses. [932]
4. Cellular Senescence: Chemotherapy and radiotherapy-induced DNA damage and the inflammatory environment activate senescence-related molecular pathways. [933] Stress-induced cellular senescence promotes the secretion of various cytokines (e.g., IL-1β, IL-6, IL-8), forming the "senescence-associated secretory phenotype", which exacerbates inflammation. [934]

GVHD patients often exhibit significant abnormalities in the meibomian glands, [935] leading to alterations in the tear film lipid layer and increased tear evaporation. [936] Corneal and conjunctival changes commonly include punctate keratopathy, filamentary keratitis, and epithelial defects, [937] accompanied by decreased corneal nerve fiber density, increased dendritic cell density, conjunctival epithelial squamous metaplasia, and reduced goblet cell density. These changes collectively contribute to the onset and progression of DED. [931,936]

10.6.3.2 Recommendations for management

GVHD-related DED can be potentially managed through a combination of preventive and treatment strategies. Preventive measures primarily include ensuring high-quality HLA matching between the donor and recipient, and the use of immunosuppressive agents such as cyclosporine, tacrolimus, and methotrexate to inhibit donor T cells.[938,939] Preservative-free and phosphate-free eye drops are recommended for tear retention and lubrication. [940]
Topical steroids, [941,942] and systemic oral immunosuppressants such as cyclosporine [943] are recommended to control inflammation. Biological tear substitutes like autologous serum eye drops provide significant anti-inflammatory and nutritional benefits. [944-946] In severe cases of DED, temporary tarsorrhaphy may be necessary to limit ocular surface exposure, and surgical interventions may be required to treat cicatricial eyelid diseases. [947] These comprehensive measures aim to maximize symptom relief and enhance the QoL for patients suffering from GVHD-associated DED. (Cross reference with management report

10.7 Future directions and conclusions

Iatrogenic causes underpin a significant proportion DED and may arise from topical and systemic medications and a range of ophthalmic and non-ophthalmic surgeries. Lifestyle choices such as contact lens wear, cosmetics and cosmetic procedures also contribute to iatrogenic DED. Improved understanding of the range of causes and how these contribute to different subtypes of DED and risk factors for more severe disease is

likely to help reduce their impact. Evaluating and managing underlying DED pre- and post-intervention will also better support these patients. A challenge in iatrogenic disease is understanding the impact of an intervention compared with progression of the primary condition. This report has also identified the need to better disaggregate the effects of topical medications alone from their preservatives and other excipients. In DED following use of systemic medications, there is greater certainty in the adverse effects of isotretinoin in DED and the persistence of effect once the medication ceases. While depression and antidepressants broadly are independent risk factors for DED, it is recognised that some classes, such as serotonin-norepinephrine reuptake inhibitors may improve ocular symptoms and there should be further exploration of the impact of systemic corticosteroids on the ocular surface. The natural history of DED associated with lifestyle choices such as contact lens wear and particularly orthokeratology remains unclear.

11. Clinical trials design

11.1 Introduction

The major international markets for medicines are the US, EU and Japan. These three regions have consistent regulatory frameworks for approval of medicines following the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use guidelines. [948] There are some variations in evidentiary standards between these jurisdictions in clinical trials designs and features. As an example, for new DED treatments, the EU requires a six-month time frame for primary efficacy studies, rather than the three-months required by the FDA. In addition, EU typically requires standard of care as one of the trial arms, in addition to vehicle as control, where using artificial tears as standard of care. Artificial tears are considered devices in the EU but not in the US or Japan. As far as possible, this report will provide examples of treatments approved from of these three regions.

The development of novel treatments for DED accelerated in the US and EU following release of both the TFOS DEWS II report in July 2017 [205] and the FDA draft guidance for industry in December 2020. [949] Five new treatments for DED were approved between August 2018 and June of 2024 in the US, one new treatment was approved in the EU and one new treatment was approved in Japan. Eight treatments were approved before July of 2017, two in the US and six in Japan. [950-952] While prior TFOS DEWS reports focused on challenges hampering the design and success of trials in DED, [953] this report will focus on how recommendations from the TFOS DEWS II report translated into innovations in trial design, an independent analysis of trial failures released in 2012, across ten therapeutic areas, [198] and how flexibility in application of the FDA draft guidance have improved success rates for bringing novel treatments to

patients. Innovations in trial design include better matching of treatment mechanism of action to patient sub-populations and more stringent control on concomitant treatments and sources of trial error. [951,952,954-958]

These innovations in trial design require a deeper understanding by both researchers and clinicians of their potential impact to support both continued approvals and to promote the continued introduction of novel treatments for patient sub-populations who may be underserved by currently approved treatments. The second point could impact the clinical utility of new market entrants as the potential consequence of designing approval programs for novel DED treatments that exclude patients with signs of DED that don't align with a drug/device mechanism of action may limit efficacy across the spectrum of patients that present with DED. For example, limiting the inclusion of patients with MGD in drug trials supporting currently approved products, knowing MGD is present in 51.3% to 70.3% of patients with DED, [359,360], may partially explain the apparent dissociation between the regulatory approval success of compounds in enriched populations and the subsequent high dropout/discontinuation rates for treatments in clinical practice. [953,959,960] Peer-reviewed literature reports discontinuation rates up to 40% within the first year of a filled prescription while certain ophthalmological practices can see estimated discontinuation rates over 60% with some anti-inflammatory treatments. [959] Newer treatments and associated clinical trial designs with enriched populations of patients will likely only expand as treatments tailored to newly identified subpopulations (e.g., Ocular Neuropathic Pain) continue to be identified over time. [205,953]

In addition to searching the published, peer-reviewed literature for information on existing methodology, detailed reviews on development programs and outcomes that are publicly available were utilized as a primary source of information for this report. The FDA website (accessdata.fda.gov) and ClinicalTrials.gov which is a publicly available registry of clinical trials maintained by the US National Library of Medicine both list publicly available information, including clinical protocols, statistical reports and outcomes. The European Medicines Agency (EMA) also issues assessment reports on products that were accepted or not approved (withdrawal assessment report) for use. These can be found at https://www.ema.europa.eu/en/documents. The PMDA publishes review reports on approved drugs and other products in English (https://www.pmda.go.jp/english/review-services/reviews/approved-information/0001.html). Beyond the major markets, global regulatory agencies generally publish assessment reports on products submitted for their review which are also readily available online (e.g., the Australian Therapeutic Good Administration (TGA) published report on a lifitegrast ophthalmic preparation: https://www.tga.gov.au/sites/default/files/auspar-lifitegrast-191107.pdf).

11.2 Approvals for DED

The recommendations for trial design from TFOS were first published in the TFOS DEWS report [961] and were followed by the TFOS DEWS II report in 2017. [953] A cyclosporine ophthalmic emulsion 0.05% was approved in 2003 in the US and is indicated for an increase in tear production in patients whose tear production is presumed to be suppressed due to ocular inflammation associated with keratoconjunctivitis sicca. Sodium hyaluronate 0.1% and Mini 0.1% and 0.3% were approved in 1995 in Japan for conjunctival epithelial disorders including sicca syndrome. The TFOS DEWS report described the approval period followed by interest in the development of additional anti-inflammatory and immunomodulatory drugs for the treatment of DED. This time was marked by the failure of numerous follow-on compounds and exploration of how the better control of trial design, endpoints and operational excellence could potentially lead to a greater success rate in DED trials.

Lifitegrast ophthalmic solution (5%) was approved in the US in 2016 and is indicated for treating the signs and symptoms of DED in adults over the age of 17 years. In Japan a sodium hyaluronate ophthalmic solution (0.3%), Diquafosol ophthalmic solution (3%), and Rebamipide ophthalmic suspension (2%) were approved between 2010 and 2012 for conjunctival epithelial disorders including sicca syndrome (dry eye) and subsequently just dry eye. The TFOS DEWS II report described the approval followed by interest in exploration of how matching the mechanism of action of a proposed therapy to a potentially responsive population in which the treatment is likely to demonstrate efficacy; the inclusion of biomarkers and/or surrogate markers of disease; and recognition of strong placebo/vehicle effects that if properly controlled could potentially lead to greater success rates in DED trials. [953] These concepts were captured in the TFOS DEWS II report, trials for novel compounds and the FDA's guidance document on trial design for DED; following which a period of successive approvals for additional DED treatments was observed (Figure 4 and Table 13).

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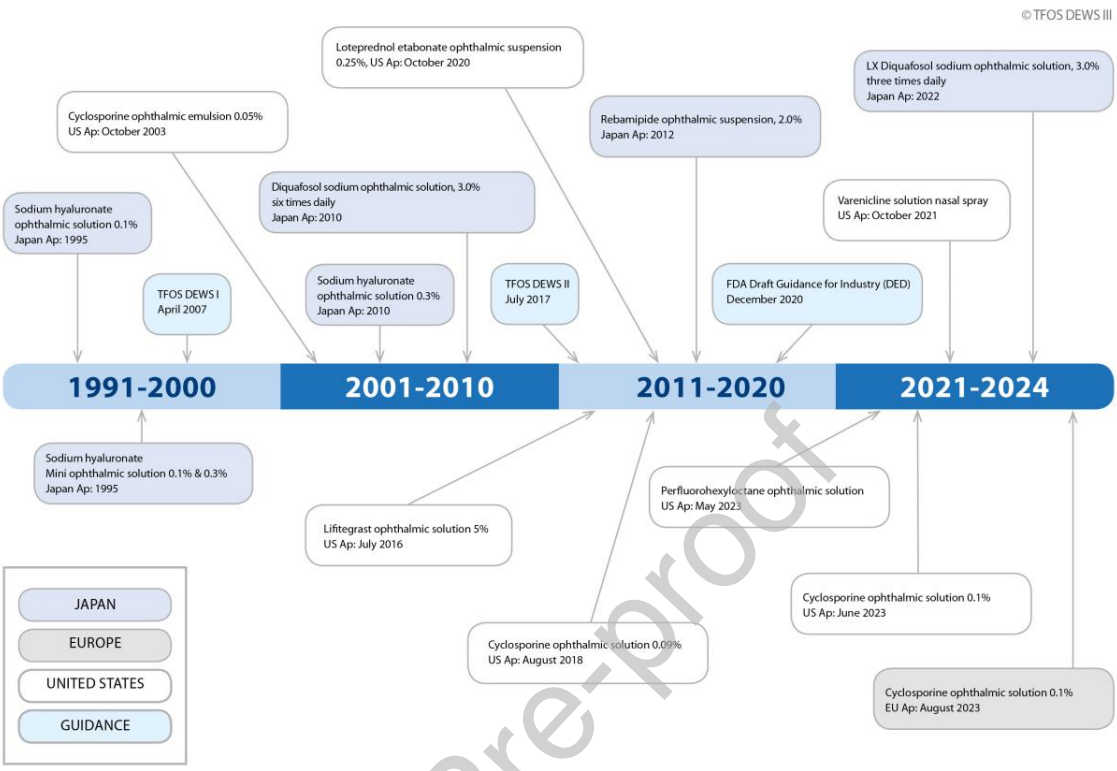


Figure 4: Approved treatments for Dry Eye Disease

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Table 13. Approved products in major markets

Generic Name	Brand Name	Year Approved	Country	MOA	Approved Indication
Cyclosporine ophthalmic emulsion, 0.05%	RESTASIS®	2003	USA	calcineurin inhibitor immunosuppressant	Indicated to increase tear production in patients whose tear production is presumed to be suppressed due to ocular inflammation associated with keratoconjunctivitis sicca.
Lifitegrast ophthalmic solution, 5%	XIIDRA™	2016	USA	lymphocyte function-associated antigen-1 (LFA-1) antagonist	Indicated for the treatment of the signs and symptoms of DED
Cyclosporine ophthalmic solution, 0.09%	CEQUA™	2018	USA	calcineurin inhibitor immunosuppressant	Indicated to increase tear production in patients with keratoconjunctivitis sicca (dry eye)
Loteprednol etabonate ophthalmic suspension, 0.25%	EYSUVIS™	2020	USA	corticosteroid	Indicated for the short-term (up to two weeks) treatment of the signs and symptoms of DED
Varenicline solution, 0.03 mg nasal spray	TYRVAYA™	2021	USA	Nicotinic acetylcholine receptor agonist	Indicated for the treatment of the signs and symptoms of DED
Perfluorohexyloctane ophthalmic solution	MIEBO™	2023	USA	semifluorinated alkane	Indicated for the treatment of the signs and symptoms of DED
Cyclosporine ophthalmic solution, 0.1%	VEVYE	2023	USA	calcineurin inhibitor immunosuppressant	Indicated for the treatment of moderate to severe DED (VEVIZYE in Europe)
Cyclosporine ophthalmic emulsion, 0.1%	IKERVIS	2023	EU	calcineurin inhibitor immunosuppressant	Treatment of severe keratitis in adult patients with DED, which has not improved despite treatment with tear substitutes

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Purified sodium hyaluronate, 0.1%	HYALEIN® ophthalmic solution 0.1%	1995	Japan	increasing tear film stability / corneal healing	Indicated for the treatment of corneal and conjunctival epithelial disorders (Keratoconjunctival epithelial disorder resulting from the following diseases: Intrinsic diseases such as Sjögren's syndrome, Stevens-Johnson syndrome and sicca syndrome (dry eye). Extrinsic diseases caused by surgery, drugs, trauma, contact lens wearing, etc.)
Purified sodium hyaluronate, 0.3%	HYALEIN® ophthalmic solution 0.3%	2010	Japan	increasing tear film stability / corneal healing	As above for 0.1% solution
Purified sodium hyaluronate, 0.1% (preservative free)	HYALEIN® Mini ophthalmic solution 0.1%	1995	Japan	increasing tear film stability / corneal healing	As above for 0.1% solution
Purified sodium hyaluronate, 0.3% (preservative free)	HYALEIN® Mini ophthalmic solution 0.3%	1995	Japan	increasing tear film stability / corneal healing	As above for 0.1% solution
Moistear/Diquafosol sodium ophthalmic solution, 3.0% (6x Daily)	DIQUAS®	2010	Japan	$P2Y_2$ purinergic receptor agonist	Treatment of DED
Rebamipide ophthalmic suspension, 2.0%	MUCOSTA®	2012	Japan	upregulates the gene expression of MUC1, MUC4, and MUC16	Treatment of DED
Diquafosol sodium ophthalmic solution, 3.0% (TID)	DIQUAS® LX	2022	Japan	$P2Y_2$ purinergic receptor agonist	Treatment of DED
Cyclosporine ophthalmic emulsion, 0.05%	RESTASIS®	2010	India	calcineurin inhibitor immunosuppressant	Not known

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Cyclosporine ophthalmic solution, 0.09%	CEQUA®	2021	India	calcineurin inhibitor immunosuppressant	Not known
Lifitegrast ophthalmic solution, 5%	SECA®	2023	India	lymphocyte function associated antigen 1(LFA-1) antagonist	Not known
Diquafosol sodium ophthalmic solution, 3.0% (6x Daily)	DIQUAS®	2017	China	$P2Y_2$ purinergic receptor agonist	Indicated for DED diagnosed with corneal and conjunctival epithelial injury accompanied by tear abnormalities
Cyclosporine ophthalmic nanoemulsion, 0.05%	Zirun Cycloome®	2020	China	calcineurin inhibitor immunosuppressant	Indicated to increase tear production in DED where tear production is presumed to be suppressed due to ocular inflammation associated with keratoconjunctivitis sicca
Diquafosol sodium ophthalmic solution, 3.0% (0.4mL:12mg)	RUNLIMING	2022	China	$P2Y_2$ purinergic receptor agonist	Indicated for DED diagnosed with corneal and conjunctival epithelial injury accompanied by tear abnormalities
Varenicline solution, 0.03 mg nasal spray	TYRVAYA™	2024	China	Nicotinic acetylcholine receptor agonist	Indicated for the treatment of the signs and symptoms of DED
Perfluorohexyloctane	HENGQIN	2024	China	Semifluorinated alkane	Indicated for the treatment of the signs and symptoms of DED

DED-dry eye disease; EU-European Union; lymphocyte LFA-1-function-associated antigen-1; TID-Three Times Daily; USA-United States of America
Disclaimer- This is not necessarily a complete list of approved prescription medications approved in China and India

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Globally, the availability of treatments for DED targeted at a sub-population of patients who present predominately with signs of inflammation or its impact (e.g., hyperemia, corneal staining, and conjunctival staining) and ADDE (e.g., decreased tear production) speaks to a potential pathway for approval of additional products. [962-966]

To date, compounds with immunomodulatory, anti-inflammatory, tear stimulatory and tear conservation mechanisms of action have all targeted DED patients**.** Approvals for these products in the major markets are generally followed by approvals in significant rest of world markets. However, a number of these compounds have failed repeatedly in clinical trials in other countries.

11.3 Defining a pathway toward approval

11.3.1 Solving industry-wide failure rates

Pfizer completed a review of its portfolio and determined that in the 2010's the pharmaceutical industry, in general, faced declining R&D success rates which reached a low in 2016 when only 10% of Phase 2 trials succeeded while Phase 3 success rates were only 30%. [198,967] By 2020, this had improved to a Phase 2 trial success rate of 53% and a Phase 3 trial success rate of 80%. [198]

Improvements in Phase II success rates were enabled by better understanding of disease heterogeneity, which led to better selection of patients for clinical trials and longitudinal sampling of mechanistic biomarkers (e.g., advanced imaging techniques and blood-borne biopsies). [198,967]

Three fundamental elements were identified to significantly improve trial success rates: 1) Exposure at the site of action (Pillar 1); 2) Binding to the pharmacological target (Pillar 2); and 3) Expression of pharmacological activity from the site of action (Pillar 3). Successful trials could achieve all 3 Pillars, achieve Pillars 1 and 3, or achieve Pillars 2 and 3. This was often combined with the inclusion of biomarkers and/or surrogate markers of disease to achieve an early proof of mechanism and/or early signal of efficacy. [198,967]

11.3.2 Overcoming disease heterogeneity with targeted sub-populations

Ocular surface diseases including EDE, [953,961,968] ADDE, [953,961,968] MGD [237,969] and Contact Lens Discomfort (CLD) [284,970] represent ocular surface disease heterogeneity that has been accounted for in more recent DED trial designs by matching the mechanism of action of a proposed therapy to a potentially responsive population in which the treatment is likely to demonstrate efficacy. [82,951,954-956,962,963,965,966,971-979]

Trials targeting patients presenting with ADDE and inflammation have consistently linked decreased tear production with more severe disease that disrupts the ocular

surface, exemplified by ocular surface staining, through the concept of the lacrimal functional unit. [951,954-957,962,963,965,966,971,972,979] The lacrimal functional unit is defined as an integrated system comprising the lacrimal glands, ocular surface (cornea, conjunctiva and meibomian glands), lids, and the sensory and motor nerves that connect them. These trials have inclusion criteria and associated primary endpoints focusing on increased ocular surface staining, decreased tear production, more severe symptoms (e.g, VAS score ≥ 40 at screening and baseline and an OSDI score of ≥23, and inflammation. [951,952,954-958,962,963,965,966,971,972,974,976,977,979-981] As a group, many of these trials also excluded patients with signs of MGD, EDE or CLD. [951,955-957,962-966,971,972]

The products targeting patients presenting with ADDE and inflammation can be broken down into two main categories based upon their mechanism of action. The first group of compounds includes immunomodulatory and anti-inflammatory agents [951,955-957,962-966,971,972] and the second can be broadly categorized as tear stimulatory or conservation agents for the various layers of the tear film (i.e., lipid layer, aqueous layer, and mucin layer) (see 13X). [952,954,958,973,974,976,977,979-981] Studies involving three of these compounds further restricted their inclusion/exclusion criteria and primary endpoints to closely match their products' proposed mechanism of action. [954,955,973,974,976,977,979]

Table 14. Mechanism of action and approval pathway

Approved Drug, Mechanism of Action	Global Regulatory Agencies Approve Drugs for DED Irrespective of Pathophysiological Subtype								
	Signs of Evaporative DED			Signs of Aqueous-Deficient/Inflammatory DED			Symptoms of DED		
	Meibomian Gland Obstruction	Meibum Quality	TBUT	Corneal Staining	Conjunctival Hyperemia	Schirmer Responder	Eye Dryness VAS Scale	Ocular Discomfort VAS Scale	Total OSDI
Meibomian Gland Dysfunction &/or Evaporative DED									
None									
Aqueous-Deficient &/or Inflammatory DED									
Calcineurin inhibitor immunosuppressant				IKERVIS		RESTASIS®, CEQUA™, VEVYE			IKERVIS
lymphocyte function-associated antigen-1 (LFA-1) antagonist				XIIDRA™			XIIDRA™		
corticosteroid					EYSUVIS™			EYSUVIS™	
Tear Replacement/Stabilser (Non-Imunomodulatory)									
Cholinergic agonist						TYRVAYA™			
Semifluorinated alkane				MIEBO™			MIEBO™		
Water retention and binding to fibronection possibly promoting adhesion and elongation of corneal epithelial cells				Sodium Hyaluronate[i]					
$P2Y_2$ purinergic receptor agonist				DIQUAS®, DIQUAS® LX					
Quinolinone derivative				MUCOSTA®					

ł HYALEIN® ophthalmic solution 0.1%; HYALEIN® ophthalmic solution 0.3%; HYALEIN® Mini ophthalmic solution 0.1%; HYALEIN® Mini ophthalmic solution 0.3%

Studies using a semifluorinated alkane to target the lipid layer included patients presenting with ADDE, inflammation and MGD defined as a total MGD score ≥3 [range 0-15] at screening and baseline (secretion of 5 central glands on the lower eyelid was

evaluated, and each was scored from 0-3: 0=normal; 1=thick/yellow, whitish, particulate; 2=paste; 3=none/occluded) and a TBUT ≤5 seconds at screening and baseline. Thus, the inclusion criteria were tailored toward the mechanism of action of the compound even though the primary endpoints were consistent with patients presenting with ADDE and inflammation. [973,974,976,977]

Studies involving a cholinergic agonist additionally targeted patients presenting with a baseline Schirmer score response to cotton swab nasal stimulation of at least 7 mm greater in the same eye that qualified as ADDE. Thus, patients with a marked response to mechanical activation of the trigeminal parasympathetic pathway in the nose were subsequently tested to determine if nicotinic acetylcholine (nACh) receptor activation of the same pathway results in increase basal tear production. [954,979]

A new topical corticosteroid additionally targeted patients presenting with bulbar conjunctival hyperemia at Screening and Day 1 of ≥ 2 as assessed using the Cornea and Contact Lens Research Unit Grading Scale. Corticosteroids cause adrenergically mediated vasoconstriction and non-competitive antagonism of vasodilation due to prostaglandin E and bradykinin. [982] Thus, targeting patients presenting with aqueous-deficient DED, inflammation and hyperemia would be consistent with a mechanism of action for a corticosteroid to reduce dilation of conjunctival blood vessels which can present as hyperemia associated with inflammation (see Table 15). [955,983]

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Table 15. Approved drugs and criteria for defining DED

		Criterion for Defining DED														
		Signs								Symptoms						
Approved Drug	Studies	Number of Functional Meibomian Glands	Meibum Quality	TBUT	Corneal Staining	Conjunctival Staining	Schirmer's Score	Eye Redness	OSDI	VAS	SPEED	Facial Expression Scale	Ocular Dryness	NEI-VFQ-25	SANDE
RESTASIS®	192731 -002, -003, and -501				Sum of corneal and conj. Staining ≥ +5 in the same eye with corneal staining ≥ +2 (Oxford scheme; 0-15)		(without anaesthesia) ≤ 5 mm/5min, if 0 mm/5min then nasal stimulation ≥ 3 mm/5 min		At least 9 responses on the OSDI other than N/A and a minimum score at screening and baseline			Score ≥ 3			
XIIDRA™	Phase 2 and Opus-1	Active lid margin disease not included			Corneal fluorescein staining score ≥ 2 in at least one region of either eye (0-4 scale) at Screening and Baseline; Change in Inferior Corneal Staining (pre to post CAE ≥ +1)		(without anaesthesia) ≥ 1 and ≤ 10 mm/5min	Conjunctiveal redness score ≥ 1 (0-4 scale) in at least one eye (any eye for OPUS-1)					ODS ≥ +3 at 2 consecutive time points intra-CAE		
	Opus-2 and Opus-3	Active lid margin disease not included			Corneal fluorescein staining score ≥ 2 in at least one region of either eye (0-4 scale) at Screening and Baseline; Inferior Corneal Staining ≥ 0.5 at screening and baseline)		(without anaesthesia) ≥ 1 and ≤ 10 mm/5min	Conjunctiveal redness score ≥ 1 (0-4 scale) in at least one eye		Eye Dryness Score ≥ 40 at Screening and Baseline					
CEQUA™		Significant eyelid irregularity not included				Lissamine green conjunctival staining sum score of 3 or more to 9 or less of a total possible score of 12 in the same eye				Global symptom score (based on symptoms of dryness, irritation, or both) rated by the patient of 40 or more (range, 0-100)					
EYSUVIS™		Meibomian gland dysfunction (MGD) not included			Corneal fluorescein staining score at Screening and Day 1 of ≥ 6 NEI (National Eye Institute)		Uanesthetized Schirmer Test score at Screening of ≤ 10 mm	Bulbar conjunctival hyperemia at Screening and Day 1 of ≥ 2 as assessed using the Cornea and Contact Lens Research Unit (CCLRU) scale							A score of > 50 mm on Modified SANDE Severity at Screening; > 40 mm on Modified SANDE Severity on the day prior to Day 1

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TYRVAYA™	Onset-1 (OPP-002)	Blepharitis not requiring treatment and mild Meibomian gland disease that are typically associated with DED were allowed.		Corneal fluorescein staining score of ≥2 in at least one corneal region OR have a sum of ≥4 for all corneal regions		baseline Schirmer's Test Score (STS; with topical anesthesia) of ≤10 mm/5 minutes with a cotton swab nasal stimulation STS at least 7 mm greater in the same eye; <20 mm difference from the study eye Schirmer Test and the fellow eye Schirmer Test Score		
	Onset-2 (OPP-101)	Blepharitis not requiring treatment and mild Meibomian gland disease that are typically associated with DED were allowed.		Corneal fluorescein staining score of ≥2 in at least one corneal region OR a sum of ≥4 for all corneal regions		baseline Schirmer Test Score (STS; with topical anesthesia) of ≤10 mm/5 minutes with a cotton swab nasal stimulation STS at least 7 mm greater in the same eye; <20 mm difference from the study eye Schirmer Test and the fellow eye Schirmer's Test	OSDI score of ≥23 with ≤3 responses of "Not Applicable") at the Screening visit	
MIEBO™	GOBI MOJAVE	MGD defined as total MGD score ≥3 at Screening and Beseline (secretion of 5 central glands on the lower eyelid was evaluated, and each was scored from 0-3: 0=normal; 1=thick/yellow, whitish, particulate; 2=paste; 3=none/occluded). Total score ranged from 0-15	TFBUT ≤5 seconds at Screening and Baseline	Total corneal fluorescein staining score between 4 and 11 (i.e., sum of inferior, superior, central, nasal, and temporal) according to the NEI scale at Screening and Baseline		unanesthetized Schirmer test I score ≥5 mm at Screening and Baseline	OSDI score ≥25 at Screening and Baseline.	
VEVYE	CYS-002	Meibomian gland dysfunction (MGD) not included		Total corneal fluorescein staining score of ≥ 6 (e.g. sum of inferior, superior, central, nasal, and temporal) according to the NEI grading at Screening and Baseline	Total lissamine green conjunctival score (sum of temporal and nasal) of ≥ 2, based on the Oxford grading at Screening and Baseline	Schirmer Test I score between ≥ 2 mm and ≤ 8 mm at Screening and Baseline		Score of ≥ 40 on the dryness VAS at Screening and Baseline

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	ESSENCE -1 (CYS-003)			Total corneal fluorescein staining score of ≥ 10 (e.g. sum of inferior, superior, central, nasal, and temporal) according to the NEI grading at Screening and Baseline	Total lissamine green conjunctival score (sum of temporal and nasal) of ≥ 2, based on the Oxford grading at Screening and Baseline	Schirmer Test I score between ≥ 1 mm and ≤ 10 mm at Screening and Baseline				Subjective complaints of poor near vision that impact activities of daily living, as defined by at least a moderate impact (score ≥ 3) on at least 1 question on the NEI VFQ-25 Questions 5 to 7 in the main questionnaire or Near Vision Subscale, Questions A3 to A5 in the Appendix of Optional Additional Questions, at the screening visit
	ESSENCE 2 (CYS-004)	Meibomian gland dysfunction (MGD) not included		Total corneal fluorescein staining score of ≥ 10 (e.g. sum of inferior, superior, central, nasal, and temporal) according to the NEI grading at Screening and Baseline	Total lissamine green conjunctival score (sum of temporal and nasal) of ≥ 2, based on the Oxford grading at Screening and Baseline	Schirmer Test I score between ≥ 1 mm and ≤ 10 mm at Screening and Baseline			Score of ≥ 50 on the dryness VAS at Screening and Baseline	
IKERVIS	Siccanove (NVG06C103)	Severe blepharitis and/or Meibomian gland disease (MGD) not included	Tear break-up time (TBUT) ≤8 seconds	Corneal fluorescein staining ≥2 and ≤4 (modified Oxford scale, scale 0-5)	Lissamine green staining >4 (Van Bijsterveld scale, scale 0-9)	Schirmer tear test without anaesthesia of ≥2 mm/5 min and <10 mm/5 min		OSDI score ≥23	NOT VAS: At least one moderate to severe symptom of dry eye with a score ≥2 (severity graded on a 4-point scale) i.e., burning/stinging, foreign body sensation, itching, eye dryness, pain, blurred vision or sticky feeling and photophobia	
	Sansika (NVG10E117)	Severe blepharitis and/or Meibomian gland disease (MGD) not included		Corneal fluorescein staining score of 4 on the modified Oxford scale		Schirmer test without anaesthesia scored ≥2 mm/5 min and		OSDI score ≥23		
HYALEIN® ophthalmi	N/A	not listed	not listed	Corneal fluorescein staining score ≥2+(0~3?) "moderate or above"	not listed	not listed	not listed			
HYALEIN® ophthalmic	N/A	not listed	not listed	not listed "refractory or severe"	not listed	not listed	not listed			
H Y	N/A	not listed	not listed	not listed "moderate or	not listed	not listed	not listed			

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				above"			
HYAL	N/A	not listed	not listed	not listed "refractory"	not listed	not listed	not listed
DIQUAS®			≤5 seconds	Corneal fluorescein staining score: ≥3 (a maximum score of 9); Rose Bengal staining score: ≥3 (a maximum score of 9); Lissamine green staining score: ≥3 (a maximum score of 9) [Staining of the temporal conjunctiva, cornea, and nasal conjunctiva, with each graded on a 0-3 scale. The scores of the three regions are summed.]	Schirmer I test: ≤5 mm	Subjective symptoms (including visual disturbance)
MUCOST				A corneal fluorescein staining score of 4 or more and a LGCS score of 5 or more	Schirmer I test: ≤5 mm		Score of 2 or more for 1 or more dry eye–related ocular symptom(s)
DIQUA			≤5 seconds	Corneal fluorescein staining score ≥ 1	Schirmer I test: ≤5 mm		Dryness score ≥ 1 using the DEQS questionnaire

OSDI = Ocular Surface Disease Index

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Obstructive MGD presents as lid telangiectasia, gland orifice capping, gland dropout, altered gland expressibility, and low TBUT). [6,984,985] This distinction from ADDE is important as newer products in development like selenium sulfide ophthalmic ointment [975,978] and devices for treating eyelid glandular abnormalities, [986] have inclusion criteria and associated primary endpoints focusing on glandular morphology and associated evaporative DED. [987-993] Such trials may exclude patients with more severe symptoms, signs of inflammation and significant ocular surface staining. [975] The inclusion/exclusion criteria and endpoints were modelled on the clinical studies performed using a thermal pulsation system. [994] These trial designs targeted at obstructive MGD and associated EDE illustrate a population-targeted trial design aimed toward a group of patients who may be underserved by some of the existing approved drug treatments.

11.3.3 US FDA clear guidance for industry and flexibility for dry eye approvals

The US FDA Draft Guidance for Industry on Dry Eye: Developing Drugs for Treatment [949] recommends the sponsor of a new DED treatment demonstrate efficacy and safety in at least two adequate and well-controlled, multicenter independent studies. The guidance covers trial design, comparator(s), trial population, demonstration of efficacy and safety database requirements in detail. The following section will summarize the key guidelines issued by the FDA. [949]

11.3.3.1 Trial design

The FDA will consider both traditional environmental exposure trials and an environmental challenge-model trial (utilizing a controlled chamber with regulated temperature, air flow, humidity, etc.) to support approval of a new drug. The FDA recommends parallel, randomized by patient, double-masked trials in which the investigational drug group demonstrates superiority over the control group (control agent can be the vehicle). Equivalence and non-inferiority trials are generally discouraged due to difficulty in defining limits for equivalence and cross-over trials will only be evaluable for the first treatment period if any carryover from one period to another is detected. Future work to establish good assay validation (sensitivity) methods (inclusion of both a positive and negative concurrent control), could change the recommendation on equivalence or noninferiority trials.

The FDA guidelines are applicable to trials designed to demonstrate the efficacy and safety of a new product in support of a new drug application and do not preclude the use of such trial designs by companies or sponsors for the purposes of supporting internal decisions related to a compound's ability to progress in development based upon measures of safety or efficacy such and proof of mechanism endpoints, biomarkers and alternative thresholds for significance (e.g., 0.1% not 0.05%).

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Daily exposure to environmental factors can impact DED endpoints: cigarette smoke, smoke from wildfires, air pollution, dust, topically instilled drugs, and allergens are just a few examples. [995-997] A model has been developed to standardize climatic, air quality and visual conditions in a controlled environment. [996,998,999] Aspects of the environment such as humidity, air flow and temperature can be manipulated to impact the ocular surface, to identify individuals who may be more responsive to external environmental factors and to create a reproducible environmental challenge. [999] This model is an attempt to reduce the number of patients, sites, and time to demonstrate the efficacy of an intervention. [996,998,999] Effects can be tested over one day in a controlled environment or longer-term natural exposure trials of 2-weeks or longer. The FDA recommends that safety trials be conducted at least 6 weeks in duration if efficacy trials are of shorter duration. Trials in which the investigational drug is used as an add-on to a standardized treatment regimen are also acceptable. [949]

While confirmatory efficacy trials have used CAE exposure to enrich study populations with patients who can change in severity secondary to environmental stress, the pre-specified primary endpoints supporting approval were environmental. [964] Trials have also used the controlled adverse environment to test an intervention's protection against the effects of an environmental challenge. [1000]

11.3.3.2 Comparator(s)

The comparators and methods for controlling variance have evolved from the first studies of cyclosporine in DED, [951] to the recent introduction of newer formulations and concentrations of the active. [951,956,957,971,972] The first studies of cyclosporine in DED evaluated cyclosporine plus artificial tears as needed to Month 4 and then less than 8 times daily after Month 4 to Month 6. [963] Numerous trials tried to replicate this methodology and failed due to large placebo effects, lack of efficacy and regression of signal which could have resulted from a therapeutic effect for the artificial tears. [963,1001] The subsequent approval of a novel integrin antagonist (lifitegrast) used only the vehicle and excluded the use of artificial tears during the treatment period (See Supplementary Figure 1 for the 0.05% ciclosporine trial design, Figure 2 for the lifitegrast controlled environmental trial design, and Figure 3 for a lifitegrast environmental exposure trial design).

Supplementary Figure 1: Cyclosporine 0.05% (192371-002: Phase 3): Environmental Exposure Study Design

Supplementary Figure 2: Opus 1 (Lifitegrast: Phase 3): CAE Study Design

Supplementary Figure 3: Opus 2 (Lifitegrast: Phase 3): Environmental Exposure Study Design

The FDA acknowledged the advancement in trial methodology and encouraged trial designers to consider that even water is known to be an effective component of topically applied treatments for DED. Vehicle responders may be 1 in 5 of the trial population suggesting this is a significant confounder in a study without a mechanism to screen for vehicle response. [1002] In general, comparative clinical trials should use only the investigational drug's vehicle as a control agent. Trials should demonstrate statistical and clinical superiority over a vehicle control or another treatment regimen. [949]

11.3.3.3 Trial population

The FDA defines a DED population as including patients with ocular complaints consistent with dry eye symptoms. Inclusion criteria should include both objective signs and subjective symptoms. [949] Examples of inclusion and exclusion criteria for approved products are included in Table 14. The FDA includes measures of meibomian gland function, meibum quality and TBUT as both a potential inclusion criterion for a DED population and a potential sign of DED therapeutic effect which can function as a primary sign efficacy endpoint. [975,978] Studies should include patients from relevant demographic subsets, including both men and women and multiple age, ethnic background, and eye color groups consistent with the US population. Confirmatory efficacy studies do not need to be completed in the US so long as the study population contains the relevant demographic subsets found within the US.

DED secondary to cicatrisation (following irradiation, alkali burns, Stevens Johnson syndrome, cicatricial pemphigoid) or the destruction of conjunctival goblet cells (vitamin A deficiency) represent a specific, severely affected patient population. In general, these are considered separate indications, and patients with these conditions should be studied separately from routine DED.

Severe blepharitis or obvious inflammation of the lid margin can interfere with the interpretation of trial results. In general, patients with these conditions should be studied separately from routine DED. [949]

11.3.3.4 Demonstration of efficacy

In general, safety and efficacy should be demonstrated in at least two adequate and well-controlled, multicenter (at least 2 sites), independent trials. Primary efficacy endpoints include one of three options:

Sign and Symptom endpoint (s): [949,950]

1. Signs and Symptoms - A statistically significant difference between the investigational treatment and vehicle from baseline for at least one objective prespecified sign of DED (mean group score of test versus vehicle) AND at least one subjective prespecified symptom of DED (mean group score)

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2. Corneal Staining - A statistically significant difference between the percentage of patients achieving a complete resolution of corneal staining, or
3. Tear Production - A statistically significant difference between the percentage of patients achieving a 10-millimeter increase or more in Schirmer score

Achieving any one of these 3 endpoints can support approval of a product for the treatment of the signs and symptoms of DED. Several marketed products have sought approval based upon sign and symptom primary endpoint(s) and have been granted US approval using pre-specified secondary endpoints of a known surrogate endpoint, a 10-millimeter increase or more in Schirmer score, when the prespecified signs and symptom endpoints failed in confirmatory efficacy studies. [951,956,957,963,971,972]

The “10-millimeter increase or more in Schirmer score” was validated as part of the new drug application for 0.05% ciclosporine and a description of the methods used to validate the endpoint are described in the “Validation of the Clinical Relevance of the Clinical Sign” section authored by Dr. William M Boyd from the US FDA as one of the medical reviewers for NDA 21-023.

11.3.3.5 Safety database

The FDA specifies the size of a drug exposure database to support a new drug application based upon Adverse Event detection rates. The clinical program should include enough patients to identify adverse drug events that occur at a rate of 1 percent or greater. To accomplish this, FDA recommends that approximately 400 or more patients using the investigational drug complete treatment with a concentration of the investigational drug at least as high as proposed for marketing and with a frequency at least as frequent as proposed for marketing. Before submission of a marketing application, the sponsor should ensure at least 300 patients have completed at least 6 weeks of follow-up after the initiation of treatment and at least 100 patients have completed 12 months of follow-up after the initiation of treatment to support chronic use. Acute use, 6 weeks or less of administration each year, does not require long-term follow-up of 12 months.

For reformulations of drug substances that are already approved in the same dosage form, same route of administration, and the same or lower concentration, the FDA recommends the sponsor ensure that a marketing application has safety information from at least 100 patients treated for at least 6 months.

FDA recommends that the following evaluations be performed in each eye and reported separately for each eye (regardless of which eye or eyes are treated):

- Best corrected, distance visual acuity (4 meters in distance or more) at every visit.
- A patient comfort examination before and after drug administration at every visit.

- A slit lamp examination of the anterior segment that includes the cornea, conjunctiva, anterior chamber, iris, lids, and lashes. At a minimum, examinations should be performed at baseline, midway through the trial, the end of treatment, and 2 weeks after treatment discontinuation.
- Endothelial cell count, systemic clinical and laboratory evaluations, and dilated fundus examinations at baseline and at the end of trial or at month 3 (whichever is later) in at least one trial.

DED also occurs in pediatric patients. [1003] Sponsors may consider a pediatric assessment waiver request when submitting their required pediatric study plans under the Pediatric Research Equity Act. [949]

11.3.4 EMA guidance for industry and flexibility for DED approvals

EMA guidelines recommend demonstration of efficacy for 6 months, as DED is often chronic and comparison against artificial tears, which are the standard of care is recommended. [1004] EMA application submissions for lifitegrast and cyclosporine were both withdrawn because of data demonstrating insufficient efficacy. [953,1005] The only EMA-approved treatment is cyclosporine 0.1%, however, the indication is not for DED but rather "severe keratitis in patients with DED, not improved despite treatment with tear substitutes." [1006]

Regarding the selection of sign endpoints, the EMA considers both the target population and the mechanism of action of the compound: Corneal staining, Schirmer score and TBUT are established clinical endpoints. A composite measure is recommended using a validated questionnaire for a symptom endpoint (e.g., OSDI) and use of single worst symptom questions is not recommended. Sponsors are encouraged to consider a relevant effect size rather than only statistical significance. For anti-inflammatory products, since inflammation is considered a secondary manifestation, biomarkers may be required to quantify treatment benefit.

The EMA will generally accept development programs completed in the US assuming the studies comply with the guidelines outlined above. The EMA only requires demonstration of safety and efficacy in one large adequately designed Phase 3 trial. The primary endpoint for a cyclosporine 0.1% Phase 3 trial was a sign and symptom composite responder rate. Specifically, the primary endpoint was the corneal fluorescein staining-OSDI composite responder rate at Month 6 (i.e. end of Part 1). A corneal fluorescein staining-OSDI responder was defined as a patient simultaneously satisfying the following conditions: [1006]

- Improvement of 2 points or more from baseline in corneal fluorescein staining based on the modified Oxford scale (i.e. change in corneal fluorescein staining ≤-2), and

- Improvement by 30% or more from baseline in OSDI (i.e. % change ≤-30%).

This is not an endpoint which has been acceptable to the FDA to date. The completed confirmatory efficacy trials (SICCANOVE and SANSIKA) failed to demonstrate consistent superiority on the primary endpoint and the product was subsequently approved on a secondary sign endpoint by showing a statistically significant reduction in corneal fluorescein staining versus vehicle. [951,1007,1008] No significant improvement in symptoms was demonstrated along with the observed significant and clinically relevant improvement in corneal staining and the indication was restricted to “severe keratitis treatment” instead of a broader DED population.

11.3.5 PMDA guidance for industry and flexibility for approvals

Historically, there has been a considerable time delay between development and approval in the United States and an ultimate launch in Japan. [1009] The Pharmaceutical and Medical Device Agency (PMDA) requirements for drug approval in Japan have been harmonized across regions with toxicology and manufacturing generally not requiring additional work or rework. [1009]

Working with the PMDA involves a multistep process. This process is initiated with the jizen mendan (“preliminary meeting”) which has no fee associated with it, is shorter in length and requires less of a briefing package. The intention of this meeting is to confirm the materials that are to be submitted and the sponsor’s questions that will be discussed at the subsequent taimen jogen (“full consultation meeting”). At the full consultation meeting, a more detailed briefing document is submitted in advance and written comments are provided by the PMDA following the meeting.

The PMDA views ethnic differences as critically important. Phase 1/pharmacokinetics studies can be completed in the United States, if done in Japanese-American patients defined as first- or second-generation Japanese Americans of pure Japanese descent. PMDA generally requires a Phase 1/pharmacokinetics study even in cases where a global development program may have proceeded into Phase 2 or 3 trials in other regions, but with a non-Japanese population. Recently the PMDA issued a notice regarding the basic approach to conducting Phase 1 studies in Japanese patients prior to global clinical trials for drugs whose clinical development has already started overseas [1010,1011], stating the following. “In general, it is not mandatory to conduct a Phase I study in each race/ethnicity or country/region before initiating multi-region clinical trials. In principle, an additional Phase I study in Japanese is not needed unless it is deemed necessary after assessing whether the safety/tolerability of the dosage to be evaluated in the multi-region clinical trials in Japanese participants can be explained and the safety in clinically acceptable/manageable on the data available prior to Japan’s participation.” This concept seems to be some progress in the development of

pharmaceuticals in Japan. The PMDA typically wants to see dose ranging established in the Japanese population specifically because of potential population differences in drug activity and practice patterns. Confirmatory Phase 3 study is required to be completed in Japan in part or whole. The PMDA generally requires a single, properly designed Phase 3 clinical trial to demonstrate safety and efficacy, similar to the EMA and discourages *post-hoc* analysis for approval.

Phase 3 trials for the approval of diquafosol sodium ophthalmic solution [952,1012-1014] and rebamipide ophthalmic suspension [952,1015] both required an active comparator (0.1% hyaluronic acid ophthalmic solution was used in both) which necessitated a dosing regimen of 6 times daily. This may have hampered interest in developing product in Japan due the risk posed by even vehicle effects at such a high dosing frequency. The introduction of long acting diquafasol sodium ophthalmic solution in 2022 which is only dosed 3 times daily may have changed this risk profile. Long acting diquafasol also employed a vehicle placebo in its Phase 3 trial. [1016]

All 3 products were tested with primary endpoints at 4 weeks and all measured primary treatment benefit using staining endpoints:

1. Diquafosol sodium ophthalmic solution: A non-inferiority margin of 0.34 for the study (between-treatment difference in the mean change in the fluorescein & rose bengal staining score [diquafosol - HA])
2. Rebamipide ophthalmic suspension: Change from baseline in corneal fluorescein staining score and the lissamine green conjunctival staining score
3. Long acting diquafosol sodium ophthalmic solution: Change in corneal fluorescein staining score from baseline to week 4

Similar to those products that were approved in the US targeting a sub-population of DED patients with ADDE and inflammation, the Phase 3 trials for these products targeted patients with evidence of ocular surface staining, decreased tear production and dry eye symptom(s). [952,958,1012-1017] Diquafosol sodium ophthalmic solution and long acting diquafosol sodium ophthalmic solution also required a TBUT ≤5 seconds which was also required as part of the development program for perfluorohexyloctane ophthalmic solution. [974,976,977] In these trials, diquafosol sodium ophthalmic solution and long acting diquafosol sodium ophthalmic solution were reported to promote the secretion of tear fluid with aqueous and mucin components and as well as to increase lipid layer thickness in normal human eyes. [1018] Thus, the Phase 3 trials were enriched based upon the known mechanism of action of a P2Y2 purinergic receptor agonist. It should be noted that a P2Y2 agonist and rebamipide failed in multiple clinical trials in the USA to assess their efficacy in treating DED. [952,958,1012-1018]

The primary endpoint for a hyaluronic acid Phase 3 study was a sign and symptom composite responder rate. [980] This is not an endpoint which has been acceptable to the FDA to date. The diquafosol ophthalmic solution Phase 3 also defined a non-inferiority margin between-treatment difference in the mean change in the fluorescein and rose bengal staining score (diquafosol compared to hyaluronic acid). [1012,1014] The FDA has not accepted a non-inferiority margin to support approval of a drug for DED. Thus, the PMDA has allowed flexibility in defining the type of comparison that will support approval: superiority or non-inferiority to an active comparator.

The PMDA does not have the same guidance for industry on developing drugs for DED as the FDA. In addition, the diagnostic criteria for DED were revised in 2016 [1019] and no longer include tear production or keratoconjunctival disorders in the diagnostic criteria. Two of three approved drugs for the treatment of dry eye, diquafosol ophthalmic solution [958,1012-1014] and rebamipide ophthalmic suspension [952,1015] were approved before the revision and the remaining drug, long acting diquafosol sodium ophthalmic solution [1016,1017] had an additional additive that reduced the frequency of eye drops, but the efficacy itself did not change, so the primary endpoint in trial was the fluorescein corneal staining score. With the revision of diagnostic criteria (ie., tear production and keratoconjunctival disorders are no longer included), it may be necessary to have a robust discussion with regulatory authorities regarding the setting of the primary endpoint based on the mechanism of action of a new drug.

11.4 Devices

11.4.1 Regulatory pathway (FDA)

The regulatory process for devices is very different from the process for pharmaceuticals. [953,1020]This section briefly describes the process utilized by the FDA, but there are likely differences in other regions of the world. [953] Similar to its role in the approval of drugs, the FDA is responsible for protecting the public by assuring the safety and efficacy of medical devices. Although both drug and device approval have increased over the last decade, there is an increasing gap favoring new devices in the USA. [1020-1022] In general, devices have a shorter time for bringing to market and significantly less cost but may have lower profitability. [1021] The key steps to FDA device approval start with determining if a product is a device and identify its purpose, and then to determine its risk classification: low, moderate or high risk (Figure 5).

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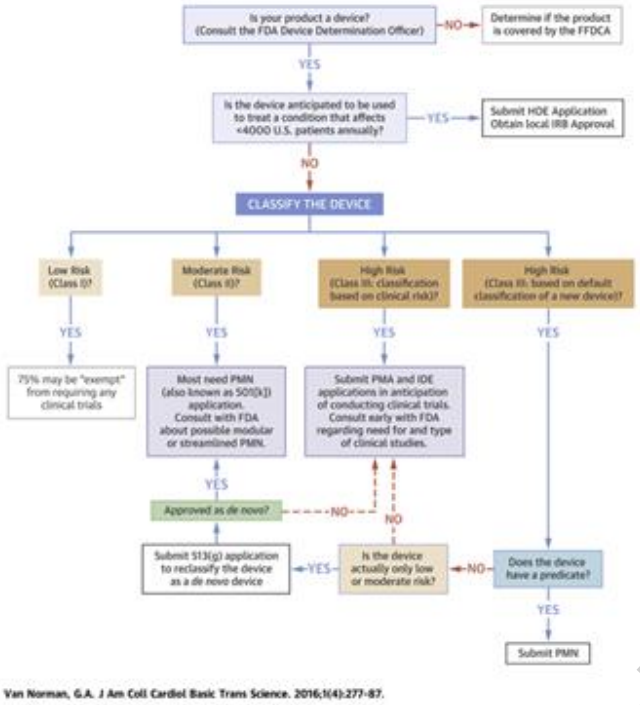


Figure 5. Determining the Regulatory Pathway for a Device (from Van Norman 2016). [1021]

ProvRisk is assessed to determine the device chance of presenting harm to patients, including from malfunction or improper use. The risk classification generally indicates the pathway and type of submission required by the FDA. The type of premarket submission depends on the risk assessment:

1. Investigational device exemption - devices used to collect safety and effectiveness data
2. Premarket notification (510k) – the device is safe and effective and substantially equivalent to a legally marketed device
3. Premarket approval application - for Class III device (high risk) that needs a clinical trial to demonstrate safety and effectiveness
4. *De novo* - for devices that have no existing classification regulation
5. Humanitarian device exemption – for devices for a disease that affects less than 4000 US patients yearly and has demonstrated reasonable assurance of safety and probable benefit
6. Low risk (Class I devices) (e.g., the Meibomian Gland Evaluator, marketed by Johnson & Johnson Surgical Vision, Inc.) and moderate (Class II devices) (e.g., iLux® System by Tear Film Innovations, Inc. and the LipiFlow Thermal Pulsation System manufactured by TearScience, Inc.) risk devices can be “exempt” or require a 510(k), while the highest risk (Class III) device requires a pre-market

approval. Class III risk devices will ultimately require a clinical trial and an Investigational Device Exemption so that an unapproved device can be used in clinical trials.

Trial design for Class III devices: Given that it is often difficult or unethical to have a sham control, there is more flexibility on what type of trial is needed and discussion with the FDA will determine what is acceptable. [1021] For the control arm of a device trial, one may be able to use an active comparator (i.e., compare with an FDA approved device), use metrics from pre- and post-treatment with a given device, or compare with registry data. Typically, a double blind randomized clinical trial cannot be done with devices, as is done with pharmaceuticals. [1022] Mixed devices, pharmaceutical and device combined, are often reviewed by both divisions of the FDA, with one taking the lead, where communications with the FDA determine the regulatory pathway. [953]

11.4.2 Regulatory pathway (Japan, PMDA)

In Japan, as in the US, the development and approval processes for medical devices differ depending on the type of device. Medical devices are classified into four classes according to the degree of risk to the human body. Medical devices in Class I require only notification, those (except new devices*) in Class II require certification by the certification body, and those in Class III and IV require approval by government authority. [953,1023]

Medical devices classified as Class III and IV are considered to have a high risk to the human body and many of them require clinical trials. On the other hand, medical devices used outside the body that are not implanted in the body and do not have a high risk to the human body are often classified as Class II. With Class II, if the probability of benefit is high when considering risk and benefit, it may be judged that evaluation is possible without the need for clinical trials, as compared to Class III and IV. Class I medical devices are those that are considered to have an extremely low risk to the human body.

For medical devices that have already been approved overseas, it is possible to extrapolate test results from overseas clinical trials, although ethnic differences must be taken into consideration. Therefore, unlike drugs, some medical devices are approved without domestic clinical trials by extrapolation to overseas clinical trials.

*New medical devices in Class II require approval by government authority (PMDA and MHLW: Ministry of Health, Labour and Welfare).

11.4.3 Artificial tears (US, FDA)

Artificial tears contain FDA-approved demulcents as active ingredient(s). Demulcents are primarily water-soluble, topically applied polymers, which may protect and lubricate

mucous membrane surfaces relieving ocular dryness and irritation. [953,1021] A list of approved demulcents and their concentration range has been provided by the FDA: [1024]

1. Cellulose derivatives:
    a. Carboxymethyl cellulose (CMC),
    b. Hydroxypropyl methylcellulose (hypromellose),
    c. Hydroxyethylcellulose,
    d. Methylcellulose,
2. Dextran 70
3. Gelatin
4. Polyols:
    a. Glycerin,
    b. Polyethylene glycol (300, 400),
    c. Polysorbate 80,
5. Polymers:
    a. Polyvinyl alcohol,
    b. Polyvinyl pyrrolidone (povidone).

Federal law does not require premarket approval for over-the-counter eye drops, including artificial tears, but does require eye drops to be sterile for safe use. The existence of the FDA's pre-approved monograph which covers artificial tears means that if a developer lists an active ingredient that is not on the monograph, they will need to provide clinical data supporting the safety of its use. Manufacturers can list potentially "active ingredients" under "inactive ingredients" to avoid clinical trials.

While clinical data demonstrate that artificial tears show benefit for the management of DED and their control is critical during drug/device trials to remove a confounding source of variance, there are no regulatory standards for study design, clinical end points, patient inclusion/exclusion criteria, or accepted methods to control for potential sponsor bias when evaluating comparative efficacy claims. [1024]

11.4.4 Artificial tears (Japan, PMDA)

Most artificial tears in Japan are classified as OTC drugs that do not require a prescription. Over-the-counter drugs require clinical trial results if they contain ingredients that are not included as active ingredients in any of the drugs specified in the Japanese Pharmacopoeia, but do not require clinical trials if they do. [953,1021]

11.4.5 Devices approved for DED/MGD

Since the first TFOS DEWS report, there have been several devices FDA approved targeting meibomian gland secretion or mechanical tear stimulation. [993] The first thermal pulsation device [990,991] was approved by the FDA for the application of localized heat and pressure therapy in adults with chronic cystic conditions of the eyelids, including

MGD. It was approved after it met the primary study effectiveness endpoint of improvement at 2 weeks from baseline in the average number of meibomian glands yielding clear liquid secretion as compared to a warm compress control. 69 subjects (138 eyes) were randomized to the thermal pulsation treatment and 70 subjects (140 eyes) were randomized to the warm compress control group. The same system is also a medical device approved without clinical trials in Japan.

Subsequent thermal pulsation devices could be approved through a single non-inferiority clinical trial that compared the device to the approved system as the predicate device. One example demonstrated non-inferiority to the original system using the primary effectiveness endpoints defined as the change from baseline to 1 month for TBUT and total Meibomian Gland Secretion Score. [993] There are numerous examples of other devices, including those using intense pulsed light, that have been evaluated as treatments of evaporative DED or MGD. [987-989]

The Warming Moist Chamber Goggle (Dr.eye®), an NMPA-approved therapeutic device, is indicated for the management of DED and MGD. It facilitates localized thermal and hygrometric conditions, which may enhance tear film stability. Empirical evidence [1025] has demonstrated that this device alleviates ocular discomfort and improves tear film parameters, beyond topical sodium hyaluronate.

A neurostimulation device received FDA approval as an electromechanical tear stimulator that is a non-implantable device intended to increase tear production via mechanical stimulation. The device underwent a prospective, open-label, single-arm, multi-center study in 108 subjects and a multi-center, nonsignificant-risk, prospective, double-masked, randomized, sham controlled, single visit clinical trial that enrolled 60 subjects. For the latter study, the primary endpoint achieved was an increase in the mean within-subject Schirmer score post- vs. pre-stimulation when compared to a sham treatment group in subjects with baseline Schirmer score < 10mm.

11.5 Design features to enhance clinical trial data quality and decisions

This section covers ways to enhance clinical trial data quality and data-supported decisions on whether to proceed with an intervention under development for a particular indication. (See TFOS DEWS III Management and Therapy Report, Section X, [2])

11.5.1 Biomarkers / proof of mechanism / early signal of efficacy

One of the most challenging aspects of randomized controlled trials for DED is the lack of objective, minimally invasive metrics for diagnosing DED for patient selection and for use as efficacy endpoints. The typical use of signs and symptoms of DED, nearly always comes with the caveat that DED affects ”a heterogenous group of subjects”. Outcome measures in RCT though called “objective,” such as corneal staining, TBUT

and Schirmer test value, are all subject to bias from the observer and /or biologic and environmental variables-reflex tearing, time of day, humidity etc. Ophthalmology does not have a lot of reliable, validated markers that correlate with clinically relevant findings in DED. [953,1026] (See [953], DEWS II Clinical Trials Design Report 2017)

Recent examples of FDA cleared biomarkers include a point of care immunoassay test for the *in vitro* detection of elevated levels of the MMP-9 protein in human tears, from patients suspected of having DED [1027] Osmolarity testing is another FDA approved biomarker.

Biomarkers were used in support of the approval of 0.1% cyclosporine and are referenced in the EU Assessment report (level of HLA-DR expression). [1028] More recently a genetic marker (TNFR1 marker) was used in evaluating anti-TNF alpha, licaminlimab, and demonstrated improved outcome measures in a genetically defined subgroup analysis. [1029] To date, no biomarker other than 10-millimeter increase or more in Schirmer scores has supported regulatory approval for a DED treatment with the US FDA where they could theoretically serve as a primary sign endpoint if validated as clinically meaningful with an associated improvement in a primary symptom endpoint.

11.5.2 Missing data considerations

See TFOS DEWS II Clinical Trials Design report. [953]

11.5.3 Global trial designs

Diagnostic tools and outcome measures have often been highly variable resulting in inconsistent observations across clinical trials and programs. This has complicated clinical development in that multiple trials may be required to achieve a successful drug approval (e.g., lifitegrast ophthalmic solution 5.0%), [962,964-966] but it has also impacted harmonization of regulatory requirements across regions. Regulatory requirements can vary considerably across the major DED markets which include US, the European Union (EU), and Japan.

For the clinical development of drug therapies, the commonly selected sign and symptom endpoints used in the trials have been similar across the major markets: corneal or conjunctival staining (although using different scales), Schirmer score, and symptom questionnaires. For symptom questionnaires there seems to be a trend favoring overall symptom questionnaires (e.g., OSDI) and Visual Analogue Scales (i.e., ocular discomfort or dryness). [951,952,955-957,962-966,972,974,977,979,1012]

Across the US, EU and Japan compounds targeted at a similar sub-population of patients who present predominately with signs of inflammation or its impact (e.g., hyperemia, corneal staining, and conjunctival staining) and ADDE (e.g., decreased tear

production) have been approved. [951,952,955-957,962-966,972,974,977,979,1012] The major difference between the regions is the duration of follow-up for the primary endpoints. The US allows multiday, natural exposure trials of 2-weeks duration or longer, EMA allows natural exposure trials of 6-months duration or longer and Japan allows natural exposure trials of 4-weeks duration or longer. [949,1004,1009,1023]

Taken together, the commonalities between the regions could allow for a single trial with a common disease population and endpoint to support approval in more than one region leveraging multiple statistical analysis plans specifying primarily endpoints at different timepoints. [975,978] This allows a single trial to potentially support approval in both the US and EU. To include Japan a sub-population of Japanese participants would need to be included. While this could work for new compounds targeting the patient populations and endpoints which have already been accepted in the major markets, it is unclear how alternative endpoints (e.g., meibomian gland morphology-related endpoints) will be accepted globally.

11.5.4 Compliance monitoring

Adherence to prescribed therapy is essential for quality clinical trials data and reliable comparison of treatment modalities. Strategies for assessing adherence to treatment vary in their applicability to ophthalmic trials but may include assessment of amount of returned investigational product, patient report by regular electronic or other means, direct observation of drug or device use, systemic or ocular monitoring of drug or metabolite levels or more technical solutions such as electronic monitoring, for example app control of reusable devices can map how long and how often they are used or sensors in topical medication containers.

11.6 Conclusions

Newer market entrants have utilized clinical trial designs with enriched populations of patients that take into consideration the mechanism of the drug under study, particularly as it pertains to defining sign endpoints. [973,974,976,977] This has increased their likelihood of technical success and is aligned with the US FDA Draft Guidance. This guidance states that a statistically significant difference between the investigational treatment and vehicle for at least one objective prespecified sign of DED (mean group score for test versus vehicle) AND at least one subjective prespecified symptom of DED (mean group score) can be used to support approval. A sign endpoint consistent with a drugs mechanism of action can be selected to demonstrate that the drug under study has its intended effect. The significance of the symptom endpoint then establishes the clinical relevance of the observed effect the drug under study has upon how a patient feels making it clinically relevant. [949] Thus, the recommendations from the TFOS DEWS II, the industry report to identify three fundamental elements needed to achieve a

significant improvement in trial success rates (Pillars 1-3) and the US FDA Draft Guidance for Industry on DED all coalesce on the same conclusion for achieving regulatory and technical success for getting a new drug approved: match the sign endpoint to the intended mechanism of the drug under study.

In conjunction with better selection of patients, trial designs have significantly evolved over the 20 years of research. They have gone from little control over confounding variables like artificial tear use [963] to careful control over comparator arms and run-in periods within confirmatory efficacy studies. [954,955,966,973,974,976,977,979] Global regulatory agencies have recognized this evolution with the FDA even encouraging trial designers to consider that even water is known to be an effective component of topically applied treatments for DED. Thus, comparative clinical trials should use only the vehicle from the investigational drug as a control agent.

The agencies have also recognized that trial endpoints could change in sensitivity along with disease severity and as a result have also evolved the requirements for replication in confirmatory efficacy studies. The US FDA Draft Guidance recommends the sponsor of a new DED treatment demonstrate efficacy and safety in at least two adequate and well-controlled, multicenter independent studies. This is because the agency recognized that different study designs, populations and timepoints may be needed to support sign and symptom endpoints in different studies. [949] Global regulatory agencies also allow pre-specified and alpha corrected secondary endpoints to support approval even if the primary endpoint miss statistical significance [951,956,957,971,972,1007,1008] While major differences still remain for trial requirements across the major markets, advances in trial design, patient identification and statistical methods will undoubtedly continue to advance allowing for more global confirmatory efficacy studies that can support approvals in more than one region.

12. Summary

This report has explored key research published since the 2017 TFOS DEWS II Workshop reports to underpin the evidence described in the TFOS DEWS III Diagnostic Methodology [1] and Management and Therapy [2] reports and to present how new findings from each of the topic areas considered, have informed the drivers of disease. Each topic area has also identified key research needs for the next short and medium time frame.

Table 16 below shows how each of the sections of the Digest report inform evidence for the drivers of disease. Figure 6 illustrates the key findings from these interdisciplinary areas, all of which have informed diagnosis and methodology and pathways towards new treatment modalities. The importance of disease subtyping and the relevance of new biomarkers has been consistently recognized.

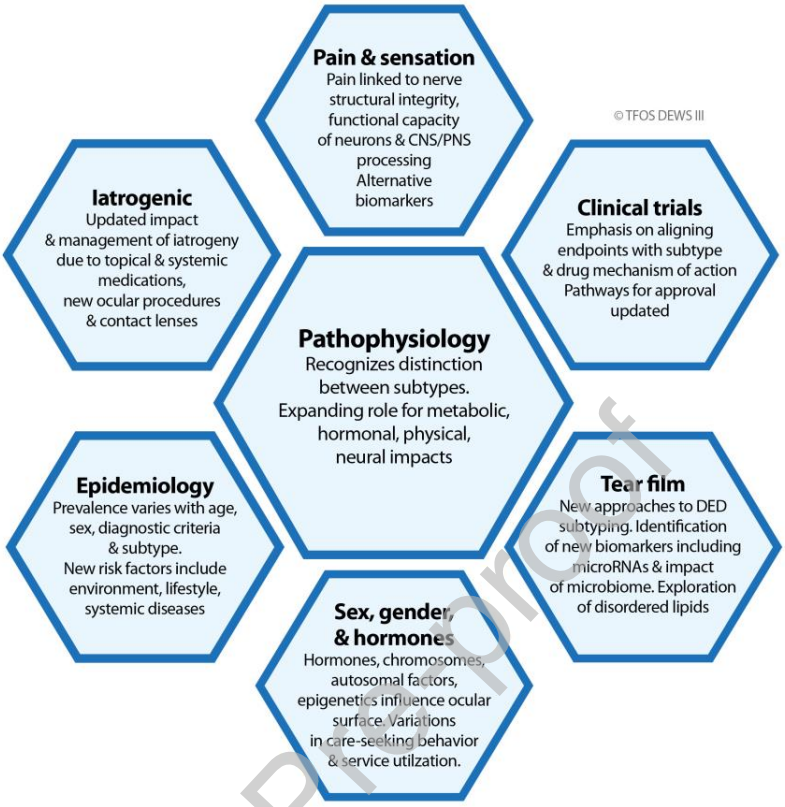


Figure 6. Key findings from interdisciplinary reports

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Table 16. Digest sections and drivers of DED

	Drivers of DED								
	Tear film lipid abnormalities	Tear film aqueous abnormalities	Tear film mucin/glycocalyx abnormalities	Anatomical juxtaposition abnormalities	Blink/lid closure abnormalities	Lid margin abnormalities	Nerve abnormalities	Ocular surface cell abnormalities	Inflammatory /oxidative stress abnormalities
Sex, gender, and hormones	✓□	✓□		□			□	✓□	□
Epidemiology	□	□	□		□		□	□	□
Pathophysiology	□	□	□	□	□	□	□	□	□
Tear film	✓□	✓□	✓□	✓□	✓□	✓□			
Pain and sensation							✓□	✓□	✓□
Iatrogenic	✓□	✓□	✓□	✓□	✓□	✓□	✓□	✓□	✓□
Clinical trials design	✓□	✓□	✓□				✓□	✓□	✓□

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14. Figures

Figure 1. Thyroid eye disease

Figure 2. Prisma flowchart of the literature search outcome for prevalence studies published between 18 September 2015 and 29 June 2024.

Figure 3. Prevalence of DED based on age and sex for different diagnostic criteria

Figure 4. Treatments for Dry Eye Disease (DED)

Figure 5. Determining the regulatory pathway for a device (from Van Norman 2016)

Figure 6. Key findings from interdisciplinary reports

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